



REVIEW ARTICLE

Canonical and noncanonical Wnt signaling: Multilayered mediators, signaling mechanisms and major signaling crosstalk

Kevin Qin ^{a,b,1}, Michael Yu ^{a,b,1}, Jiaming Fan ^{b,c},
Hongwei Wang ^b, Piao Zhao ^{b,d}, Guozhi Zhao ^{b,d}, Wei Zeng ^{b,e},
Connie Chen ^b, Yonghui Wang ^{b,f}, Annie Wang ^{b,h},
Zander Schwartz ^{b,g}, Jeffrey Hong ^b, Lily Song ^b,
William Wagstaff ^b, Rex C. Haydon ^b, Hue H. Luu ^b,
Sherwin H. Ho ^b, Jason Strelzow ^b, Russell R. Reid ^{b,h},
Tong-Chuan He ^{b,h,*}, Lewis L. Shi ^{b,*}

^a Chicago Medical School, Rosalind Franklin University of Medicine and Science, North Chicago, IL 60064, USA

^b Molecular Oncology Laboratory, Department of Orthopaedic Surgery and Rehabilitation Medicine, The University of Chicago Medical Center, Chicago, IL 60637, USA

^c Ministry of Education Key Laboratory of Diagnostic Medicine, and Department of Clinical Biochemistry, The School of Laboratory Medicine, Chongqing Medical University, Chongqing 400016, China

^d Departments of Orthopaedic Surgery and Urology, The First Affiliated Hospital of Chongqing Medical University, Chongqing 400016, China

^e Department of Interventional Neurology, The First Dongguan Affiliated Hospital, Guangdong Medical University, Dongguan, Guangdong 523475, China

^f Department of Clinical Laboratory Medicine, Shanghai Jiaotong University School of Medicine, Shanghai 200000, China

^g School of Biomedical Engineering, Vanderbilt University, Nashville, TN 37235, USA

^h Laboratory of Craniofacial Biology and Development, Department of Surgery Section of Plastic Surgery, The University of Chicago Medical Center, Chicago, IL 60637, USA

Received 3 August 2022; received in revised form 1 November 2022; accepted 29 January 2023

Available online 24 March 2023

* Corresponding author. Molecular Oncology Laboratory. The University of Chicago Medical Center. 5841 South Maryland Avenue, MC3079, Chicago, IL 60637, USA. Fax: +1 (773) 834 4598.

E-mail addresses: tche@uchicago.edu (T.-C. He), lshi@bsd.uchicago.edu (L.L. Shi).

Peer review under responsibility of Chongqing Medical University.

¹ These authors contributed equally to the work.

KEYWORDS

β -catenin;
Canonical Wnt;
Noncanonical Wnt;
Signal transduction;
Signaling crosstalk

Abstract Wnt signaling plays a major role in regulating cell proliferation and differentiation. The Wnt ligands are a family of 19 secreted glycoproteins that mediate their signaling effects via binding to Frizzled receptors and LRP5/6 coreceptors and transducing the signal either through β -catenin in the canonical pathway or through a series of other proteins in the noncanonical pathway. Many of the individual components of both canonical and noncanonical Wnt signaling have additional functions throughout the body, establishing the complex interplay between Wnt signaling and other signaling pathways. This crosstalk between Wnt signaling and other pathways gives Wnt signaling a vital role in many cellular and organ processes. Dysregulation of this system has been implicated in many diseases affecting a wide array of organ systems, including cancer and embryological defects, and can even cause embryonic lethality. The complexity of this system and its interacting proteins have made Wnt signaling a target for many therapeutic treatments. However, both stimulatory and inhibitory treatments come with potential risks that need to be addressed. This review synthesized much of the current knowledge on the Wnt signaling pathway, beginning with the history of Wnt signaling. It thoroughly described the different variants of Wnt signaling, including canonical, noncanonical Wnt/PCP, and the noncanonical Wnt/ Ca^{2+} pathway. Further description involved each of its components and their involvement in other cellular processes. Finally, this review explained the various other pathways and processes that crosstalk with Wnt signaling.

© 2023 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Introduction

Wnt signaling is an important evolutionarily conserved pathway that regulates a diverse range of cellular activities. The importance of these pathways in cell proliferation, differentiation, and migration has led to extensive study of its various components, but there is still much to be learned and discovered about this pathway and its various interactions with other pathways. The extent of the research on this pathway has led to it becoming the subject of many evolving therapies. Many of its components have been implicated in the development of treatments for various conditions, such as cancer, neurodegenerative diseases, congenital disorders, and even diabetes and heart disease. Wnt signaling is truly a fundamental pathway to much of human development and health.

The discovery of the first Wnt gene dates back 40 years ago to 1982 during experiments intended to discover proto-oncogenes via activation by proviruses.^{1,2} These experiments led to the discovery of *int1*, which causes a tumor of mammary epithelial cells when activated.¹ Eventually, it was discovered that mice and human *int1* homologs shared 99% of their amino acid sequences, elucidating the high degree of conservation of the *int1* proto-oncogene.³ Prior to this, Sharma and Chopra identified a gene that codes for the development of the wings in *Drosophila melanogaster*, naming the gene wingless (*Wg*).⁴ Additionally, molecular hybridization identified a similar *int1* homologue in *Drosophila*.⁵ This gene was initially called *Dint1*, but isolation and sequencing of clones of the *Dint1* gene determined that it was identical to the *Wg* gene.^{5–7} Later, it was decided that the *int1* and *Wg* genes would be named together as *Wnt1* to reduce confusion with the other *int* genes. *Int2* was renamed to *FGF3*, *int3* was renamed to

Notch4, and *int4* is now called *Wnt3A*.⁸ The Wnt ligands are now known to be a group composed of 19 glycoproteins that each bind receptors at the cell surface to trigger intracellular signaling cascades to modulate gene expression.^{9–13} Each of the Wnt ligands is a cysteine-rich protein that is 350–400 amino acids in length, with an N-terminal signal sequence targeting them for secretion (Table 1).¹⁴

Wnt ligand family: canonical vs. noncanonical

Canonical and noncanonical Wnt ligands

Wnt signaling can be categorized into two pathways, the β -catenin-dependent pathway (canonical) and the β -catenin-independent pathway (noncanonical).^{15,16} The canonical pathway is important for inducing cell proliferation, differentiation, and maturation.¹⁶ It is also vital in producing proper body-axis specifications.¹⁷ The pathway is activated via the Wnt1 class ligands, which include Wnt2, Wnt3, Wnt3a, and Wnt8a.¹⁷ The canonical pathway is associated with the transport of β -catenin to the nucleus upon Wnt binding to the Frizzled (Fz or Fzd) receptor and the coreceptors LDL-receptor-related proteins 5 and 6 (LRP5 and LRP6)^{18,19} (Fig. 1). The Fz receptor contains a cysteine-rich domain (CRD) that is used to bind Wnt.²⁰ In mammals, 10 Fz receptors have been identified.^{17,21}

If Wnt ligand binding does not occur, a destruction complex that is normally inhibited by Wnt removes the β -catenin. This destruction complex is composed of adenomatous polyposis coli (APC) protein, Axin, serine/threonine kinase glycogen synthase kinase 3 (GSK-3), casein kinase 1 (CK1), the E3-ubiquitin ligase β -TrCP, and protein phosphatase 2A (PP2A).²² While PP2A can impact Wnt signaling

Table 1 List of Wnt genes and orthologs in humans and model organisms.

Human		Mouse		Xenopus	Zebrafish	Drosophila
Wnt Gene	Chr	Ortholog	Chr	Ortholog	Ortholog	Ortholog
WNT1	12	Wnt1	15	Wnt1	Wnt1	Wg
WNT2	7	Wnt2	6	Wnt2a	Wnt2	
WNT2B/13	1	Wnt2b/13	3	Wnt2b	Wnt2b	
WNT3	17	Wnt3	11	Wnt3	Wnt3	
WNT3A	1	Wnt3a	11	Wnt3a		
WNT4	1	Wnt4	4	Wnt4	Wnt4a & Wnt4b	
WNT5A	3	Wnt5a	14	Wnt5a		
WNT5B	12	Wnt5b	6	Wnt5b	Wnt5b	
WNT6	2	Wnt6	1	Wnt6		DWnt6
WNT7A	3	Wnt7a	6	Wnt7a	Wnt7 & Wnt7a	DWnt2
WNT7B	22	Wnt7b	15	Wnt7b		
WNT8A	5	Wnt8a	18	Wnt8a	Wnt8a	DWnt8/WntD
WNT8B	10	Wnt8b	19	Wnt8b	Wnt8b	
WNT9A	1	Wnt9a/14	11			DWnt4
WNT9B		Wnt9b/15	11			
WNT10A	2	Wnt10a	1	Wnt10a	Wnt10a	DWnt10
WNT10B/12	12	Wnt10b	15	Wnt10b	Wnt10b	
WNT11	11	Wnt11	7	Wnt11 & Wnt11R	Wnt11	
WNT14	1					
WNT16	7	Wnt16	6		Wnt16	

either positively or negatively in a cellular context-specific fashion,²³ CK1 phosphorylates β -catenin at the Ser45 residue first, in a process called “priming”. This enables GSK-3 to phosphorylate the Ser33, Ser37, and Thr41 residues, which ultimately creates the binding site for the β -TrCP protein.^{22,24} The β -TrCP protein functions as an adaptor protein that complexes with Skp1/Cullin machinery to ubiquitinate β -catenin, enabling the destruction of β -catenin by the proteasome.^{22,24,25}

The binding of Wnt to the Fz receptors and LRP 5/6 transports disheveled protein (Dvl) to the cell membrane, leading to phosphorylation of the cytoplasmic tails of LRP 5/6. The LRP 5/6 can then bind Axin, removing it from the destruction complex, thus causing the complex to disassemble and release the β -catenin.^{14,26,27} This results in the stabilization and accumulation of β -catenin in the cytoplasm, and then its translocation to the nucleus and binding with the transcription factors TCF/LEF (T-cell factor/Lymphoid enhancer factor) and thus gene expression.^{14,26,28} Without β -catenin, the TCF/LEF complex is joined with the transducing-like enhancer protein (TLE/Groucho), which recruits HDACs, leading to transcriptional repression. However, β -catenin binding to TCF/LEF displaces the TLE/Groucho complexes and leads to the recruitment of activators to modify the interacting proteins, such as CBP/p300, Pygo, BCL9, and BRG1²⁹ (Fig. 1).

In contrast with the canonical pathway, noncanonical Wnt signaling is β -catenin-independent, as previously mentioned, and involves the Wnt5a type ligands, which include Wnt 4, Wnt5a, Wnt5b, Wnt6, Wnt7a, and Wnt11^{15,17}. Additionally, the noncanonical pathway involves different functions in comparison to the canonical pathway,

such as dictating cellular polarization and migration.^{17,30} Noncanonical Wnt signaling follows two distinct pathways, the Wnt/planar cell polarity (PCP) pathway and the Wnt/calcium (Ca^{2+}) pathway.^{17,30}

The Wnt/PCP pathway, like the canonical pathway, predominantly uses Fz receptors to bind Wnt (Fig. 2). These Fz receptors utilize several coreceptors, including protein tyrosine kinase 7 (PTK7),³¹ muscle-skeletal receptor tyrosine kinase (MUSK),³² tyrosine kinase-like orphan receptor (ROR1/ROR2),³³ tyrosine kinase related receptor (RYK),³⁴ syndecan,^{35,36} and glypican.^{37,38} However, the Wnt/PCP pathway also uses Celsr1 and Vangl2 receptors, although the ligand-receptor binding interaction is still relatively unknown³⁹ (Fig. 2). The Fz receptors in this pathway bind Wnt ligands and phosphorylate Dvl, leading to the recruitment of Inversin (Invs).⁴⁰ The polarity protein Par6 interacts with Dvl, and Smad ubiquitination regulatory factor (Smurf) is recruited by the phosphorylated Dvl and binds to Par6. Smurf then ubiquitinates Prickle, a protein that normally inhibits Wnt/PCP signaling, targeting it for proteasomal destruction.⁴¹ The breakdown of Prickle enables Dvl to associate with the Dvl-associated activator of morphogenesis (DAAM). This complex can then activate Ras homologue gene-family member A (RhoA), but not Rac1 or Cdc24.^{15,39} DAAM also activates Profilin.⁴² Rac1 activates JNK, which phosphorylates and activates c-Jun to go to the nucleus and initiate gene expression.^{43,44} JNK also activates CapZ-interacting protein (CapZIP) via phosphorylation.⁴⁵ RhoA activates RHO-associated coiled-coil-containing protein kinase (ROCK) and diaphanous 1 (DIA1).^{46,47} ROCK activates the myosin II regulatory light chain (MLRC).^{48,49} CapZIP, MLRC, DIA1, and profilin all contribute to actin

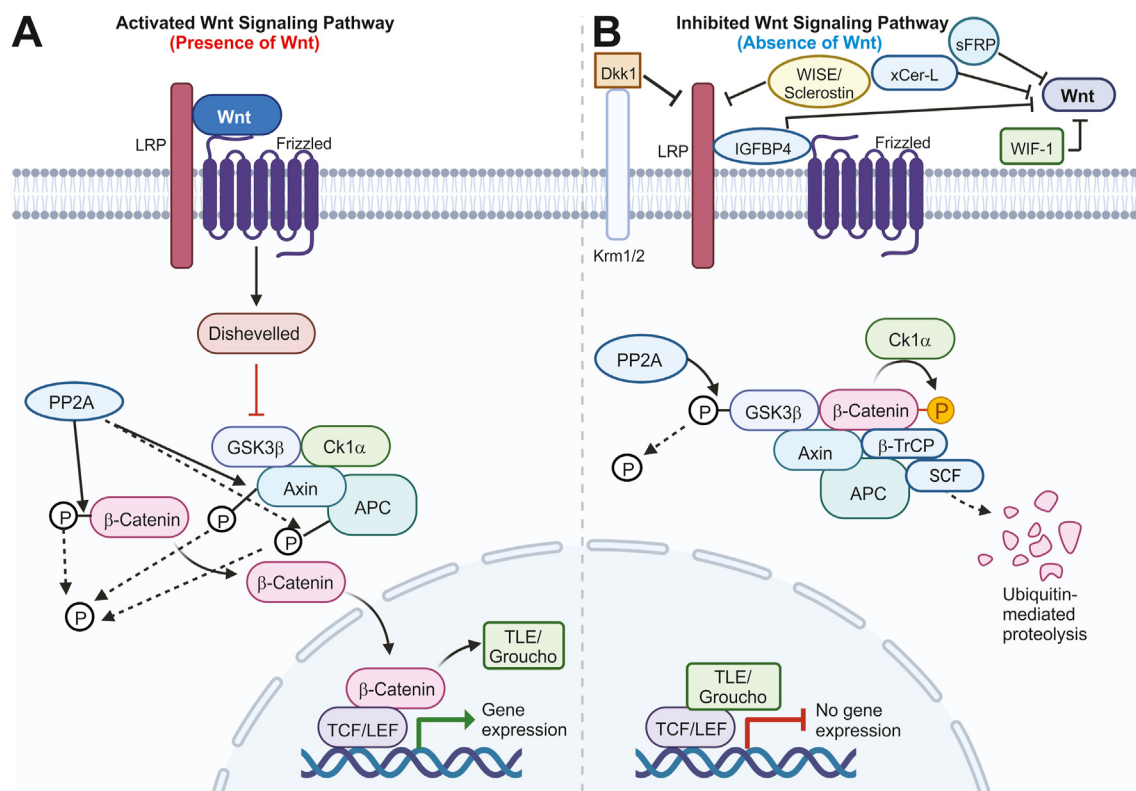


Figure 1 The canonical Wnt signaling pathway. The left panel (A) demonstrates the activated Wnt signaling cascade, while the right side portrays the inhibited Wnt signaling cascade. Wnt binds to the Fz receptor and LRP5/6 co-receptor. This activates Dvl to cause the dissociation of Axin from the destruction complex, causing β-catenin to be stabilized and enter the nucleus. β-catenin can then displace the inhibitory TLE/Groucho complexes, enabling TCF/LEF to transcribe the target genes. PP2A can also enhance Wnt signaling by dephosphorylating β-catenin, APC, and Axin. The result is the preservation of β-catenin by preventing ubiquitination and proteasomal breakdown. In the absence of Wnt signaling (B), the destruction complex breaks down β-catenin and inhibits gene transcription. Several other proteins also contribute to the inhibition of Wnt signaling. Dkk1 associates with Krm1 or Krm2 and LRP5/6, causing endocytosis of the LRP5/6 co-receptor. Wise/sclerostin binds to LRP5/6 to inhibit proper Wnt association with the coreceptor. xCer-L and WIF-1 both bind to Wnt ligands to inhibit signaling. IGFBP-4 functions as a competitive inhibitor of Wnt signaling by associating with LRP6 and Fz8, while sFRPs complex with Fz receptors to prevent Wnt ligand binding. The illustration was inspired by and created in BioRender.

polymerization, which is vital to cell polarity and migration.³⁹ The Celsr1 receptor appears to function similarly to the Fz receptor, where Wnt binding ultimately causes activation of Dvl to stimulate the same signaling cascade. On the other hand, the binding of Wnt on Vangl2 causes dissociation of a complex of Dvl, Prickle, and Inturned (Intu), enabling Dvl to complex with Invs³⁹ (Fig. 2).

The Wnt/ Ca^{2+} pathway is predominantly activated by the Wnt5a ligand and Fzd2 receptor^{50,51} (Fig. 3). The binding of the Wnt5a ligand to Fzd2 triggers G protein to activate phospholipase C (PLC).^{52,53} PLC then cleaves phosphatidylinositol-4,5-bisphosphate (PtdInsP2 or PIP2) into diacylglycerol (DAG) and inositol-1,4,5-trisphosphate (InsP3 or IP3). DAG, along with Ca^{2+} , activates protein kinase C (PKC) to stimulate cell-division cycle 42 (Cdc42), which causes actin polymerization to contribute to cell polarization and migration.³⁹ Meanwhile, IP3 binds to inositol-1,4,5-trisphosphate receptors (InsP3Rs) on the membrane of the ER, stimulating Ca^{2+} release through the Ca^{2+} channels and increasing cytoplasmic Ca^{2+} levels.^{39,54} Stromal interaction molecule 1/2 (STIM1/2) detects the

decrease in Ca^{2+} in the endoplasmic reticulum (ER) and activates Orai family proteins (Orai1, Orai2, or Orai3) on the plasma membrane to mediate store-operated Ca^{2+} entry (SOCE).^{39,54} Sarcoplasmic/ER Ca^{2+} ATPases (SERCAs) then pump Ca^{2+} back into the ER.^{39,54} The increased cytosolic Ca^{2+} from IP3 binding to InsP3R not only stimulates PKC but also stimulates calcineurin and CAMKII. Calcineurin activates the nuclear factor of activated T cells (NFAT), which causes gene transcription.^{55,56} On the other hand, CAMKII stimulates TGF-β-activated protein kinase 1 (TAK1), which then activates Nemo-like kinase (NLK).⁵⁷ NLK phosphorylates TCF, inhibiting the β-catenin/TCF complex and preventing gene transcription⁵⁸ (Fig. 3).

Biological conservation of Wnt signaling

The Wnt signaling pathway is highly conserved. Wnt was first identified in *Drosophila* as the Wg protein due to mutations in the gene causing lack of wing and haltere development.⁴ The Wg gene is also called Dint1, as it was

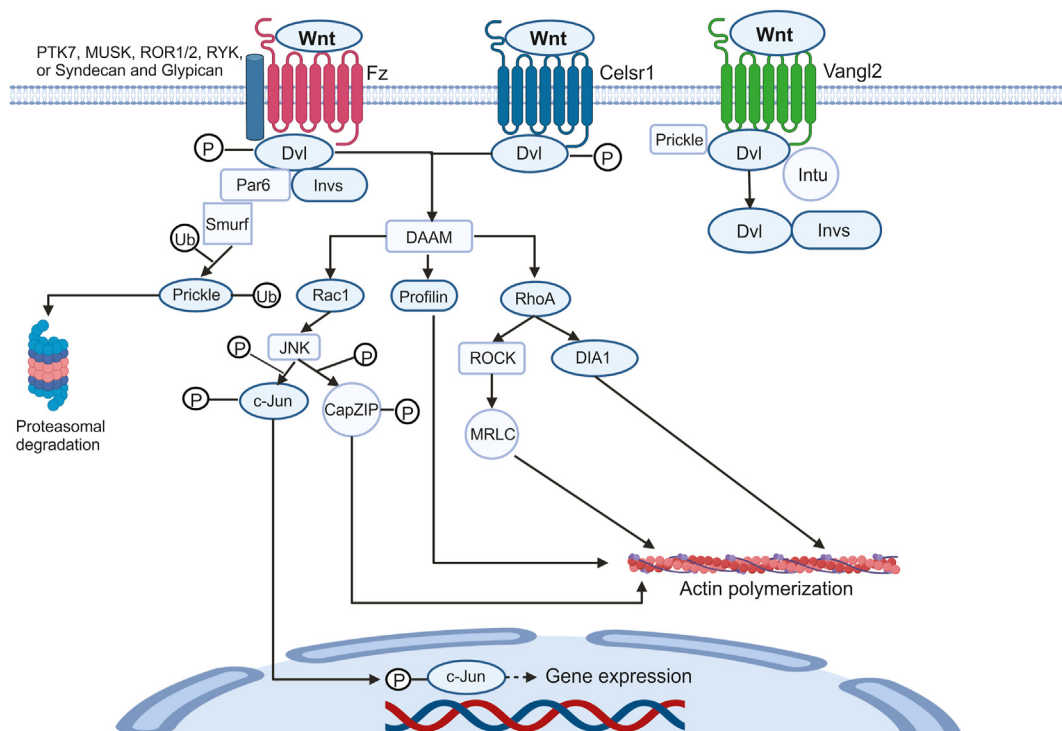


Figure 2 The noncanonical Wnt/PCP pathway. The binding of Wnt ligands leads to the phosphorylation of Dvl, which recruits Invs, Par6, and Smurf. Smurf ubiquitinates the inhibitory protein Prickle, targeting it for destruction. Dvl can then associate with DAAM, activating Rac1, profilin, and RhoA. Rac1 activates JNK, which phosphorylates c-Jun and CapZIP. c-Jun then goes to the nucleus to stimulate gene transcription. RhoA activates ROCK and DIA1, with the latter activating MRLC. CapZIP, MRLC, DIA1, and profilin all stimulate actin polymerization. Celsr1 stimulates Dvl due to Wnt binding like the Fz receptor. Wnt binding to the Vangl2 receptor causes dissociation of Prickle and Intu from Dvl, which can then bind to Invs. The illustration was inspired by and created in BioRender.

shown to be homologous to Int1 found in mice. This was the first connection discovered between Wnt found in *Drosophila* and Wnt found in vertebrates.⁷ In an experiment that proved the Wg gene is homologous to Int1 found in mice, van Ooyen et al sequenced the human and mouse Int1 genes in blocks of 50 nucleotides³ (Table 1). Comparison between the mouse and human Int1 genes revealed conservation of the splice sites, TATA box, and polyadenylation signal.³ Both proteins were also 370 amino acids long, with the only differences being found in the hydrophobic N-terminus.³ Homology was also established in the non-coding sequences.³ The overall homology between mouse and human Int1 homologs was 99%, revealing the conservation of the protein across species.³

Studies of cnidarians and sponges have also revealed the conservation of the Wnt signaling pathway, as the pathway was shown to be involved in the control of axis polarity.⁵⁹ Petersen and Reddien further clarified that Wnt signaling is broadly used in primary body axis development.⁵⁹ The Porcupine gene (Porc) codes for a transmembrane protein in the ER, allowing the processing and distribution of *Drosophila* Wg *in vitro*.⁶⁰ Porc is an acyltransferase that palmitoylates the Wnt protein in the ER for secretion.⁶¹ Porc is also important for localizing *Drosophila* Wnt3 in the embryonic CNS.⁶⁰ Tanaka et al isolated mouse Porc (Mporc) and *Xenopus* (Xporc) and analyzed them alongside homologous human MG61, *C. elegans* Mom-1, and *Drosophila*

Porc. It was discovered that the Porc homologs among the vertebrates were well conserved via comparison of amino acid sequences, while the *Drosophila* Porc had an additional hydrophilic N-terminal sequence. This study revealed the conservation of Porc across species.⁶⁰ Furthermore, injection of Mporc RNA into *Drosophila* embryos with *Drosophila* Porc resulted in some rescue of the embryos, albeit to a reduced extent compared to *Drosophila* Porc RNA. This study revealed the conservation of Porc and Porc function across multiple species.⁶⁰

Biological assays for canonical vs. noncanonical Wnt classification

Various biological assays are used to identify components of the canonical and noncanonical pathways. From a broader perspective, the secondary-axis formation (*i.e.*, axis duplication) analysis is one of the prototypes of canonical Wnt assays and can be used to assay activators and inhibitors of the canonical pathway, due to its role in proper axis specification.⁶² Luciferase reporter assays revealed that PTK7, a part of the noncanonical pathway, inhibits canonical Wnt signaling by precipitating Wnt3a and Wnt8 in the canonical pathway.⁶³ Luciferase reporter assays can also be used to detect TCF/LEF activation,^{64,65} which are proteins that are mainly involved in the canonical

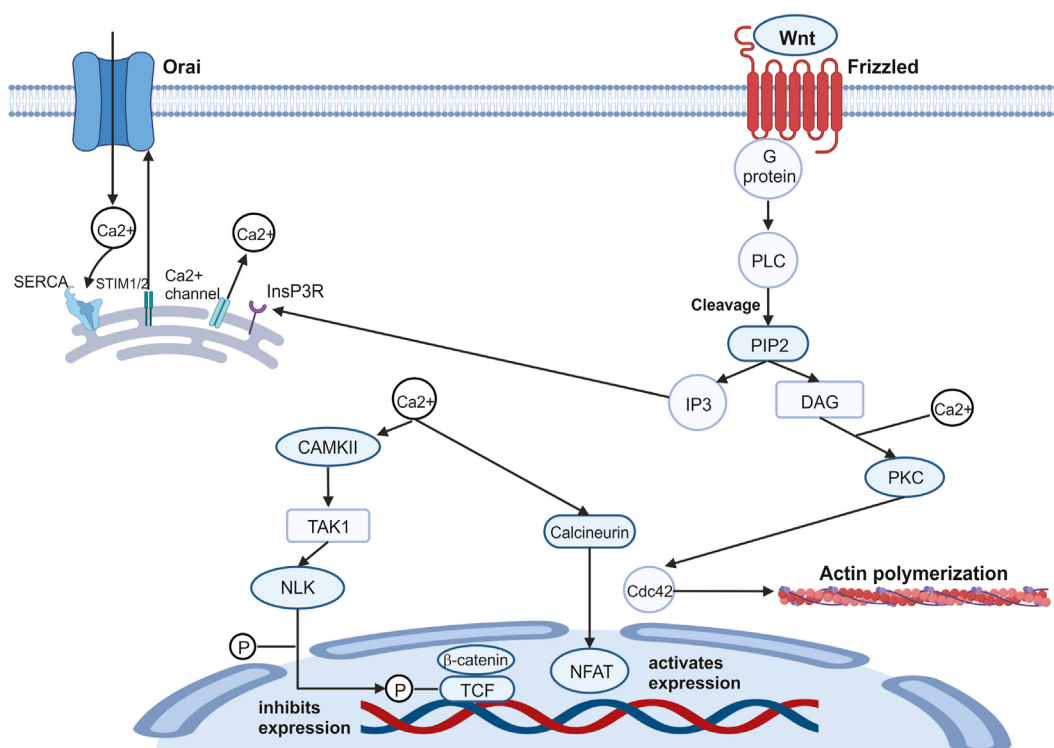


Figure 3 The noncanonical Wnt/ Ca^{2+} pathway. Wnt binding to the Fz receptor leads to G protein-mediated activation of PLC. PLC cleaves PIP2 into IP3 and DAG. IP3 binds to IP3 receptors (InsP3R) on the ER membrane to stimulate Ca^{2+} release. STIM1/2 detects this decrease in ER Ca^{2+} levels and activates Orai proteins in the plasma membrane to bring more Ca^{2+} into the cell, where SERCA can pump Ca^{2+} back into the ER. DAG can then activate PKC in the presence of Ca^{2+} and PKC can stimulate Cdc42 to enhance actin polymerization. The elevated intracellular Ca^{2+} level also stimulates calcineurin and CAMKII. Calcineurin activates NFAT, causing gene transcription. CAMKII activates TAK1, which activates NLK, when then phosphorylates TCF, preventing β -catenin-mediated gene transcription. The illustration was inspired by and created in BioRender.

pathway.²⁸ Luciferase reporter assay could be used to form a high throughput screen for Wnt/ β -catenin signaling.⁶⁵ Enzyme-linked immunosorbent assay (ELISA) is an assay that can be used to detect Dickkopf-1 (DKK-1), an inhibitor of the canonical pathway.⁶⁶ Another type of analysis is the Western blot analysis, which can be used to detect the up-regulation of the canonical proteins β -catenin, Dvl, APC, and GSK-3 in the retina of mice.⁶⁶ β -Catenin levels can also be detected using immunodetection assays.⁶⁷

The Wnt inhibitory factor-1 (WIF-1), which is a member of the secreted Frizzled-related protein (sFRP) family, can inhibit both the canonical and noncanonical pathways by binding directly to Wnts, preventing them from binding to the Wnt receptor.⁶⁸ Soft agar assay and Western blotting can be used to detect inhibition of osteosarcoma cell growth as a result of WIF-1 overexpression, providing insight into the regulation of both pathways.⁶⁴

In the noncanonical Wnt pathway, Wnt5a is the most prominent ligand, using the ROR family of tyrosine kinases as receptors.⁶⁹ A member of the kinesin family, Kif26b, is a downstream effector of the Wnt5a-ROR pathway, mediating cell migration during embryonic development.⁷⁰ Hence, a Wnt5a-ROR-Kif26b (WRK) reporter assay was developed and could be used to measure the degree of Wnt5a-ROR signaling in real-time, utilizing a combination of flow cytometry, Western blot, and time-lapse microscopy.⁷⁰ A similar test involves a GFP-Kif26b reporter cell line

combined with flow cytometry to detect the levels of Wnt5a in the cells.⁷¹ The GFP signal enables quantitative analysis of the cells to detect Wnt5a expression, although the assay is sensitive to cell density.⁷¹

Controlled secretion of Wnt proteins

Wnt ligands are highly lipidated in the ER, limiting the range of diffusion to localize the ligand to its recipient cell.⁷² Specifically, the acyltransferase Porc palmitoylates the Wnt in the ER.⁶¹ The mom-1 gene codes for a similar acyltransferase in *C. elegans*.⁷³ The Wnt ligands are secreted via a Wntless (WLS) transporter, which is a conserved transmembrane protein in the Golgi apparatus that was characterized in Wg-sending cells in *Drosophila*.⁷⁴ Banziger et al transfected embryonic kidney cells (HEK-293T) with the Wnt3a expression vector and cultured them with siRNA to knock down the expression of the human WLS (hWLS) gene. Assaying the level of Wnt3a protein revealed that siRNA-treated hWLS (sihWLS) cells could not activate the Wnt pathway due to, at least in part, the lack of secretion of Wnt3a in the absence of the hWLS gene. This study established the importance of WLS for the secretion of Wnt ligands. Additionally, cell surface heparan sulfate proteoglycans (HSPGs) are also involved in Wnt signaling. Glypicans and syndecans compose the protein core of

HSPGs and are covered in heparan sulfate chains.⁷⁵ In *Drosophila*, a glypican called division abnormally delayed (dally) functions as a coreceptor for the *Drosophila* frizzled 2 (Dfz2) receptor.⁷⁶ The dally gene specifically encodes the protein core of the HSPGs in Wg/Wnt signaling.⁷⁶ Glypicans like dally also participate in transporting Wnt ligands toward target cells.⁷⁴

Wnt receptors, co-receptors, and accessory proteins

Cognate receptors: the frizzled (Fz) proteins

Frizzled (Fz) is a seven-pass transmembrane receptor that binds Wnt ligands.⁷⁷ The Fz genes generate Fz receptors, each of which ranges from 500 to 700 amino acids long, with a CRD on the N-terminus and a 40 to 100 amino acid-long hydrophilic linker region.⁷⁸ The seven transmembrane domains are hydrophobic alpha-helices.⁷⁸ Fz protein localizes in the plasma membrane, with the cysteine-rich N-terminus oriented extracellularly and the carboxy-terminus oriented intracellularly towards the cytoplasm.^{21,79,80} The intracellular carboxy-terminus is variable in length and not well conserved between the different Fz receptor types.⁷⁸ Fz is also glycosylated⁷⁹ and there are 10 Fz genes organized into four clusters.⁷⁸ By amino acid sequence, Fzd1, Fzd2, and Fzd7 are 75% similar; Fzd5 and Fzd8 are 70% similar; Fzd4, Fzd9, and Fzd10 are 65% similar; and Fzd3 and Fzd6 are 50% similar.⁸¹ Comparison of amino acid sequences between clusters yields only 20%–40% similarity.⁷⁸ The relative lack of sequence similarity suggests a lack of genomic conservation between the Fz receptors.

The importance of Fz receptors in the Wnt/PCP pathway was characterized by Fz mutations in *Drosophila* that caused abnormal wing hair patterns and polarity disruptions, revealing the importance of Fz in the Wnt/PCP pathway.^{82,83} Loss of function and overexpression mutations in Fz altered the assembly location of F-actin in wing development, indicating that Fz is important in cytoskeletal development.⁷⁹ The misorientation of hairs also revealed the importance of Fz in not only receiving the Wnt signal but also propagating it in a proximal-distal direction.⁸⁴ Mutations in Fz receptors also cause the defective orientation of the ommatidia (individual units composing a compound eye).⁸⁵

Mutations in the frizzled 4 (*Fzd4*) receptor causes familial exudative vitreoretinopathy (FEVR), a disease that generally results in the lack of retinal angiogenesis. This leads to a variety of symptoms, including retinal fibrosis, detachment, and dysplasia.⁸⁶ Mutations in Fzd1 and Fzd2 in mice caused cleft palate and ventricular septal defects (VSD). These mutations also affected neural tube closure and inner ear development. *Fzd7* mutations also caused VSDs but were more commonly associated with a kinked tail.⁸⁷ Knockout of *Fzd5* in mice led to embryonic lethality due to placental insufficiency.⁸⁸ Loss of *Fzd5* also caused retinal cell death and excessive mesenchymal cells in the vitreous cavity among other issues.⁸⁹ Loss of *Fzd8* alone causes no phenotypic change, although the loss of even a single *Fzd8* allele can increase the severity and penetrance of the ocular effects from loss of *Fzd5*.⁹⁰ Homozygous loss

of *Fzd9* has yielded B-cell developmental abnormalities, defective visuospatial learning, and bone mass reduction compounded by osteoblast dysfunction.²¹ Homozygous *Fzd6* deletions cause randomized hair follicles, generating waves, whorls, and tufts.⁹¹ Homozygous deletion of *Fzd3* in mice causes inappropriate development of peripheral and central axons, leading to the inability to detect thermal and mechanical stimuli from the feet.⁹² Concomitant deletions in *Fzd3* and *Fzd6* cause neural tube closure defects and misorientation of inner ear hair cells, indicating their similar functions.⁹³

Co-receptors: LRP5 and LRP6 proteins

Low-density-lipoprotein receptor-related proteins 5/6 (LRP5/6) are single-pass transmembrane proteins that function as coreceptors for Fz receptors.^{19,94} The binding of the Wnt ligand triggers the dimerization of Fz and LRP5/6.⁹⁴ This leads to the phosphorylation of the cytoplasmic tail of LRP5/6 at five conserved PPPSP (also called the PPPSPXS) motifs.^{95,96} Several protein kinases are involved in this phosphorylation, including GSK-3, which is part of the destruction complex that normally binds β -catenin in the absence of Wnt signaling.^{94,97} The PPPSP motifs bind Axin from the destruction complex, inhibiting β -catenin phosphorylation and subsequent ubiquitination that leads to proteasomal breakdown.^{95,96,98} LRP5 and LRP6 are homologous and expressed in embryogenesis.¹⁹ However, LRP6 is more important during embryogenesis, as evidenced by mice with homozygous deletion of *Lrp6* demonstrating axial skeleton truncation, neural tube closure defects like spina bifida, as well as midbrain and hindbrain malformations.⁹⁹ On the other hand, homozygous deletion of *Lrp5* caused low bone mass due to decreased osteoblast activity, defective capillary cell apoptosis in the eye, defective clearance of chylomicrons, and impaired insulin secretion.^{100,101}

Accessory Wnt binding proteins at the cell membrane

R-spondins

Roof plate specific-spondins (R-spondins) are four members of a larger family that contain thrombospondin type 1 repeats (TSR-1).¹⁰² The R prefix was given based on the first R-spondin due to its expression in the boundary of the roof plate and neuroepithelium in the dorsal neural tube.¹⁰³ The R-spondins also contain an N-terminal signal peptide, two furin-like (FU1 and FU2) cysteine-rich domains near the N-terminus, and a C-terminal region with many positively charged amino acids.^{102,104} The R-spondins can bind to leucine-rich repeat-containing G-protein coupled receptors 4–6 (Lgr4-6).¹⁰⁵ R-spondins bind Lgr4-6 via their FU2 domain and can bind ZNRF3 and RNF43 via their FU1 domain.¹⁰⁶

ZNRF3 and RNF43

ZNRF3 and RNF43 are E3 ubiquitin ligases that target Wnt receptors for destruction to decrease Wnt signaling responses.¹⁰⁷ ZNRF3 is a member of the ZNRF protein family, a family of E3 ubiquitin ligases that contain a zinc finger and a RING domain.¹⁰⁸ R-spondins can bridge ZNRF3/RNF43 and

LGR4/5/6, inhibiting ZNRF3/RNF43 activity via auto-ubiquitination and membrane clearance.^{106,109} In this manner, R-spondins act as Wnt agonists. RNF is a homolog of ZNRF43 that also contains a RING domain for its function as an E3 ubiquitin ligase.¹⁰⁹

Derailed/Ryk

Derailed/Ryk (related to tyrosine kinase) is a member of the atypical tyrosine kinase family consisting of a Wnt inhibitory factor (WIF) domain extracellularly, an atypical kinase domain intracellularly, and a PDZ binding motif. Derailed is the *Drosophila* homolog, while Ryk is found in mammals.¹¹⁰ Interestingly, the tyrosine kinase domain is considered atypical due to sequence variations in the normally conserved tyrosine kinase residues. The domain lacks tyrosine kinase activity.¹¹¹ The WIF domain of Ryk binds to Wnt1 and Wnt3a to activate TCF for the transcription of target genes.¹¹⁰ Ryk also forms a ternary complex with Wnt1 and Fz using its extracellular WIF domain, while the intracellular kinase domain binds to Dvl using its PDZ binding motif to activate TCF in response to Wnt3a stimulation.¹¹⁰ In *Drosophila*, Derailed was found to be important in learning and memory.¹¹² Derailed/Ryk was also found to be important in axon guidance via interaction with Wnt5.¹¹³ The *C. elegans* homolog, lin-18, is important in determining vulval cell fate patterning.¹¹⁴

Receptor tyrosine kinase-like orphan receptors (RORs)

RORs are members of the receptor tyrosine kinase (RTK) family that are highly conserved and consist of two members, ROR1 and ROR2.¹¹⁵ ROR1 and ROR2 are single-pass transmembrane receptors with an intracellular tyrosine kinase domain and a proline-rich domain (PRD) that is flanked by two serine–threonine rich domains.¹¹⁶ The extracellular side of the ROR contains an immunoglobulin (Ig)-like domain, a CRD, and a Kringle domain (KRD). The CRD of ROR1 and ROR2 is like those on Fz receptors.¹¹⁷ ROR1 and ROR2 play crucial roles in embryonic development. Mice lacking ROR2 expression have shortened limbs and tails, facial abnormalities, and dwarfism among other issues.¹¹⁸ Mutations in ROR2 also cause Robinow syndrome and Brachydactyly type B in humans.^{119–121} Wnt5a binds ROR2, causing heterodimerization of ROR2 with Fz2 using its CRD. RORs can activate the noncanonical Wnt/JNK-PCP pathway and inhibit the canonical β -catenin/TCF pathway.¹²² However, the components of the Wnt/ROR pathways are still mostly unknown.¹²² ROR1 is expressed in B-lymphocyte precursors and can rarely cause precursor-B acute lymphoblastic leukemia (B-ALL).¹²³ ROR1 and ROR2 in mice have been known to play an important role in the development of the nervous system.¹²⁴

Gpr124 and Reck

Gpr124 is a G protein-coupled receptor (GPCR) and Reck is a glycosylphosphatidylinositol-anchored glycoprotein.¹²⁵ Reck binds to Wnt7, creating the Reck/Wnt7 complex that binds to Gpr124. This complex then joins with the Fz receptor and LRP5/6 coreceptor to stimulate canonical Wnt/ β -catenin signaling.^{125–127} The Gpr124 and Reck coactivators are vital in the development of the blood–brain barrier (BBB).¹²⁸ Gpr124 knockout mice demonstrated microvascular hemorrhage and lethality in the embryo.

Interestingly, Gpr124 deletion did not affect BBB integrity in adult mice.¹²⁸

Intracellular mediators of Wnt signaling

The intracellular component of the Wnt signaling pathway is composed of several proteins for each pathway. This section will give a brief overview of each component of the canonical and two noncanonical Wnt pathways, looking further at the additional functions of the components outside of Wnt signaling. A description of the functions in Wnt signaling and the signaling cascade will be described in the canonical vs. noncanonical Wnt section. The inhibitors will also be discussed later. For the canonical pathway, the binding of Wnt to Fz and LRP5/6 mediates signal transduction to the nucleus via β -catenin (Fig. 1).

Intracellular mediators of the canonical pathway

β -Catenin was discovered to have two functions, one of them being its association with α - and γ -catenin to link Ca^{2+} -dependent cell adhesion molecules (CAMs) to cytoskeletal structures.¹²⁹ The term catenin was given to these three molecules due to its linkage of the CAM, E-cadherin, to cytoskeletal structures.¹²⁹ The second function of β -catenin is its role in Wnt signaling, which was discovered through the analysis of its *Drosophila* homolog, Armadillo (Arm). Seminal screens for mutations that caused altered segmentation of *Drosophila* embryos revealed the signaling potential of β -catenin.¹³⁰ Additionally, mutations in Wg in *Drosophila* caused a corresponding decrease in Arm, leading to altered segment polarity, further delineating the link between Wnt signaling and β -catenin.¹³¹ Further studies would later reveal that β -catenin mediates its effects via the TCF/LEF transcription factors, stimulating the transcription of target genes.¹³²

β -Catenin is a 781 amino acid long protein with 12 Arm repeats, which forms a super-helical structure composed of multiple α -helices with a hydrophobic core.¹³³ The super-helix contains a large, positively charged groove that allows β -catenin to interact with cadherins, TCF, and APC.^{133–139} β -Catenin then binds TCF and LEF to mediate gene transcription (Fig. 4). Interestingly, TCF is considered a transcriptional repressor, while LEF is considered a transcriptional activator.¹⁴⁰ TCF/LEF binds to DNA via its HMG box domain, which recognizes a sequence called the Wnt/Wg response element (WRE) on the DNA.^{141,142} The HMG domain binds to the WRE in the minor groove of the DNA, causing the DNA to bend.¹⁴² In addition to the HMG domain, TCF/LEF also contains a nuclear localization signal (NLS) that makes nonspecific contacts with the phosphate backbone of the DNA, increasing TCF/LEF affinity for the DNA.¹⁴²

Intracellular mediators of noncanonical Wnt/PCP pathway

Dvl proteins

There are two noncanonical Wnt signaling pathways, the Wnt/PCP pathway, and the Wnt/ Ca^{2+} pathway. The Wnt/PCP pathway involves several additional receptors and co-

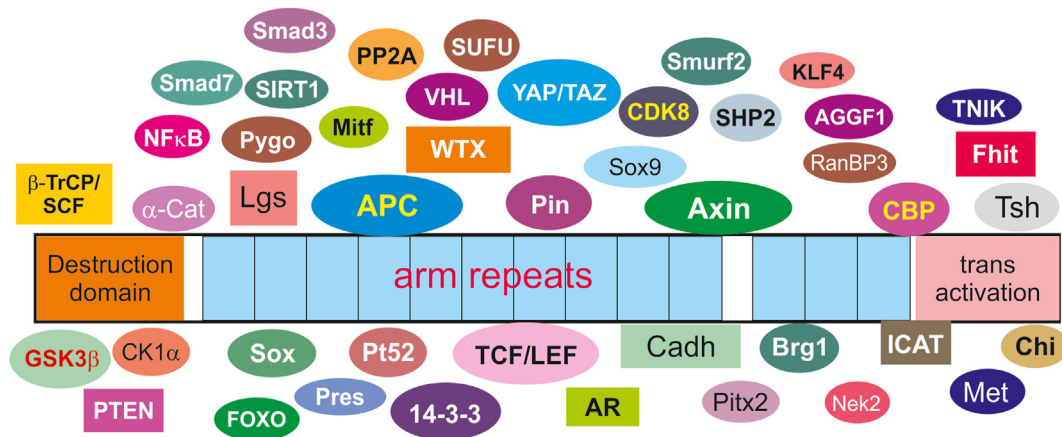


Figure 4 β -Catenin protein interactions. Some of these proteins were not discussed in this paper due to space constraints. Among those that have been discussed in this paper, there are notable inhibitors and activators. GSK-3 β , CK1, APC, Axin, and β -TrCP are inhibitors of β -catenin as part of the destruction complex. PP2A has dual effects; it can dephosphorylate β -catenin to prevent ubiquitination and stabilize β -catenin, while it can also dephosphorylate GSK-3 β , which can then inhibit β -catenin. YAP/TAZ is another inhibitor, as it can either bind to and suppress β -catenin without affecting its levels or associate with the destruction complex. Smad7 and Smurf2 can complex with β -catenin to ubiquitinate and degrade the protein. SUFU can export β -catenin from the nucleus. On the other hand, there are several activators of β -catenin, proteins that are activated by β -catenin, and proteins that assist with the functions of β -catenin. Smad3 is a chaperone protein that transports β -catenin into the nucleus. TCF/LEF are transcription factors that are activated by β -catenin. α - and γ -catenin join with β -catenin to link CAMs like E-cadherin to cytoskeletal structures, strengthening cell adhesion. It is noteworthy that some of the protein interactions are species-, tissue-, and/or context-dependent. The illustration was inspired by the Wnt homepage created and maintained by the Nusse Lab at Stanford University (http://web.stanford.edu/group/nusselab/cgi-bin/wnt/protein_interactions) and reference ¹³⁷.

receptors that will be discussed in later sections. The first intracellular mediator of Wnt/PCP signaling is Dvl. All Dvl proteins contain three conserved domains, a DIX domain at the N-terminus, a central PDZ domain, and a DEP domain at the carboxy-terminus.¹⁴³ In between the DIX and PDZ domains is a "basic region" composed of conserved Ser and Thr residues. Between the PDZ and DEP domains is a "proline-rich region".¹⁴³ The DIX domain primarily activates the canonical pathway, enabling dynamic polymerization of Dvl to form puncta, which can then interact with Axin to prevent it from mediating the destruction of β -catenin.¹⁴⁴ Although further research needs to be conducted to delineate the exact process of this interaction between Dvl and Axin, it has been theorized that the DIX domain of Dvl interacts with the similar DIX domain on Axin to induce a conformational change in Axin or relocates Axin.¹⁴⁴ The PDZ domain is involved in both the canonical and noncanonical pathways.¹⁴⁵ It binds the Fz receptor at its C-terminal conserved Lys-Thr-X-X-X-TRP (KTXXXW) motif.¹⁴⁶ This motif is required for the activation of the canonical pathway, although the molecular mechanisms are poorly understood.¹⁴⁷ The DEP domain activates the noncanonical pathway by mediating the interaction between Dvl and DAAM1.¹⁴³ The DEP domain also translocates the Dvl protein to the plasma membrane after Wnt stimulation.¹⁴⁸

Dvl has an NLS located between the PDZ and DEP domain (aside from the proline-rich region), and an NES (nuclear export signal) located between the DEP and C-terminus.¹⁴³ Increasing evidence indicates that nuclear Dvl protein is critical for β -catenin and TCF factors to form a complex.¹⁴⁹ Dvl has been shown to interact with many other transcriptional factors including FOXK1/2, TAZ, and HIPK1, as well as

gene promoters such as CYP19A1.¹⁴⁹ Furthermore, Dvl1 was shown to interact with EZH2 while Dvl3 interacts with chromatin-modifying enzymes such as KMT2D in cancer cells. In addition, Dvl proteins are extensively modified post-translationally by phosphorylation, lysine acetylation, and methylation, as well as ubiquitination, although the functional roles of Dvl post-translational modifications remain to be fully investigated.¹⁴⁹

Inversin (Inv) protein

Inversin (Inv) is a 1062 amino acid long protein with characteristic 15 successive ankyrin repeats. It was first identified in Inv mutant mice that had situs inversus (reversed left/right polarity), underdeveloped tubules, and cyst formation in the kidneys.¹⁵⁰ This discovery linked Inv to Wnt signaling. Simons et al used glutathione S-transferase (GST) fusion protein containing the PDZ domain of Dvl to characterize the effect of Inv. The Inv directly interacted with the GST fusion protein. Inv also formed a protein complex with Dvl, indicating that it could inhibit the canonical pathway.¹⁵¹ Inv also participates in the noncanonical pathway, interacting with the planar cell polarity (PCP) pathway proteins Strabismus (Stbm) and Prickle (Pk) like the *Drosophila* PCP protein Diego.¹⁵¹ Inv mutant mice developed renal cysts due to unopposed canonical Wnt signaling, which causes overgrowth of cells without terminal differentiation of renal tubular epithelial cells.¹⁵¹

Par6 protein

Par6 serves as a polarity protein and a scaffolding protein with other molecules.¹⁵² The scaffolding function is beneficial in complexing with Dvl and Smurf in the noncanonical

pathway.⁴¹ Its cell polarity function is mediated by the G-protein-activated phospholipase C- β (PLC- β) interacting with Par proteins like Par6 through multiple PDZ domains that can bind the extreme C-terminal S/TXL motifs of PLC- β .¹⁵² This activates the PLC- β to hydrolyze PIP2 into IP3 and DAG, two important secondary messengers. IP3 and DAG play important roles in regulating cell polarity and asymmetric cell division.¹⁵²

Smurf1 and Smurf2 proteins

Smurfs (Smurf1 and Smurf2) are E3 ubiquitin ligases of the C2-WW-HECT family of proteins.¹⁵³ They were first identified in the ubiquitination and degradation of R-Smads in the BMP pathway to antagonize TGF- β /BMP signaling.¹⁵⁴ Smurf1 contains a phospholipid/Ca²⁺ binding domain on its N-terminus, two WW domains for binding to PPXY (also called PY) motifs on other proteins, and a HECT domain on the C-terminus for ubiquitination of target proteins.¹⁵⁴ Smurf1 targets Smad1 and Smad5 using its WW domains, which bind to the PY motifs on Smad1 and Smad5, enabling their degradation.¹⁵⁴ Smurf2 uses the same mechanism to degrade Smad1 and Smad2.¹⁵⁵ Smurf1 can also be recruited by Par6 to target RhoA for degradation, establishing the proper cell polarity needed for cell movement.¹⁵⁶ Smurf1-induced RhoA degradation in tight junctions leads to their dissolution and enables TGF- β dependent epithelial–mesenchymal transition.¹⁵⁷ These results indicate the importance of Smurfs in noncanonical Wnt/PCP signaling.

Prickle (Pk) protein

Prickle (Pk) is also involved in the Wnt/PCP pathway. Pk is considered a type 1 polarity gene along with Dvl and Fz because it affects the body surface and is believed to directly establish tissue polarity. On the other hand, type 2 and 3 tissue polarity genes affect specific body regions and are believed to interpret the polarity that the type 1 genes have established.¹⁵⁸ All Pk proteins contain three LIM motifs and a conserved domain called Prickle Espinas Testin (PET).¹⁵⁸ The LIM motifs are cysteine-rich domains with two zinc fingers that are joined by an amino acid spacer.¹⁵⁹ The LIM domains bind target proteins and enable protein–protein interactions.¹⁵⁸ The PET domain is monomeric and works with the LIM motifs to target Dvl to the cell membrane to facilitate its function in the Wnt/PCP pathway.¹⁶⁰ Pk also has a prenylation motif on its C-terminus that is important but not required for its localization to the plasma membrane.¹⁶¹

Dvl-associated activator of morphogenesis (DAAM)

DAAM is a conserved actin nucleator that is a member of the formin family.¹⁶² DAAM members contain a GTPase binding domain (GBD), a diaphanous inhibitory domain (DID), an N-terminal dimerization domain (DD), a coiled-coil (CC), an FH1 domain, an FH2 domain, and a diaphanous autoregulatory domain (DAD). DAAM is highly expressed in tissues like the CNS, somites, dermomyotomes, and the heart and is important in organ symmetry. Specifically, it causes left-right (LR) symmetry due to its modulation of the myosin 1D (Myo1D) function. *Drosophila* Myo1D induces dextral twisting, which is important for orienting the native LR organs in larvae. The absence of Myo1D leads to situs

inversus.¹⁶³ DAAM nucleates F-actin, promoting the assembly of an F-actin network that enables Myo1D to induce chirality.¹⁶² Interference of DAAM with RNAi suppressed the expected 180-degree dextral rotation of the larval body, reducing it to just 90°. This experiment revealed the importance of DAAM in inducing proper chirality in *Drosophila* larvae.¹⁶² The FH2 domain is particularly important in actin nucleation and polymerization, while the FH1 domain interacts with profilin-actin during actin filament elongation.⁴²

Profilin protein

Profilin is a protein associated with non-muscle actin (β - and γ -actin) and is involved in the control of actin polymerization. It was first isolated from calf spleens as a small protein that accompanied an actin-containing complex.¹⁶⁴ When complexed with actin, profilin is called profilactin.¹⁶⁵ Profilin contains a core of seven-stranded β -pleated sheets with α -helical N- and C-termini on one side, and two shorter α -helices on the other. The N- and C-termini form the poly-L-proline (PLP) binding surface, which enables profilin to bind actin and factors that participate in actin nucleation and elongation.¹⁶⁶ Profilin inhibits spontaneous actin nucleation and polymerization by sequestering G-actin. However, it can also promote actin filament elongation using its PLP binding domain to interact with other proteins such as formins, Ena/vasodilator-stimulated phosphoprotein (VASP), Arp2/3-dependent Wiskott–Aldrich syndrome protein (WASP), and WASP family verprolin-homologous protein (WAVE) family.^{166,167} Profilin also facilitates the exchange of ADP for ATP on actin monomers, further enhancing polymerization.^{166,167} Interaction with formins allows profilin to associate with microtubules and increases the rate of depolymerization.¹⁶⁸ With profilin as a mediator, DAAM1 can mediate cytoskeletal reorganization and cell movement for important processes such as gastrulation.¹⁶⁹

Rac1 and RhoA proteins

Rac1 and RhoA are members of the Rho family of small GTPases.¹⁷⁰ Both proteins are involved in a wide variety of cell processes, such as motility, proliferation, migration, and polarity via their regulation of actin polymerization.¹⁷¹ Both have conserved GDP/GTP binding domains called the G domain and a C-terminal region that contains a CAAX motif.¹⁷² The binding of GTP to Rac1 causes two regions (amino acids 25–40 and 60–76) called switch I and II to undergo a conformational change to interact with specific downstream effectors in the signaling cascade.¹⁷¹ Rac1 and RhoA can also undergo post-translational modification to prenylate the C-terminal CAAX motif, enabling membrane interaction.¹⁷¹ This association with the membrane is what enables Rac1 and RhoA to stimulate downstream signaling cascades to modulate cellular functions.¹⁷⁰ The exchange of GDP for GTP is mediated by Guanine Nucleotide Exchange Factors (GEFs) and GTPase activating proteins (GAPs), triggering the active state of Rac1 and RhoA to interact with downstream proteins.^{171,173} Guanosine dissociation inhibitors (GDI) can conceal the C-terminal isoprenyl motif in a hydrophobic pocket, sequestering Rac1 in the cytoplasm and preventing downstream activation.^{173,174} GDI also prevents GDP/GTP exchange to keep both in their inactive states.^{171,174} RhoA and Rac1 are

spatially separated, with Rac1 active towards the leading edge of the cell, while RhoA is active towards the lagging edge.¹⁷⁰ The two are also temporally segregated, with RhoA activity peaking before Rac1 in a coordinated cycle of protrusion and retraction. Rac1 and RhoA can antagonize each other mutually to coordinate this effect.¹⁷⁵ Dysregulation of Rac1 and RhoA have been linked to cancer, as well as cognitive and cardiovascular diseases.¹⁷²

Rho-associated coiled-coil kinase (ROCK)

ROCK (also called RhoA/Rho kinase) is a Ser/Thr kinase that is stimulated by Rho.^{46,47,176,177} ROCK comes in two isoforms, ROCK1 and ROCK2. ROCK1 is 1354 amino acids long, while ROCK2 is 1388 amino acids long.¹⁷⁸ They share 64% of their primary amino acid sequences, 92% homology in the kinase domains, and only 55% homology in the coiled-coil domains.¹⁷⁸ The N-terminal region contains the ROCK kinase domains.¹⁷⁸ The C-terminal region contains the coiled-coil domain and a pleckstrin homology (PH) domain; this region binds to the catalytic kinase domain to inhibit its activity. The coiled-coil domain contains a Rho-binding region that enables GTP-bound RhoA to disrupt the binding of the C-terminus and the kinase domain.¹⁷⁹ The C-terminal region can also be cleaved by caspases during apoptosis to activate ROCK.^{180,181} Activated ROCK can associate with mammalian DIA (mDia) to stimulate actin cytoskeletal reorganization.^{177,182} ROCK proteins are generally expressed in many tissues and phosphorylate target proteins on R/KXXS/T or R/KXS/T amino acid motifs.¹⁷⁸ ROCK proteins are involved in forming stress fibers composed of bundles of F-actin and myosin II.¹⁸³ Focal adhesion complexes bind these fibers to the inner plasma membrane.¹⁸⁴ The activation of ROCK by caspases gives ROCK a crucial role in forming membrane blebs during apoptosis.^{180,181} ROCK also plays a role in embryonic development, inducing cell migration, differentiation, and axis formation through its expression in the cardiac mesoderm, lateral plate mesoderm, and neural plate.¹⁷⁸

c-Jun N-terminal kinase (JNK)

JNK is a member of the three mitogen-activated protein kinase (MAPK) pathways that control cell proliferation, migration, and differentiation.¹⁸⁵ The MAPK pathways are all activated by a series of phosphorylation reactions. JNK is activated by the MAP2K enzymes MKK4 and MKK7.¹⁸⁶ Scaffold proteins just as JNK interacting protein 1 (JIP1) facilitate rapid activation of the JNK pathway.¹⁸⁷ The JNK pathway can be inactivated by dual-specificity phosphatases (DUSPs).¹⁸⁸ The JNK family contains three genes that can be spliced into 10 isoforms, namely, JNK1, JNK2, and JNK3. JNK1 and JNK2 are expressed in a variety of tissues, while JNK3 is expressed in the brain, heart, and testis.¹⁸⁵ The protein products of these three genes are about 400 amino acids long, with a canonical Ser/Thr kinase domain.¹⁸⁹ JNK is best known for inducing apoptosis via stimulating the mitochondrial release of cytochrome C to activate caspase and trigger apoptosis.¹⁹⁰ However, the effects of JNK are context-specific, as JNK can phosphorylate anti-apoptotic Bcl-2 to promote apoptosis, but it can also phosphorylate pro-apoptotic BAD protein to prevent apoptosis.¹⁸⁵ The dysregulation of JNK is linked to multiple diseases, such as neurodegenerative disorders, cancer, and

autoimmune diseases. JNK3 is being investigated as a potential target for the treatment of CNS disorders.¹⁹¹

c-Jun is the major substrate for JNK.¹⁸⁵ c-Jun is a subunit of the transcription factor, activator protein 1 (AP-1).¹⁹² Anti-c-Jun antibodies caused partial G0 arrest in the cell cycle,¹⁹³ while overexpression produced a greater transition into the S, G2, and M phases.¹⁹⁴ Indeed, c-Jun is crucial in the regulation of the G1/S phase transition.¹⁹⁵ c-Jun also plays an important role in both inducing and inhibiting apoptosis.¹⁹⁶ Endogenously, c-Jun inhibits the expression of apoptosis-inducing genes and maintains cell survival, particularly p53.^{196,197} However, JNK signaling can cause c-Jun to induce apoptosis via survival factor removal.¹⁹⁸ JNK phosphorylates the Ser63 and Ser73 residues of the c-Jun activation domain, causing AP-1 transcriptional activity to increase.¹⁹⁹ This can cause the induction of apoptosis, although AP-1 is also involved in the inhibition of apoptosis as well depending on the tissue and developmental stage.¹⁹⁶ As one might assume, c-Jun, along with JNK has been implicated in many cancers.^{185,197}

CapZ-interacting protein (CapZIP)

CapZIP is a protein detected in muscle extracts that interacts with the F-actin capping protein CapZ. Capping proteins normally inhibit actin polymerization, preserving the actin monomer pool.²⁰⁰ In humans, CapZIP is phosphorylated at Ser-179 and Ser-244 by mitogen-activated protein kinase-activated protein kinase 2 (MAPKAP-K2).⁴⁵ CapZIP serves as a substrate for stress-activated protein kinases (SAPKs), which can also phosphorylate it at several sites (Ser-68, Ser-83, Ser-108, and Ser-216).⁴⁵ Cell exposure to various stressful events causes the activation of several MAPKs, including SAPKs, JNK1, and JNK2. When under stress, SAPK3 and SAPK4 will phosphorylate CapZIP, triggering the dissociation of CapZIP from CapZ and enabling CapZ to modify actin filaments.⁴⁵ JNK1 can also phosphorylate CapZIP, although slower and less extensively.⁴⁵ Northern blot analysis reveals that CapZIP is expressed highly in skeletal muscle but to a lesser extent in cardiac muscle. In other organs such as the brain, lung, and liver, it is hardly detectable at all.⁴⁵ Multiple-tissue expression arrays revealed that CapZIP is expressed in immune organs like the thymus, spleen, lymph nodes, and bone marrow.⁴⁵ CapZIP is also expressed in immune cells and has been isolated from B cells, as well as leukemia and lymphoma cell lines.⁴⁵ CapZIP also regulates ciliogenesis via its upstream regulator Dvl. Dvl binds to ERK7 and CapZIP, functioning as a scaffold for a MAPK family member, ERK7 (also called MAPK15), to phosphorylate CapZIP, enabling ciliogenesis.²⁰¹

Myosin II regulatory light chain (MRLC)

MRLC is a regulatory component of myosin II. Myosin light chain kinase (MLCK) phosphorylates the Ser19 residue of MRLC, and sometimes can even phosphorylate the Thr18 residue.²⁰² This enhances the activity of the actin-activated Mg²⁺ ATPase in myosin II, increasing the assembly and stability of myosin II filaments. The dephosphorylation of MRLC increases myosin II activity and stability more so than monophosphorylation.^{203,204} However, both forms of MRLC are required for organizing stress fibers during interphase and forming the contractile ring in cell division.²⁰² Dephosphorylation is also important for the disassembly of

previous myosin filaments to form new ones.²⁰⁵ ROCK1 can phosphorylate MRLC to facilitate the generation of traction for cell motility.²⁰⁶ ROCK proteins also dephosphorylate MRLC,²⁰⁵ enabling the colocalization of MRLC with actin filaments in mitotic and interphase cells.²⁰² Diphosphorylated MRLC also induces the formation of thick actin bundles containing myosin II, while unphosphorylated MRLC inhibits actin bundle formation.²⁰² These findings emphasize the importance of MRLC in actin organization.

Diaphanous 1 (DIA1)

DIA1 (also called mDia in mammals) is a member of the diaphanous-related formins (DRFs) that are Rho-GTPase binding proteins.⁴⁶ DIA1 contains a novel formin homology (FH) 2 domain that protects the barbed end of the actin filament from capping proteins and enables rapid assembly of actin subunits.²⁰⁷ DIA1 also has an FH1 domain that binds profilin-bound G-actin to bring it closer to the barbed end for elongation of the actin filament.^{207,208} DIA1 contains a Rho-GTPase binding domain in the N-terminal region and a Diaphanous-autoregulatory domain (DAD) in the C-terminal region. The N-terminal regulatory region is usually bound to DAD, causing autoinhibition. However, the binding of RhoA relieves this inhibition.^{209,210} DIA1 is involved in a variety of processes, such as mechanotransduction, cell polarity, migration, and even exocrine vesicle secretion.⁴⁷ DIA1 mutations have been associated with deafness, cancer, and mental retardation.⁴⁷ Mutations in formins like DIA1 have also been implicated in cancer metastasis due to the loosening of cellular adhesion. Expression levels of DIA1 have correlated with the stage and metastasis of cancer cells.²¹⁰

Intracellular mediators of the Wnt/Ca²⁺ pathway

Phospholipase C (PLC)

PLC is activated by G protein due to Wnt ligand binding in the Wnt/Ca²⁺ pathway.^{52,53} The PLC family contains 13 different members with various functions and different structures. Although these members usually have little amino acid sequence homology, they have conserved EF-hand domains, PH domains, and C2 domains.²¹¹ They also all have catalytic X and Y domains.²¹¹ The PH domain is located towards the N-terminus and mediates the recruitment of PLC to the plasma membrane via binding to PIP₂.²¹² The EF-hand motifs are part of the catalytic core of PLC along with X, Y, and C2.²¹³ The EF-hand motif undergoes a conformational change upon binding of Ca²⁺ to PLC, revealing binding sites for other ligands.²¹⁴ The X and Y domains form a triosephosphate isomerase (TIM) barrel-like structure composed of alternating α - and β -pleated sheets.²¹³ The X domain contains all the catalytic residues, while the Y domain is important in modulating the preference of PLC for PIP₂, as well as two other ligands PIP and PI.²¹⁵ The C2 domains are formed from an eight-stranded β -pleated sandwich.²¹³ Upon Ca²⁺ binding, C2 domains can mediate PLC binding to phospholipids to mediate signal transduction and membrane trafficking.²¹⁶ Hokin et al used P³² to detect how phospholipid levels changed during enzyme secretion in pancreas slices with the addition of acetylcholine or carbamylcholine. They discovered that phospholipid activity increased five-to nine-fold.²¹⁷

Unbeknownst to them at the time, the enzyme responsible for the increased phospholipid level was PLC.²¹⁴ PLC cleaves PIP₂ into IP₃, which can then induce the intracellular release of Ca²⁺.²¹⁸ PLC can be activated by a wide variety of receptors, such as B-cell receptors, T-cell receptors, Fc receptors, tyrosine kinase receptors, and G-protein coupled receptors. Due to the variety of receptors, this means that PLC can be activated by a wide variety of ligands, such as neurotransmitters, hormones, and histamine.²¹⁴ Aside from IP₃, the other product of PIP₂ cleavage is DAG, which serves as a secondary messenger to activate Ca²⁺-dependent protein kinase C (PKC) to phosphorylate numerous downstream effectors and activate a wide array of cellular functions, such as cell polarization, proliferation, as well as learning and memory.^{219,220} Through Ca²⁺ release, PLC can regulate cell proliferation, differentiation, motility, gene expression, and other functions.²¹⁴

Protein kinase C (PKC)

PKC is a family of protein kinases involved in a wide variety of diseases, such as diabetes, cancer, and heart disease.²²¹ There have been 518 protein kinase genes identified.²²² PKC, in particular, phosphorylates Ser and Thr residues.²²¹ PKC is activated by DAG, but can also be activated by phorbol esters, which are tumor promoters that mimic the action of DAG.^{221,223} Specifically, DAG and phorbol esters bind to the C1 domain on PKC, which contains a cysteine-rich sequence that resembles a DNA-binding zinc finger domain.²²⁴ PKC contains a regulatory region in the N-terminal half and a catalytic region in the C-terminal half. The C1 and C2 domains are located in the regulatory region and bind to the catalytic region, inhibiting its activity.²²¹ The catalytic kinase region contains C3 and C4 domains.²²¹ An important aspect of PKC is that, upon activation, it translocates to various cellular locations, joining with specific anchoring proteins at each site of action.²²⁵ With phorbol esters that irreversibly activate PKC isoenzymes in a nonselective manner, PKC was discovered to regulate many cellular functions.²²¹ Some of these include cell proliferation, cell death, regulation of ion channels and receptors, cell-to-cell contact, and increasing gene transcription.²²¹ However, this early work using phorbol esters does not reflect the effects of DAG due to the irreversibility of phorbol ester binding. The lack of selectivity of phorbol esters also means that they cannot identify the function of each PKC isoenzyme.²²¹ Isoenzyme-specific inhibitors are still under development and undergoing trials, but this endeavor has proven to be difficult.^{221,223}

Cdc42 protein

Cdc42 is a member of the Rho family of small GTPases with roles in actin cytoskeleton regulation, cell motility, cell polarity, and cell cycle progression among other functions.²²⁶ Cdc42 was characterized in *Saccharomyces cerevisiae*, a species of yeast, with G25K being its mammalian and human homolog.^{227,228} Cdc42 has a P-loop, two switch regions (switch I and switch II), a polybasic region at the C-terminus, and a CAAX box for posttranslational geranylgeranylation.²²⁶ Cdc42 has been found in distinct pools in the Golgi apparatus, ER, and plasma membrane.^{229,230} The Golgi pool functions in three ways; it serves as a reservoir of Cdc42, functions independently from the

plasma membrane pool to control protein transport from the Golgi apparatus, and coordinates with the pool to dictate cell polarity.²³¹ Cdc42 modulates the Golgi-to-ER transport via actin regulation.²³² The plasma membrane pool serves to regulate cell polarity via actin cytoskeletal rearrangement.²³³ At the ER, Cdc42 is necessary for tubule fission during ER remodeling.²³⁴ The activation of Cdc42 is regulated by GEFs, but unlike other Rho GTPases, Cdc42 is a hydrolase that can hydrolyze the GTP to GDP in the presence of GAP, even though GAP normally exchanges GDP for GTP.^{235,236} GDI inhibits Cdc42 like other Rho GTPases, via preventing GDP/GTP exchange.¹⁷⁴ However, GDI also functions as a chaperone, delivering Cdc42 to its proper location and preventing its degradation.²³⁰ Cdc42 is currently under investigation as a therapeutic target for the treatment of cancer, although there are few Cdc42 mutations and no driver mutations that have been linked to cancer.²²⁶

Ca²⁺/calmodulin (CAM)-dependent kinase II (CAMKII)

CAMKII is a Ser/Thr protein kinase²³⁷ that was first characterized as a Ca²⁺-dependent regulator by Schulman and Greengard in 1978.²³⁸ There are over 80 known CAMKs.²²² CAMKII is encoded by four different genes (α , β , γ , δ) in eukaryotes.²³⁹ CAMKII monomers form 12 subunit holoenzymes, with a C-terminal association domain that brings the N-terminal catalytic kinase domains together to fold into two rings of six subunits each.^{237,239} There is a regulatory segment that follows the kinase domain and is joined to a linker region that then connects to the association domain.^{237,240} The regulatory domain forms an α -helix to block the catalytic domain of each subunit.²³⁷ Additionally, the T286 phosphorylation site is sequestered in a hydrophobic groove, preventing autophosphorylation that would up-regulate CAMKII activity. Upon Ca²⁺/CAM binding to the regulatory domain, the regulatory segment is removed, enabling progressive autophosphorylation of the T286 site and increasing CAMKII activity.²³⁷ Normally, in the autoinhibited (compact) state, CAM cannot bind^{237,240,241}; it is only when the compact state is in equilibrium with the non-autoinhibited (extended) state that CAM can bind the regulatory domain.²³⁷

Between the four CAMKII genes found in humans, the enzymes produced share 95% of their amino acid sequences in the kinase domains and 80% homology in the hub domains, with the linker region being the primary variable component.²⁴⁰ When two adjacent subunits are activated by Ca²⁺/CAM, they can phosphorylate one another on Thr286, causing increased Ca²⁺-independent activity.²⁴⁰

CAMKII has a role in adaptive contractive response during aerobic exercise.²⁴² It is also involved in glucose production, cell cycle progression, and vascular smooth muscle function.²⁴³ Continuous activation of CAMKII can cause cardiac myocyte apoptosis, heart failure, and cardiac arrhythmia.²⁴⁴ CAMKII has also been identified as being able to spread the inflammatory response caused by damage to heart muscle.²⁴³ The progression of cardiomyopathy and even Chagas disease caused by *Trypanosoma cruzi* is mediated by CAMKII signaling.^{243,245}

TGF- β activated kinase 1 (TAK1)

TAK1 is a member of the MAPK kinase kinase (MAPKKK) family. It is a Ser/Thr kinase that was originally discovered

to be a mediator of BMP and TGF- β signaling.^{246,247} TAK1 can be activated by many cytokines, such as TNF- α , TGF- β , TLRs, and IL-1.^{248,249} TAK1 activation causes the phosphorylation of TAK1, which leads to the activation of NF- κ B, JNK, ERK, p38, and MAPKs.^{248,249} TAK1 is also involved in T- and B-cell signaling,²⁴⁸ as well as angiogenesis during embryonic development.²⁴⁹ Due to its role in immune and inflammatory processes, TAK1 has been implicated in multiple cancers, such as lymphoma and neuroblastoma, as well as colon, ovarian, and pancreatic cancers.²⁵⁰ In addition, a blockade of TAK1 leads to p53 up-regulation, indicating the importance of TAK1 in controlling cellular stresses.²⁵¹ Activation of TAK1 requires three proteins, TAK1-binding protein 1 (TAB1), TAB2, and TAB3. TAB1 serves as an adaptor protein located on the N-terminal kinase domain of TAK1, while TAB2 and TAB3 will only bind the C-terminal TAK-binding domain after stimulation.²⁴⁹ TAB1 overproduction leads to increased TAK1 activity, but TAB1 deficiency has minor downstream effects.²⁴⁹ TAB2 and TAB3 are not required early on in TAK1 activation but are required for sustained TAK1 activation.²⁴⁹ In mice, TAK1 has been demonstrated to be a regulator of TNF signaling in the skin and modulates skin inflammation. It was found that a lack of TAK1 caused keratinocyte death due to absent NF- κ B and JNK-mediated cell survival signaling.²⁵² Recently, a selective TAK1 inhibitor called Takinib has been developed; it is activated by ATP and competitively inhibits TAK1 by binding to its ATP-binding pocket.²⁵³ Another TAK1 inhibitor called piperidylmethoxyxalcone (PMOC) also functions in the same manner.²⁵⁴

Nemo-like kinase (NLK)

NLK is a conserved Ser/Thr MAPK.²⁵⁵ It is the mammalian homolog of the *Drosophila* nemo gene, which was identified in *Drosophila* as important for the rotation of photoreceptor clusters in eye morphogenesis.²⁵⁶ NLK has a longer N-terminal region that is rich in histidine, proline, alanine, and glutamine.²⁵⁷ NLK can phosphorylate TCF4 to prevent the β -catenin/TCF complex from binding to DNA and initiating gene transcription. TAK1 can also stimulate NLK to phosphorylate TCF4, so NLK functions as a downstream effector of TAK1 in the repression of Wnt/ β -catenin signaling.⁵⁸ Although TAK1 and NLK can repress canonical Wnt signaling, CAMKII from the noncanonical Wnt/Ca²⁺ pathway activates TAK1 in order to do so, indicating crosstalk between the two pathways.⁵⁷ Furthermore, TAB2 can serve as a scaffold for TAK1 and NLK, enabling cooperative interaction for the inhibition of canonical Wnt signaling.²⁵⁸ NLK can also associate with NLK-associated RING finger protein (NARF), which is an E3 ubiquitin ligase whose action is up-regulated by NLK kinase activity.²⁵⁹ NARF can ubiquitinate TCF/LEF for degradation by the proteasome.²⁵⁹ NLK is involved in other pathways as well, such as Notch signaling and STAT protein signaling.²⁵⁹ Due to the function of NLK in Wnt signaling and other pathways, it is crucial in regulating cell proliferation, apoptosis, migration, and other functions.²⁵⁹ For instance, NLK was found to inhibit the growth and migration of non-small cell lung cancer by restoring the expression of E-cadherin, which normally suppresses migration and invasion.²⁶⁰ However, analysis of colorectal cancer (CRC) cells found that NLK activity was increased, and only half of the CRC

cells were apoptotic compared to non-CRC cells, indicating that NLK may have an anti-apoptotic function.²⁶¹ NLK has also been identified as a pathological effector in mouse hearts, leading to progression toward heart failure and other cardiac conditions.²⁶² This makes NLK a potential therapeutic target for various diseases.

Calcineurin (CaN)

CaN is a conserved Ser/Thr phosphatase that is activated by increased intracellular Ca^{2+} .²⁶³ It was first discovered by Wang and Desai in 1976 as a protein that counteracted the activation of bovine brain cyclic nucleotide phosphodiesterase.²⁶⁴ CaN is a heterodimer consisting of calcineurin A (the catalytic subunit) and calcineurin B (the regulatory subunit).²⁶⁵ The catalytic domain of calcineurin A is towards the N-terminal end, while the C-terminus contains three regulatory domains, the calcineurin B binding domain, the calmodulin-binding domain, and the auto-inhibitory domain that binds to the active site when Ca^{2+} /calmodulin is absent.²⁶⁵ Calcineurin B contains four Ca^{2+} -binding EF-hand motifs.²⁶⁵ The mechanism of calcineurin B activation is still unknown.²⁶⁶ It is also structurally homologous to CAM.²⁶³ CaN along with CAM binds to Ca^{2+} , then CAM binds CaN. This complex is what forms the active phosphatase.²⁶³ Although there are multiple CAM-modulated kinases, CaN is the only phosphatase that is directly activated by Ca^{2+} .²⁶³ CaN is best known for regulating the transcription of IL-2, dephosphorylating the transcription factor NF-ATp in response to an increase in intracellular Ca^{2+} via T cell receptor activation, and stimulating NF-ATp. CaN also controls cellular Ca^{2+} sequestration and cytokinesis.²⁶⁶ The immunosuppressive drugs FK506 and cyclosporin A can both target CaN and were key to identifying its functions.²⁶⁶ CaN is also important in the regulation of neurotransmitter release in neuromuscular junctions (NMJs).²⁶⁷ Hypertrophic and dilated cardiomyopathy have both been linked to CaN overactivation.²⁶⁸

Antagonists and inhibitory regulators of Wnt signaling

Extracellular antagonists of Wnt signaling

Secreted frizzled-related proteins (sFRPs)

The largest family of secreted Wnt inhibitors are the sFRPs, which are structurally like the CRD ligand-binding domain of the Fz receptors.²⁶⁹ The first sFRP discovered was the Frizzled motif associated with bone development (Frzb).²⁷⁰ The sFRPs are about 295–346 amino acids long, with a CRD at the N-terminus.²⁶⁹ These CRDs share 30%–50% sequence similarity with the CRDs of Fz receptors, with 10 cysteine residues linked by disulfide bridges. The C-terminus has a netrin-related motif (NTR) that functions as a heparin-binding domain.²⁷¹ Studies of *Xenopus* embryos revealed that Frzb binds to Wnt1 and Wnt8, sequestering them away from the Fz and LRP5/6 receptor complex.^{272–274} sFRPs can also form a nonfunctional complex with Fz receptors to prevent Wnt binding and signaling.^{275,276} Both the CRD and NTR domains are important in Wnt inhibition.²⁷⁷ The CRD domain binds to Wnt,²⁷³ while the NTR domain mimics the

function of the entire sFRP1 to bind to Wnt ligands and inhibit Wnt signaling.²⁷⁸

Wnt-inhibitory factor 1 (WIF-1)

WIF-1 was first identified as an expressed sequence tag in the human retina²⁷⁹ found in fish, amphibians, and mammals.²⁷⁹ It contains a 150 amino acid long N-terminal signal sequence called the WIF domain, five epidermal growth factor (EGF)-like repeats, and a 45 amino acid hydrophilic domain at the carboxy-terminus.²⁷⁹ The complete WIF-1 protein is 379 amino acids long.²⁷⁹ Analysis of *Xenopus* embryo assays found that WIF-1 binds Wnt8, while in *Drosophila* clone-8 cells, WIF-1 binds Wg, although not as strongly as with Wnt8.²⁷⁹ WIF-1 inhibits the binding of Wnt8 to Dfz2, blocking Wnt signaling.²⁷⁹

Dickkopf (Dkk) proteins

The Dickkopf (Dkk) proteins contain four members in vertebrates (Dkk1–4). Dkk1 was the first protein of its family to be discovered through experiments determining its importance in embryonic head formation and as a Wnt antagonist.²⁸⁰ The Dkk proteins are glycoproteins composed of 255–350 amino acids.²⁸¹ Interestingly, although the Dkks all have a signal sequence and two conserved CRDs, there is little sequence similarity between them otherwise.²⁸⁰ Dkks primarily modulate the canonical Wnt signaling pathway via binding to the LRP5/6 co-receptor. Dkk1 associates with one of two single-pass transmembrane proteins called Kremen 1 or 2 (Krm1 or Krm2) while binding to LRP5/6. This interaction allows the endocytosis of the LRP5/6 receptor.^{19,281} Analysis of Dkk1 in cardiogenesis suggests that Dkk1 can activate JNK, indicating that Dkk1 may be involved in the noncanonical Wnt/PCP pathway as well.²⁸²

Wise and sclerostin (SOST) proteins

Wise and SOST are both members of a subfamily of cysteine knot proteins as they both contain cysteine knot motifs.²⁸³ Both Wise and SOST are also BMP antagonists.^{283,284} Wise protein is composed of 206 amino acids with 38% sequence homology with sclerostin.²⁸⁴ Wise is expressed in a wide variety of tissues, including branchial arches, rat endometrium, developing testes, and more.²⁸⁵ The sclerostin polypeptide is about 190 amino acids long with a cysteine knot formed from its flexible N- and C-terminal regions.²⁸⁶ Sclerostin is expressed in osteocytes to regulate bone formation. Deletions in the SOST gene that codes for sclerostin can result in van Buchem disease, which is characterized by a high bone mass due to loss of inhibition of bone formation.²⁸⁶ Reporter assays demonstrate that Wise can block Wnt1, Wnt3a, and Wnt10b.²⁸⁵ Wise also binds to the LRP6 coreceptor, preventing the binding of Wnt8.²⁸⁴ Additionally, Wise can function intracellularly to prevent LRP6 trafficking to the cell surface.²⁸⁷ Wise can also bind LRP4 to inhibit Wnt signaling.²⁸⁸ Sclerostin binds LRP5/6, specifically binding to LRP5 at the YWTD-EGF repeat domains, inhibiting Wnt signaling.^{289,290}

Cerberus

Cerberus is an abundant organizer-specific gene that was isolated from *Xenopus* and can induce ectopic head formation.²⁹¹ It also plays an important role in cardiogenesis during vertebrate embryonic development.²⁹² Cerberus,

like Wise and SOST, is in the cysteine knot superfamily.^{293,294} Long-form Cerberus (xCer-L) binds to Wnt8 and could inhibit Wnt signaling, but short-form Cerberus (xCer-S) cannot do the same.²⁹⁴ Further study is required to better understand the effects of Cerberus on the Wnt pathway.

IGFBP-4

IGFBP-4 is a member of the insulin-like growth-factor-binding proteins (IGFBPs) that bind and modulate insulin-like growth factors (IGFs). IGFBP-4 is important in cardiomyogenesis *in vitro*.²⁹⁵ IGFBP-4 interacts with LRP6 and Fz8, functioning as a competitive inhibitor of Wnt3a binding.²⁹⁵ In this manner, IGFBP-4 inhibits the canonical Wnt pathway.^{295,296} Furthermore, there are six IGFBP members. Although IGFBP-4 is the most powerful Wnt inhibitor, IGFBP-1, -2, and -6 also have Wnt inhibitor activity, albeit modestly. On the other hand, IGFBP-3 and -5 have no Wnt inhibition activity.²⁹⁵ Although some cancer cell lines *in vitro* saw a decrease in cell proliferation when treated with IGFBP-4, decreased levels of the protein increased the risk of breast cancer, and overexpression of IGFBP-4 caused prostate cancer growth *in vivo*.²⁹⁵

Intracellular negative regulators of Wnt signaling

Adenomatous polyposis coli (APC)

APC is a tumor suppressor gene that is heavily associated with colorectal cancer (CRC).²⁹⁷ The APC protein is 2843 amino acids long,²⁹⁷ with a central region spanning about 1000 amino acids containing motifs that bind β -catenin or Axin.²² In humans, APC contains four 15-mers and seven 20-mers. The 20-mer repeats are phosphorylated by GSK-3 β and CK1, enhancing the affinity of APC for β -catenin.²² Interspersed with the 20-mer repeats are three Ser-Ala-Met-Pro (SAMP) repeats that bind to Axin.^{298,299} Full-length APC or APC with SAMP repeats was found to protect β -catenin from dephosphorylation by PP2A, ensuring the destruction of β -catenin.²⁹⁹ The N-terminus of APC has a dimeric coiled-coil domain, with a heptad-repeat region that is crucial for the dimerization of APC.³⁰⁰ APC also contains an armadillo repeat domain (APC-arm) that enables APC to regulate the actin cytoskeleton and microtubules during cell polarization and migration.^{301,302} The C-terminal region of APC binds several proteins, such as microtubulin, indicating its role in microtubule assembly.³⁰³ APC mutation is best known for causing familial adenomatous polyposis (FAP), a disease characterized by a family history of colorectal polyps and cancer. FAP is inherited in an autosomal dominant manner via a germline mutation.³⁰⁴

Axin proteins

Axin is a scaffold protein that was initially discovered as a protein product of the mouse Fused (Fu) gene, inhibiting Wnt signaling and regulating embryonic axis formation.³⁰⁵ There are two Axin homologs in eukaryotic organisms, Axin1 and Axin2, which are often collectively referred to as Axin. Axin1 is vital for embryonic viability and is widely expressed, while Axin2 is limited in its distribution to certain tissues.^{306,307} Axin1 was identified in mice as a locust causing kinky tail phenotypes, while Axin2 was

identified due to its interactions with GSK-3 β and β -catenin, as well as its homology to Axin1. The N-terminus contains the RGS (regulation of G-protein signaling) domain, which binds to APC.²⁹⁸ It is important to note that this region is homologous to regulators of G-protein signaling, but does not actually regulate any known G-protein.²⁹⁸ The RGS domain binds the third SAMP repeat and is highly conserved among Axins but is not conserved in other RGS proteins.²⁹⁸ Axin contains a C-terminal DIX domain that enables it to form homodimers with other Axin proteins, as well as heterodimers with Dvl.^{308,309} This interaction via the DIX domains is how Dvl can inhibit Axin and enable transduction of Wnt/ β -catenin signaling.¹⁴⁴ In between the RGS and DIX domains are regions that allow Axin to bind to β -catenin, GSK-3 β , and CK1, forming a complex that will target β -catenin for degradation.³⁰⁷ Axin is important in many developmental processes, including anterior–posterior axis formation, organogenesis, neuronal proliferation and differentiation, synapse formation, and several others.³⁰⁷ Both Axin1 and Axin2 can function as the scaffold of the β -catenin destruction complex. They share the RGS and DIX domains, as well as the Tankyrase binding domain that maintains the Axin protein's stability.³⁰⁶ Interestingly, Axin2 is a target of β -catenin mediated gene transcription, but not Axin1. Due to this fact, Axin2, in particular, has been of interest in cancers caused by aberrant Wnt signaling activation, which leads to high levels of Axin2 expression.³⁰⁶ Axins also interact with many other pathways. Its interactions with p53 are important in stimulating transcription in p53-dependent target genes, shedding light on the importance of Axin as a tumor suppressor.³¹⁰ Axin1 and Axin2 both inhibit Wnt signaling and stimulate TGF- β signaling. However, TGF- β signaling can inhibit Axin1 and Axin2 expression, leading to enhanced Wnt signaling that increases chondrocyte maturation.³¹¹

Glycogen synthase kinase-3 (GSK-3) proteins

GSK-3 is a conserved Ser/Thr kinase that was first identified in rabbit skeletal muscle.^{312,313} There are two forms of GSK-3 in mammals, GSK-3 α and GSK-3 β , which are 98% homologous in the internal kinase domain, albeit with different N-terminal regions³¹⁴ and C-terminal regions.³¹⁵ Defects in GSK-3 α in mice caused impaired locomotion and coordination, as well as psychiatric disorders.³¹⁶ GSK-3 β defects, on the other hand, are embryonically lethal,^{317,318} indicating that GSK-3 β may be more important than GSK-3 α . GSK-3 inhibits glycogen synthase via phosphorylation, thereby inhibiting glycogen synthesis.³¹⁹ GSK-3 is inactivated by insulin, so GSK-3 dysregulation has been implicated in type II diabetes.³¹⁸ Since both Wnt and insulin signaling pathways involve GSK-3 β , this suggests that there is potential crosstalk between them. AKT phosphorylates and inhibits GSK-3³²⁰ and Wnt causes GSK-3 dislocation from the destruction complex.³²¹ GSK-3 β is rich in the brain and is involved in neurogenesis, neurotransmission, and regulation of synaptic plasticity.³²² However, it has also been implicated in the formation of neurofibrillary tangles in Alzheimer's disease.³²² GSK-3 β can be inhibited by lithium, which is used as a mood stabilizer for treating bipolar disorder.³²³ GSK-3 β also enhances apoptosis via up-regulation of p53.³²⁴ GSK-3 can also modulate inflammation downstream of TLR signaling; activation of GSK-3 leads to the

production of the pro-inflammatory cytokines IL-6, IL-1B, and IFN γ , while inhibition of GSK-3 leads to the production of anti-inflammatory IL-10.³²⁵

Casein kinase 1 (CK1)

Casein kinase 1 (CK1) is a large family of conserved Ser/Thr kinase found in eukaryotes, with seven isoforms in humans (α , γ 1, γ 2, γ 3, δ , ϵ , and α -like).³²⁶ CK1 is involved in a diverse range of cellular processes, such as vesicular trafficking, DNA repair, cell proliferation, and apoptosis.³²⁷ The kinase domain is conserved, but the N- and C-terminal domains vary in length among the CK1 family members.³²⁶ The N-terminal domain of CK1 is its catalytic domain, while the C-terminal domain has been associated with an inhibitory function, as CK1 with a truncated C-terminus led to an increase in catalytic domain activity.^{328,329}

CK1 is generally considered constitutively active.³³⁰ The reason is unknown, but phosphatases have been inferred to be involved in this activity.^{329,330} Phosphorylation of CK1 either by itself or by other kinases inhibits its catalytic activity.³³⁰ Autophosphorylation on the C-terminus results in a phosphopeptide that binds to the catalytic domain.³³⁰ CK1 isoforms are regulated by scaffold proteins, which can sequester CK1 at several subcellular locations or bind CK1 allosterically to promote or inhibit CK1 activation.³³⁰ Although CK1 is also involved in the Hedgehog, NF- κ B, and Hippo signaling pathways,^{326,327,330} Wnt signaling is perhaps its best-characterized process.³³⁰ CK1 phosphorylates β -catenin on Ser45, priming it for subsequent phosphorylation on Thr41, Ser37, and Ser33 by GSK-3.³³¹ β -TrCP can then ubiquitinate β -catenin for proteasomal destruction.^{22,24}

CK1 also has a role in modulating p53 due to its inhibitory interacting proteins, mouse double minute homologue 2 and 4 (MDM2 and MDM4), being substrates of CK1. Phosphorylation of these substrates by CK1- δ and CK1- ϵ leads to p53 activation.³³² However, other studies have demonstrated that the knockdown of CK1- α can also activate p53.³³³ This suggests that different isoforms could have opposite effects on p53 signaling. CK1 was also the first kinase discovered to regulate the circadian rhythm, where the positive regulatory complex CLOCK-BMAL activates the transcription of PER orthologs PER1-3, and cryptochrome proteins CRY1-2. The PER and CRY proteins then inhibit the CLOCK-BMAL complex.³³⁰ This completes the circadian cycle, but then CK1 can phosphorylate PER proteins for degradation, starting the cycle again.³³⁰ Additionally, CK1 is involved in mitosis, regulating spindle positioning upon being delivered to spindles by the CK1-specific binding protein FAM83D.³³⁴ CK1 has been prominently linked to neurodegeneration and cancer.³³⁰

β -Transducin repeat-containing proteins (β -TrCP)

β -TrCP is the substrate recognition subunit for the Skp1/Cullin1/F-box protein (SCF) E3 ubiquitin ligases.³³⁵ β -TrCP (also called FWD1 in mammals) is part of the Fbw (F-box/WD40 repeat containing) protein family, whose members share an N-terminal F-box motif and seven C-terminal WD40 repeats. Other F-box proteins include Fbl (F-box/leucine-rich repeats) and Fbx (F-box/unknown motifs).³³⁵ These motifs are highly conserved, and humans have two isoforms, β -TrCP1 and β -TrCP2.³³⁵ β -TrCP functions as an

intracellular receptor to bind phosphorylated β -catenin at the phosphorylated sites, but this interaction is more efficient in the presence of Axin.³³⁶ In addition, Axin does not bind well with β -catenin without β -TrCP, suggesting that the three form a ternary complex that requires all three members for optimal stability.³³⁶ The F-box motif is used to bind β -TrCP to Skp1, which can then ubiquitinate β -catenin for degradation.³³⁷ However, the F-box domain is not required to bind β -catenin, as the binding to β -catenin and Axin is mediated by the WD40 repeat domains.³³⁶

SCF uses β -TrCP to determine substrate specificity.³³⁶ SCF ubiquitinates substrates like β -catenin at DSGXXS destruction motifs, with both Ser residues needing to be phosphorylated for SCF-mediated ubiquitination.³³⁵ β -TrCP/SCF is also involved in the NF- κ B signaling pathway. NF- κ B is usually retained in the cytoplasm by a family of inhibitory molecules called I κ Bs. These can then be phosphorylated by I κ B kinase (IKK) and subsequently ubiquitinated by SCF for degradation, releasing NF- κ B to translocate to the nucleus.³³⁸ β -TrCP has been implicated in cell division due to β -TrCP knockout mice having impaired progression through mitosis, with spermatocytes accumulating at metaphase I.³³⁹ This effect is attributed to the buildup of early mitotic inhibitor 1 (Emi1), which inhibits the activity of the anaphase-promoting complex/cyclosome (APC/C).

Normally β -TrCP/SCF would degrade Emi1 to allow progression through mitosis, but this does not happen without β -TrCP recognizing the DSGXXS destruction motif.³⁴⁰ Due to the oncogenic risks of NF- κ B in inducing gene products controlling signaling proliferation and suppressing apoptosis,³⁴¹ β -TrCP/SCF is linked to oncogenesis and anticancer therapy resistance.³³⁵ β -TrCP/SCF is also under investigation for targeting disease-causing proteins for degradation.³³⁵

Protein phosphatase 2A (PP2A)

PP2A is a member of the phosphoprotein phosphatase (PPP) family and is involved in cell cycle regulation.³⁴² It is a Ser/Thr phosphatase.³⁴³ PP2A is a heterotrimeric enzyme consisting of A, B, and C subunits. A is the scaffolding subunit, B is the regulatory subunit, and C is the catalytic subunit.³⁴³ The core of PP2A is composed of A and C subunits, which both have α and β isoforms that are homologous to each other, although the α isoform is the more commonly expressed variant.³⁴³ The A subunit contains 15 huntingtin-elongation-A subunit-TOR (HEAT) tandem repeats, with each HEAT repeat consisting of a pair of antiparallel helices, which assemble to form a curved structure.³⁴⁴ These helices form a hydrophobic inner ridge that facilitates B and C binding. They also stack with a hinge region between HEATs 12 and 13 to give the A subunit flexibility.³⁴³ The catalytic activity of the C subunit is thought to be mediated by two Mn²⁺ ions,³⁴³ coordinated by six conserved residues (two aspartate, one asparagine, and three histidine residues) along with a catalytic water molecule.³⁴² Modifications of the C-terminal tail of the C subunit are needed for B subunit binding, although the B subunit has different classes and each requires its own unique modification.³⁴⁵

There are four classes of B subunits, namely, the B55 family (B), the PR72 family (B'), the B56 family (B''), and the

Striatin family (B^{'''}). They all vary widely in structure, allowing for diverse substrate specificity for a wide range of cellular functions.³⁴³ Furthermore, although A and C subunits are expressed ubiquitously, B subunit expression levels vary significantly across different cells and tissues.³⁴³ PP2A has dual roles in Wnt signaling. It can dephosphorylate β -catenin, APC, and Axin, enhancing Wnt signaling.³⁴² However, it also inhibits Wnt signaling by dephosphorylating the Ser9 residue on GSK-3 β after being recruited by heat shock cognate 40 (HSC40), activating GSK-3 β .³⁴⁶

PP2A can also activate GSK-3 β indirectly by inhibiting protein kinase B (AKT), which initially phosphorylates GSK-3 β to inhibit it.^{320,346} Due to the different PP2A holoenzymes that exist, PP2A has a wide variety of functions. For instance, PP2A-B55 is crucial to the prevention of entry into mitosis and must be inactivated for cell division to begin.³⁴² However, PP2A holoenzymes are also involved in the disassembly and reassembly of the Golgi apparatus mediated by phosphorylation and dephosphorylation during mitosis. Specifically, dephosphorylation of the Golgi matrix protein GM130 by PP2A induces Golgi reassembly at the end of mitosis.³⁴⁷ Mutations in PP2A have been associated with defects in brain development and are causally linked to intellectual disability and neurodevelopmental disorders.³⁴⁸ Abnormalities in PP2A have also been linked to Alzheimer's disease.³⁴⁹ It is also a factor in cancer onset, such as the progression of COPD in lung cancer when inactivated.³⁵⁰

Groucho

Groucho is a transcriptional corepressor that was first identified in 1968 by a mutation in *Drosophila* that caused clumps of extra bristles to form above the eyes. Transducin-like enhancer of split (TLE) protein is the human homolog.³⁵¹ TLE/Groucho is involved in a variety of processes, such as eye development, segmentation, and sex determination.³⁵¹ TLE/Groucho is characterized by a conserved glutamine-rich (Q) domain on the N-terminus, and a conserved WD-repeat domain on the C-terminus, with a less conserved central region dividing the two.³⁵¹ In the nucleus, TCF/LEF are bound to TLE/Groucho via the N-terminal glutamine-rich (Q) domain on TLE/Groucho.³⁵² The Q domain has an α -helical coiled-coil motif that enables TLE/Groucho to tetramerize, which is required for transcriptional repression.³⁵² The Q domain binds to the HMG domain of TCF/LEF.³⁵² The C-terminal WD-repeat domains function to interact with transcription factors and are also important in chromatin condensation.³⁵³ TLE/Groucho also contains a central glycine/proline (GP)-rich domain that recruits histone deacetylases (HDACs) to repress transcription.^{351,354}

TAK1 and NLK

As mentioned previously, from the noncanonical pathway, TAK1 and NLK can inhibit canonical Wnt signaling via TAK1 activation of NLK to phosphorylate TCF4.⁵⁸ NARF can also inhibit canonical Wnt signaling by ubiquitinating TCF/LEF for proteasomal degradation.²⁵⁹ Additionally, the coexpression of the noncanonical Wnt ligand, Wnt5a, in hepatocellular carcinoma (HCC) cells, resulted in a three-fold decrease in TCF activity. However, the mechanism that caused this is unknown.³⁵⁵

Crosstalk between Wnt and major signaling pathways

Crosstalk with bone morphogenetic protein (BMP) signaling

BMPs are members of the transforming growth factor-beta (TGF- β) family of cytokines.³⁵⁶ In TGF- β /BMP signaling, the binding of a ligand to type I and II receptors brings them together, where receptor II phosphorylates the receptor I kinase domain, propagating the signal via phosphorylation of the receptor-regulated Smad (R-Smad) signal transducing proteins.³⁵⁷ Of the five R-smads (Smad1, -2, -3, -5, and -8), Smad1, Smad5, and Smad8 are activated by BMP signaling, while Smad4 is the co-mediator Smad (Co-Smad).³⁵⁷ The R-Smads form complexes with Smad4 to translocate into the nucleus and regulate gene transcription.^{357,358} BMPs are vital in embryogenesis and homeostasis and are known to induce bone formation.^{11,356,359–363}

BMP receptors determine the strength of BMP signaling through the C-terminal phosphorylation on Smad1, which contains four GSK-3 phosphorylation sites. GSK-3 can phosphorylate Smad1, leading to its ubiquitination and destruction, which reduces the strength of BMP signaling. Wnt signaling can also inhibit GSK-3 β , stabilizing Smad1 and enabling BMP signaling.³⁵⁶ Smad7 is an inhibitory Smad, primarily down-regulating R-Smad activation via the recruitment of the E3 ubiquitin ligases: Smurf1 and Smurf2. Smurf1 and Smurf2 target Smads and the TGF- β receptors for degradation.³⁵⁷ Smad7 can complex with β -catenin and Smurf2, which causes ubiquitination and subsequent degradation of β -catenin³⁶⁴ (Fig. 5).

Additionally, Smad7 can interact with Axin, and this Smad7-Axin complex can cause GSK-3 β and β -catenin to dissociate from Axin while also inhibiting the recruitment of Smurf2 to β -catenin. The overall effect is to stabilize β -catenin, which complexes with E-cadherin to strengthen cell-to-cell adhesion.³⁶⁵ BMP receptor type 1a (BMPR1a) induces the expression of SOST to limit cancellous bone accrual by inhibiting Wnt signaling. However, BMPR1a-deficient osteocytes experience a decreased expression of SOST, indicating the role of BMPR1a in modulating bone growth through Wnt signaling.³⁵⁸ Loss of BMPR1a in mouse rib bones also caused decreased levels of Dkk1, indicating Dkk1 serves as another downstream target of BMPR1a.³⁶⁶ BMP9 induces the formation of ectopic bone via the recruitment of Runx2 and β -catenin to the osteocalcin promoter. Knockdown of β -catenin prevented osteogenic differentiation of MSCs via BMP9, establishing the importance of Wnt signaling in mediating BMP9 activity.³⁶⁷

Wnt3a and BMP9 both induce alkaline phosphatase (ALP) activity in MSCs while enhancing one another's ALP induction, suggesting that they act synergistically to induce bone formation.³⁶⁷ Between the two, Wnt3a induces ALP activity to a stronger degree and earlier than BMP9, demonstrating the crosstalk between the two pathways in osteogenic differentiation.³⁶⁷ Furthermore, Wnt3a and BMP9 both up-regulate CTGF/CCN2, a gene involved in osteogenic differentiation.³⁶³

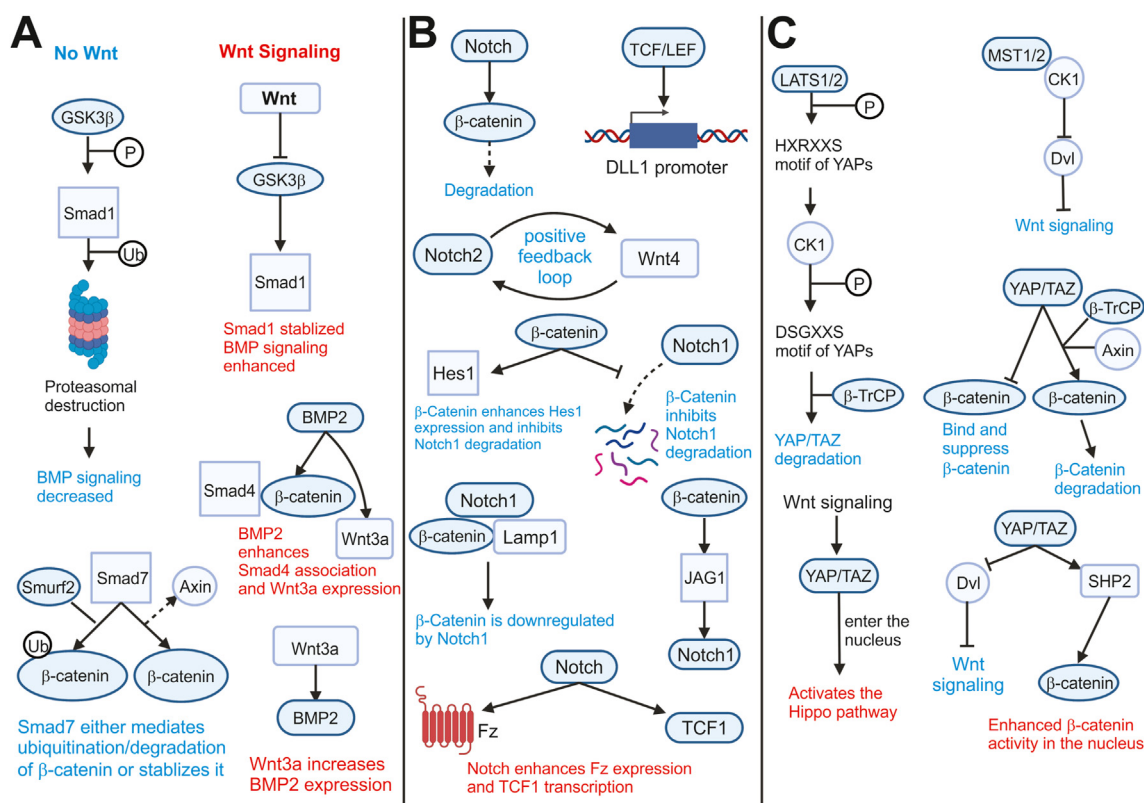


Figure 5 Crosstalk between Wnt signaling and other major signaling pathways. The (A) left, (B) middle, and (C) right panels show the Wnt/BMP crosstalk, Wnt/Notch crosstalk, and the Wnt/Hippo crosstalk, respectively. Note that not all crosstalk interactions between each of these pathways are represented in the image due to space constraints. In Wnt/BMP crosstalk, the absence of Wnt signaling allows GSK-3 β to phosphorylate Smad1, which is ubiquitinated and degraded. The activation of Wnt signaling inhibits GSK-3 β , stabilizing Smad1 and enhancing BMP signaling. BMP2 can up-regulate Wnt3a and enhance β -catenin binding with Smad4. Wnt3a can also up-regulate BMP2 expression. Smad7 can either recruit Smurf2 to ubiquitinate β -catenin for degradation or cause Axin to dissociate from β -catenin, stabilizing it. For Wnt/Notch crosstalk, Notch can complex with β -catenin and promote its degradation. TCF/LEF can regulate the expression of the DLL1 Notch receptor ligand. Notch2 and Wnt4 are involved in a positive feedback loop. β -Catenin can enhance Hes1 transcription and inhibit the degradation of Notch1. β -Catenin can also up-regulate transcription of the JAG1 Notch ligand. Notch1 can complex with β -catenin and Lamp1 to promote lysosomal degradation of β -catenin. Notch can increase both Fz receptor expression and transcription of TCF1. Finally, in Wnt/Hippo crosstalk, there are many interactions as well. LATS1/2 can phosphorylate the HXRXXS motifs of YAP to signal to CK1 to phosphorylate the DSGXXS destruction motifs, leading to the recruitment of β -TrCP-mediated degradation of YAP. MST1/2 can sequester CK1, preventing phosphorylation of Dvl and inhibiting Wnt signaling. YAP/TAZ can bind to and suppress β -catenin while preserving its stability, although other reports indicate that YAP/TAZ is associated with Axin as part of the destruction complex. YAP/TAZ can also inhibit Dvl, reducing Wnt signaling. However, YAP can also interact with SHP2 to enhance β -catenin activity in the nucleus. Wnt signaling can cause YAP/TAZ to transcribe Hippo pathway target genes. The illustration was inspired by and created in BioRender.

BMP2 stimulates LRP5 expression and inhibits β -TrCP expression. This leads to an increase in β -catenin levels in osteoblasts and promotes osteogenic differentiation.³⁶⁸ BMP2 also increased the expression of the canonical Wnt ligands, Wnt1, Wnt3a, and Wnt4.³⁶⁸ The BMP2-mediated up-regulation of Wnt3a and enhancement of the interaction of Smad4 with β -catenin reveal the importance of BMP2 in the modulation of Wnt signaling.³⁶⁹ Interestingly, Wnt3a signaling or overexpression of β -catenin/TCF4 can increase transcription of BMP2, adding further complexity to the interplay of BMP and Wnt signaling.³⁷⁰ BMP2 and Wnt3a exhibit a cooperative effect to increase the transcription of osteogenic genes (Id1, Dlx5, Msx2, Osx, and Runx2).³⁷¹

BMP2 can also antagonize Wnt signaling by promoting Smad1 binding to Dvl1, preventing the activation of β -

catenin.³⁷² BMP2, along with BMP4, can also induce Wnt8 expression in *Xenopus* mesoderm.³⁷³ Perhaps unsurprisingly, mutations in BMP genes can lead to several bone disorders, such as skeletal developmental defects and inappropriate ossification.³⁷⁴ BMP signaling has also been a therapeutic target in cardiovascular diseases such as atherosclerosis, pulmonary arterial hypertension, and anemia of chronic disease.³⁷⁵

Crosstalk with TGF- β signaling

TGF- β signaling coordinates with Wnt signaling in development and homeostasis. The presence of both signaling pathways is necessary for the initial inductive activity of

BMPs.³⁷⁶ TGF- β and Wnt signaling regulate each other's ligand production, and the Smad/ β -catenin/LEF complex regulates a group of shared target genes, further delineating the cooperativity of the two pathways.³⁵⁶ Eger et al conducted a study of polarized mouse mammary cells that underwent epithelial–mesenchymal transition (EMT) induction by an inducible c-fos estrogen receptor (FosER) oncoprotein. The inhibition of both TGF- β and Wnt signaling caused a reversal of cells that underwent EMT back to their polarized forms, while inhibition of only one pathway led to a partial rescue of the epithelial features. This indicates the cooperativity of TGF- β and Wnt signaling in the EMT.³⁷⁷ Furthermore, Nawshad et al demonstrated in palate medial-edge epithelial cells (MEE) that TGF- β 3 signaling forms Smad2-P-Smad4-LEF1 transcription complexes that repress the E-cadherin gene during EMT.³⁷⁸ The mediation of the EMT via β -catenin and TGF- β is dependent on cyclic AMP-responsive element-binding protein (CREB)-binding protein (CBP). One target gene in EMT is α -smooth muscle actin (α -SMA), which requires β -catenin for the induction of transcription via TGF- β signaling. CBP promotes the interaction between Smad3 and β -catenin after TGF- β 1 activation, leading to the transcription of α -SMA. This signaling mechanism has been implicated in pulmonary fibrosis and is being explored as a therapeutic target.³⁷⁹ TGF- β 1 signaling also causes the translocation of β -catenin to the nucleus via Smad3 in MSCs. The stimulation of the TGF- β receptor causes phosphorylation of Smad3, disrupting its interactions with GSK-3 β . Smad3 can then function as a chaperone to shuttle β -catenin into the nucleus.³⁸⁰

Crosstalk with Notch signaling

Notch signaling is a conserved cell–cell communication mechanism involved in determining cell fate and cell lineage.³⁸¹ In mammals, there are four Notch receptors (Notch1–4) that are homologous to the single Notch receptor in *Drosophila*, and five Delta/Serrate/Lag-2 (DSL) ligands (DLL1, DLL3, DLL4, JAG1, and JAG2).^{382,383} Ligand binding leads to the opening of the negative regulatory domain (NRR). This enables the cleavage of the Notch intracellular domain (NICD) by the multiprotein γ -secretase complex. The NICD can then translocate to the nucleus and form a Notch transcriptional activation complex (NTC) with RBPJ and co-activators of the Mastermind-like (MAML) family. Without Notch, RBPJ interacts with transcriptional repressors, while the binding of Notch enables transcription.^{382,383}

In *Drosophila*, Notch signaling promotes Wg expression at the wing margin, while Wg promotes Serrate and Delta expression in wing patterning. Notch has also been demonstrated to down-regulate Wnt signaling by complexing with β -catenin and promoting its degradation.³⁸⁴ Notch2 is involved in a positive feedback loop with Wnt4, up-regulating Wnt4 expression to promote differentiation of progenitor cells in the nephrons of mice.³⁸⁵ The Wnt pathway transcription factors LEF1/TCF regulate the expression of DLL1, establishing the importance of Wnt signaling in the regulation of Notch signaling.³⁸⁶ Additionally, Wnt/ β -catenin signaling also regulates the transcription of the Notch target gene Hes1, which encodes a basic helix-loop-helix transcriptional repressor.³⁸⁷

β -Catenin can bind to Notch1 and the NICD, which enhances the transcription of Hes1 while also preventing ubiquitin-dependent degradation of Notch1 and NICD.³⁸⁸ On the other hand, Notch1 can down-regulate β -catenin by forming a complex with β -catenin and then colocalizing with the lysosomal protein Lamp1. Lysosomal activity can then decrease the levels of β -catenin.³⁸⁹ GSK-3 β phosphorylates NICD, stabilizing Notch and positively regulating its signaling activity.³⁹⁰ CAMKII induced by Wnt5a signaling can phosphorylate the Notch signaling corepressor, silencing mediator or retinoic acid, and thyroid hormone receptor (SMRT). This causes SMRT to dissociate from RBPJ and get degraded by the proteasome. CAMKII also enhances binding between NICD and RBPJ. The overall effect of CAMKII is to increase Notch signaling.³⁹¹ Dvl down-regulates Notch signaling by inhibiting the CSL (RBPJ) transcription factors that mediate Notch signaling³⁹² (Fig. 5).

Notch-regulated ankyrin repeat protein (NRARP) is a negative regulator of Notch signaling but can also up-regulate Wnt signaling via stabilizing LEF1.³⁹³ In both colorectal and renal cancers, Wnt/ β -catenin signaling up-regulates transcription of the Notch ligand Jagged1 (JAG1), leading to Notch activation.^{394,395} In hematopoietic progenitor cells (HPCs), Notch signaling up-regulates the expression of Fz receptors to enhance dendritic cell (DC) differentiation.³⁹⁶ Notch signaling also increases transcription of TCF1, increasing mediation of Wnt signaling to stimulate T-cell specification.³⁹⁷ Notch dysregulation can lead to kidney fibrosis and chronic kidney disease (CKD).³⁸¹ Defective Notch signaling can also cause congenital heart disease but has become a potential therapeutic target for heart regeneration.³⁹⁸ Liver fibrosis, chronic liver disease, and osteosarcoma are also attributable to aberrant Notch signaling.^{399,400}

Crosstalk with the Hippo/Yap pathway

The Hippo pathway plays an important role in regulating cell fate and tissue structure.⁴⁰¹ The Hippo pathway in *Drosophila* contains four core components, namely, the NDR family protein kinase Warts (Wts), the WW domain-containing protein Salvador (Sav), the Ste20-like protein kinase Hippo (Hpo), and the adaptor protein Mob as a tumor suppressor (Mats). The mammalian orthologs for each of these are LATS1/2 kinase (Wts), SAV1 (Sav), MST1/2 (Hpo), and MOB1 (Mats). MST1/2 forms heterodimers with SAV1 for the phosphorylation of SAV1, MOB1, and LATS1/2. This enables LATS1/2 to phosphorylate yes-associated protein (YAP) and WW domain-containing transcription regulator protein 1 (TAZ). YAP/TAZ are orthologs of the transcriptional coactivator called Yorkie (Yki), which normally stimulates transcription. The phosphorylation of YAP/TAZ by LATS1/2 and the Hippo kinase cascade inhibits transcription.⁴⁰¹ The Hippo pathway interacts with Wnt signaling in several ways.

Phosphorylation of the HXRXXS motif of Ser381 of YAP by LATS1/2 can provide a signal for CK1 to phosphorylate a phosphodegron, which is a DSGXXS destruction motif. This causes the recruitment of β -TrCP, which leads to YAP ubiquitination and eventual degradation, inhibiting YAP and preventing gene transcription.⁴⁰² TAZ is also degraded via

phosphorylation by CK1 in the phosphodegron and subsequent ubiquitination and degradation.⁴⁰³ CK1 can also be bound by MST1/2, causing sequestration of CK1 and preventing it from phosphorylating Dvl, leading to inhibition of Wnt signaling.⁴⁰⁴ An analysis of overexpression of YAP/TAZ in cells found that YAP/TAZ bound and suppressed β -catenin without affecting β -catenin levels, suggesting the preservation of β -catenin stability.⁴⁰⁵ However, other reports indicate that when Wnt signaling is inactivated, YAP/TAZ normally binds β -catenin as part of the destruction complex. This association of YAP/TAZ to the destruction complex (via Axin, specifically) recruits β -TrCP, degrading β -catenin.⁴⁰⁶ Further experimentation may be needed to determine whether YAP/TAZ affects the stability of β -catenin or not.

When Wnt signaling is activated, YAP/TAZ is released from this complex and moves to the nucleus to activate the Hippo pathway.^{406,407} YAP/TAZ can also bind to Dvl, suppressing its phosphorylation and downstream effects on Wnt signaling.⁴⁰⁵ YAP can also interact with Src homology 2 domain tyrosine phosphatase (SHP2), an amino acid phosphatase that enhances β -catenin activity in the nucleus. This enables YAP to dampen Wnt signaling strength.⁴⁰⁷ These findings indicate that YAP/TAZ can serve as mediators or antagonists of Wnt signaling, where activation of Wnt signaling enables YAP/TAZ to transcribe Hippo pathway target genes, while inactivation leads to the antagonism of β -catenin.^{405–407} The noncanonical Wnt5a/b and Wnt3a ligands can activate YAP/TAZ as mediators of noncanonical Wnt signaling.⁴⁰⁸ Dysregulation of the Hippo pathway causes resistance to apoptosis and even chemotherapeutic treatments, leading to tumorigenesis, metastasis, and cancer relapse.⁴⁰¹ The inhibition of Wnt signaling by Hippo is also crucial in controlling heart size by limiting cardiomyocyte proliferation.⁴⁰⁹

Crosstalk with Hedgehog (Hh) signaling

The Hedgehog (Hh) pathway can be divided into canonical and noncanonical pathways. The canonical pathway is activated by the Shh ligand binding to the transmembrane protein Patched (Ptch1), which normally inhibits another transmembrane protein called Smo. The binding of Shh leads to Ptch1 degradation, causing Smo accumulation at the primary cilium (PC) which stimulates a signaling cascade. This leads to the Gli family of proteins translocating to the nucleus and transcribing genes, including Ptch1 for negative feedback and Gli1 for positive feedback.⁴¹⁰ Noncanonical signaling is independent of Gli and can be further subdivided into two types, type I which modulates Ca^{2+} and the actin cytoskeleton via downstream actions of Smo, and type II which increases cell proliferation and survival and is Smo-independent.⁴¹⁰

Hh signaling can recruit CK1 to phosphorylate Smo to increase its cell-surface accumulation, leading to stronger Hh signaling.^{411,412} CK1 can also stabilize Gli by phosphorylating Ser/Thr-rich motifs in the N- and C-terminal regions of Gli to prevent degradation by HIB.⁴¹³ Additionally, CK1 can phosphorylate and activate the Fused (Fu) protein kinase, a kinase that phosphorylates Gli for activation.⁴¹⁴ Further regulation of Hh signaling can be induced by GSK-

3 β , which can phosphorylate suppressor of fused (SUFU), which normally binds the Gli transcription factors and prevents them from entering the nucleus. Phosphorylation of SUFU by GSK-3 β prevents SUFU from binding to the Gli transcription factors, presenting a positive regulatory role for GSK-3 β in Hh signaling.⁴¹⁵ One of the downstream genes that are transcribed by the Hh signaling cascade is sFRP1. Specifically, Gli1 and Gli2 are responsible for the transcription. In other words, the activation of Hh signaling inhibits Wnt signaling via the production of sFRP1, indicating a regulatory role of the Hh pathway for the Wnt pathway.⁴¹⁶ Furthermore, SUFU can also decrease β -catenin levels via nuclear export and in turn down-regulate TCF-dependent transcription.⁴¹⁷

Crosstalk with fibroblast growth factor (FGF) signaling

Fibroblast growth factors (FGFs) are a family of 22 polypeptide growth factors in seven subfamilies that are involved in diverse cellular processes. The binding of FGFs to FGF receptors (FGFRs) causes receptor dimerization that is promoted by heparan sulfate. This leads to autophosphorylation of FGFR that can activate the MAPK, phosphatidylinositol-3 kinase/Akt (PI3K/Akt), and phospholipase C- γ (PLC- γ) pathways.⁴¹⁸ They also have interactions with the Wnt signaling pathway. Akt from the PI3K/Akt pathway phosphorylates Ser9 on GSK-3 β , decreasing its activity. This reduces the ability of GSK-3 β to phosphorylate SNAIL for degradation. SNAIL usually represses transcription of the E-cadherin gene, CDH1, which maintains cell adhesion. In this manner, FGF signaling can stimulate the EMT via decreasing E-cadherin levels by GSK-3 β inhibition.⁴¹⁹ Activation of the PI3K/Akt pathway by FGF also releases β -catenin from a complex composed of E-cadherin, β -catenin, and α -catenin, enabling β -catenin to translocate to the nucleus.⁴¹⁹ FGF18 and FGF20 are target genes of the canonical Wnt pathway, indicating that Wnt signaling can mediate FGF signaling activation.⁴¹⁹ In the development of the anterior heart field (AHF) in mice, Wnt/ β -catenin signaling regulates the expression of FGF3, 10, 16, and 20 in AHF progenitor cells.⁴²⁰ Mesenchymal FGF signaling is vital in Wnt2a expression stabilizing β -catenin, and Wnt/ β -catenin signaling sustains expression of FGFR1c/2c and responsiveness to FGF9, creating a positive feedback loop for mesenchymal proliferation in the lung.⁴²¹

Crosstalk with parathyroid hormone (PTH) signaling

Parathyroid hormone (PTH) is a protein implicated in bone remodeling and acts through two G-protein coupled pathways, the Gs/cAMP/protein kinase A (PKA) pathway and the Gq/PLC/ Ca^{2+} /protein kinase C (PKC) pathway.⁴²² The bone repair effects of PTH are partly mediated by Wnt signaling, as PTH-treated bones had increased numbers of β -catenin-expressing osteoblastic cells in the fracture callus. These fracture calluses also demonstrated increased levels of Wnt4, 5a, 5b, 10, and 11.⁴²³ PTH also enhances the transcriptional activity induced by β -catenin by promoting the expression of Smad3.⁴²⁴ Additionally, PTH appears to be able to stimulate Wnt/ β -catenin signaling to prevent

apoptosis, but this does not require PTH, suggesting the need for further studies to clarify this role.⁴²⁴ PTH can stimulate osteoclastogenesis in an opposing manner to Wnt signaling, which inhibits osteoclastogenesis. PTH and Wnt signaling modulates β -catenin in different manners to cause these opposite effects.⁴²⁵ The PTH receptor (PTH1R) can recruit Dvl and activate β -catenin signaling without Wnt or LRP5/6. This association of PTH1R and Dvl is what enables PTH-induced osteoclastogenesis.⁴²⁵

Crosstalk with other pathways

Wnt signaling can interact with multiple other pathways in addition to the BMP, Notch, and Hippo signaling pathways. Analysis of mouse hepatocytes found that high insulin levels can stimulate the Wnt/ β -catenin pathway to promote lipogenic gene expression. Insulin activates stearyl-CoA-desaturase 1 (SCD1), an enzyme involved in lipogenesis that supplies palmitoleate for Porc, enabling it to acylate Wnt ligands for secretion.⁴²⁶ Insulin can also enhance Wnt signaling by activating Akt to phosphorylate and inactivate GSK-3 β , which leads to increased glycogen storage, glucose clearance, and insulin sensitivity. As such, GSK-3 β inhibitors have become a therapeutic target in the treatment of type II diabetes.³¹⁸

PP2A is another component of Wnt signaling that is involved in other pathways. For instance, it is involved in the mechanistic target of the rapamycin (mTOR) pathway, which is vital in cell growth and metabolism.³⁴² PP2A can negatively regulate the mTOR signaling pathway by dephosphorylating and inactivating Akt, which is normally activated by mTOR. PP2A can also inhibit IRS1, which is upstream of Akt. However, mTOR can also inhibit PP2A, suggesting a complex interaction between Wnt signaling and mTOR signaling.³⁴² PP2A also has a major role in both positively and negatively regulating the MAPK pathway. In terms of negative regulation, PP2A can bind the phosphotyrosine-binding (PTB) domain of Shc, an adaptor protein that acts as a signal transducer in the Ras/MAPK pathway. PP2A can inhibit MAPK activation, but this association of PP2A to Shc can be alleviated by growth factor stimulation or small-t antigen expression.⁴²⁷ For positive regulation, the PR130 regulatory subunit of PP2A can form a holoenzyme that complexes with SRC homology 2 domain-containing inositol polyphosphate phosphatase (SHIP2). This complex stabilizes the EGF receptors (EGFR) that are necessary for the ERK/MAPK signaling pathway.⁴²⁸

Conclusions and future directions

The study of Wnt signaling has progressively provided more insight into this complex pathway. The structures and many interactions of the Wnt signaling components have been established, providing a breadth of knowledge about the crosstalk capability of Wnt signaling with a heavy emphasis on its vast effects in different organs. The conservation of the Wnt ligands, Fz receptors, and coreceptors, has further driven home the fundamental importance of the Wnt signaling pathway. Additionally, the antagonists of Wnt signaling also have functions beyond just the Wnt pathway, adding further complexity to the interactions of Wnt signaling with other pathways.

However, many details are still needed to be clarified. For the 19 Wnt ligands, it is not completely clear how each ligand interacts with Fz receptors and LRP5/6 coreceptors. The specificity of the interaction between these receptors and the Wnt ligands needs to be further clarified. Further research continues to uncover more and more pathways with which Wnt signaling interacts, yet the number of pathways and proteins that crosstalk with Wnt signaling is still not fully determined. Furthermore, the mechanisms of many Wnt signaling interactions with other pathways are not fully elucidated. The functions of each component of the Wnt signaling pathway are diverse and may still not be completely known yet.

Due to the implication of dysregulated Wnt signaling in diseases such as cancer, therapeutics have been developed. However, like the Wnt signaling pathway and components, the therapeutics also require caution in their function and use. Both activators and inhibitors of the Wnt signaling pathway can come with their own risk of tumorigenesis or unintentional inhibition of important cellular functions, which can lead to organ dysfunction. Furthermore, if an activator or inhibitor is affected by treatment, it may be unknown what potential effects can occur if that activator or inhibitor is also involved in another pathway. Additional research is needed to further clarify the therapeutic effects of various treatment options, as well as ensure their safety.

Although the current analysis of Wnt signaling is not completely exhaustive, the broad amount of research on the Wnt signaling pathway has provided us with significant insight into the system and its many functions in living organisms over the last 40 years. Hopefully, in the future, more research will be performed to provide more knowledge on the Wnt signaling pathway, leading to further advancements in biology, medicine, and human health.

Conflict of interests

Tong-Chuan He is the Editor-in-Chief of *Genes & Diseases*. To minimize bias, he was excluded from all editorial decision-making related to the acceptance of this article for publication. The remaining authors declare no conflict of interests.

Funding

The reported work was supported in part by research grants from the National Institutes of Health (No. CA226303 to TCH and No. DE030480 to RRR) and the American Shoulder and Elbow Surgeons PJI Research Grant (LLS). JF was supported in part by research grants from the Natural Science Foundation of China (No. 82102696), the 2019 Science and Technology Research Plan Project of Chongqing Education Commission (China) (No. KJQN201900410), and the 2019 Funding for Postdoctoral Research (Chongqing Human Resources and Social Security Bureau No. 298). WW was supported by the Medical Scientist Training Program of the National Institutes of Health (No. T32 GM007281). This project was also supported in part by The University of Chicago Cancer Center Support Grant (No. P30CA014599) and the National Center for Advancing Translational Sciences (NCATS) of the National Institutes of Health through Grant Number 5UL1TR002389. TCH was also

supported by the Mabel Green Myers Research Endowment Fund and The University of Chicago Orthopaedics Alumni Fund. Funding sources were not involved in the study design, in the collection, analysis, and interpretation of data, in the writing of the report, and in the decision to submit the paper for publication.

Acknowledgements

The authors apologize to the investigators whose work cannot be cited in the manuscript due to space constraints. We here acknowledge the investigators for their contribution to this work.

References

- Nusse R, Varmus HE. Many tumors induced by the mouse mammary tumor virus contain a provirus integrated in the same region of the host genome. *Cell*. 1982;31(1):99–109.
- Nusse R, Varmus H. Three decades of Wnts: a personal perspective on how a scientific field developed. *EMBO J*. 2012;31(12):2670–2684.
- van Ooyen A, Kwee V, Nusse R. The nucleotide sequence of the human int-1 mammary oncogene; evolutionary conservation of coding and non-coding sequences. *EMBO J*. 1985;4(11):2905–2909.
- Sharma RP, Chopra VL. Effect of the *Wingless* (*wg*¹) mutation on the wing and haltere development in *Drosophila melanogaster*. *Dev Biol*. 1976;48(2):461–465.
- Uzvolgyi E, Kiss I, Pitt A, et al. *Drosophila* homolog of the murine Int-1 protooncogene. *Proc Natl Acad Sci U S A*. 1988;85(9):3034–3038.
- Baker NE. Molecular cloning of sequences from wingless, a segment polarity gene in *Drosophila*: the spatial distribution of a transcript in embryos. *EMBO J*. 1987;6(6):1765–1773.
- Rijsewijk F, Schuermann M, Wagenaar E, et al. The *Drosophila* homolog of the mouse mammary oncogene *int-1* is identical to the segment polarity gene *wingless*. *Cell*. 1987;50(4):649–657.
- Nusse R, Brown A, Papkoff J, et al. A new nomenclature for *int-1* and related genes: the Wnt gene family. *Cell*. 1991;64(2):231.
- He TC, Sparks AB, Rago C, et al. Identification of c-MYC as a target of the APC pathway. *Science*. 1998;281(5382):1509–1512.
- He TC, Chan TA, Vogelstein B, et al. PPAR δ is an APC-regulated target of nonsteroidal anti-inflammatory drugs. *Cell*. 1999;99(3):335–345.
- Kim JH, Liu X, Wang J, et al. Wnt signaling in bone formation and its therapeutic potential for bone diseases. *Ther Adv Musculoskelet Dis*. 2013;5(1):13–31.
- Mohammed MK, Shao C, Wang J, et al. Wnt/ β -catenin signaling plays an ever-expanding role in stem cell self-renewal, tumorigenesis and cancer chemoresistance. *Genes Dis*. 2016;3(1):11–40.
- Yang K, Wang X, Zhang H, et al. The evolving roles of canonical WNT signaling in stem cells and tumorigenesis: implications in targeted cancer therapies. *Lab Invest*. 2016;96(2):116–136.
- MacDonald BT, Tamai K, He X. Wnt/ β -catenin signaling: components, mechanisms, and diseases. *Dev Cell*. 2009;17(1):9–26.
- Hayat R, Manzoor M, Hussain A. Wnt signaling pathway: a comprehensive review. *Cell Biol Int*. 2022;46(6):863–877.
- Fan J, Wei Q, Liao J, et al. Noncanonical Wnt signaling plays an important role in modulating canonical Wnt-regulated stemness, proliferation and terminal differentiation of hepatic progenitors. *Oncotarget*. 2017;8(16):27105–27119.
- Chae WJ, Bothwell ALM. Canonical and non-canonical Wnt signaling in immune cells. *Trends Immunol*. 2018;39(10):830–847.
- Logan CY, Nusse R. The Wnt signaling pathway in development and disease. *Annu Rev Cell Dev Biol*. 2004;20:781–810.
- He X, Semenov M, Tamai K, et al. LDL receptor-related proteins 5 and 6 in Wnt/ β -catenin signaling: arrows point the way. *Development*. 2004;131(8):1663–1677.
- Janda CY, Waghray D, Levin AM, et al. Structural basis of Wnt recognition by frizzled. *Science*. 2012;337(6090):59–64.
- Wang Y, Chang H, Rattner A, et al. Frizzled receptors in development and disease. *Curr Top Dev Biol*. 2016;117:113–139.
- Stamos JL, Weis WI. The β -catenin destruction complex. *Cold Spring Harbor Perspect Biol*. 2013;5(1):a007898.
- Thompson JJ, Williams CS. Protein phosphatase 2A in the regulation of Wnt signaling, stem cells, and cancer. *Genes*. 2018;9(3):121.
- Shang S, Hua F, Hu ZW. The regulation of β -catenin activity and function in cancer: therapeutic opportunities. *Oncotarget*. 2017;8(20):33972–33989.
- Xue J, Chen Y, Wu Y, et al. Tumour suppressor TRIM33 targets nuclear β -catenin degradation. *Nat Commun*. 2015;6:6156.
- Duan P, Bonewald LF. The role of the Wnt/ β -catenin signaling pathway in formation and maintenance of bone and teeth. *Int J Biochem Cell Biol*. 2016;77:23–29.
- Silva-García O, Valdez-Alarcón JJ, Baizabal-Aguirre VM. Wnt/ β -catenin signaling as a molecular target by pathogenic bacteria. *Front Immunol*. 2019;10:2135.
- Doumpas N, Lampart F, Robinson MD, et al. TCF/LEF dependent and independent transcriptional regulation of Wnt/ β -catenin target genes. *EMBO J*. 2019;38(2):e98873.
- Zhan T, Rindtorff N, Boutros M. Wnt signaling in cancer. *Oncogene*. 2017;36(11):1461–1473.
- Chen Y, Chen Z, Tang Y, et al. The involvement of noncanonical Wnt signaling in cancers. *Biomed Pharmacother*. 2021;133:110946.
- Berger H, Wodarz A, Borchers A. PTK7 faces the Wnt in development and disease. *Front Cell Dev Biol*. 2017;5:31.
- Messéant J, Dobbertin A, Girard E, et al. MusK frizzled-like domain is critical for mammalian neuromuscular junction formation and maintenance. *J Neurosci*. 2015;35(12):4926–4941.
- Katoh M, Katoh M. Molecular genetics and targeted therapy of WNT-related human diseases (Review). *Int J Mol Med*. 2017;40(3):587–606.
- Bovolenta P, Rodriguez J, Esteve P. Frizzled/Ryk mediated signalling in axon guidance. *Development*. 2006;133(22):4399–4408.
- Pataki CA, Couchman JR, Brábek J. Wnt signaling cascades and the roles of syndecan proteoglycans. *J Histochem Cytochem*. 2015;63(7):465–480.
- van Andel H, Kocemba KA, Spaargaren M, et al. Aberrant Wnt signaling in multiple myeloma: molecular mechanisms and targeting options. *Leukemia*. 2019;33(5):1063–1075.
- Kolluri A, Ho M. The role of glypican-3 in regulating Wnt, YAP, and Hedgehog in liver cancer. *Front Oncol*. 2019;9:708.
- de Moraes GFA, Listik E, Justo GZ, et al. The Glypican proteoglycans show intrinsic interactions with Wnt-3a in human prostate cancer cells that are not always associated with cascade activation. *BMC Mol Cell Biol*. 2021;22:26.
- Xu X, Zhang M, Xu F, et al. Wnt signaling in breast cancer: biological mechanisms, challenges and opportunities. *Mol Cancer*. 2020;19:165.
- Jenny A, Reynolds-Kenneally J, Das G, et al. Diego and Prickle regulate Frizzled planar cell polarity signalling by competing for Dishevelled binding. *Nat Cell Biol*. 2005;7(7):691–697.

41. Narimatsu M, Bose R, Pye M, et al. Regulation of planar cell polarity by Smurf ubiquitin ligases. *Cell*. 2009;137(2):295–307.
42. Barkó S, Bugyi B, Carlier MF, et al. Characterization of the biochemical properties and biological function of the formin homology domains of *Drosophila* DAAM. *J Biol Chem*. 2010;285(17):13154–13169.
43. Coso OA, Chiariello M, Yu JC, et al. The small GTP-binding proteins Rac 1 and Cdc 42 regulate the activity of the JNK/SAPK signaling pathway. *Cell*. 1995;81(7):1137–1146.
44. Yang Y, Mlodzik M. Wnt-Frizzled/planar cell polarity signaling: cellular orientation by facing the wind (Wnt). *Annu Rev Cell Dev Biol*. 2015;31:623–646.
45. Evers CE, McNeill H, Knebel A, et al. The phosphorylation of CapZ-interacting protein (CapZIP) by stress-activated protein kinases triggers its dissociation from CapZ. *Biochem J*. 2005;389(Pt 1):127–135.
46. Kitzing TM, Sahadevan AS, Brandt DT, et al. Positive feedback between Dia 1, LARG, and RhoA regulates cell morphology and invasion. *Genes Dev*. 2007;21(12):1478–1483.
47. Kühn S, Geyer M. Formins as effector proteins of Rho GTPases. *Small GTPases*. 2014;5(3):e983876.
48. Hamao K, Ono T, Matsushita M, et al. ZIP kinase phosphorylated and activated by Rho kinase/ROCK contributes to cytokinesis in mammalian cultured cells. *Exp Cell Res*. 2020;386(1):111707.
49. Oyama H, Nukuda A, Ishihara S, et al. Soft surfaces promote astrocytic differentiation of mouse embryonic neural stem cells via dephosphorylation of MRLC in the absence of serum. *Sci Rep*. 2021;11:19574.
50. Slusarski DC, Corces VG, Moon RT. Interaction of Wnt and a frizzled homologue triggers G-protein-linked phosphatidylinositol signalling. *Nature*. 1997;390(6658):410–413.
51. De A. Wnt/Ca²⁺ signaling pathway: a brief overview. *Acta Biochim Biophys Sin*. 2011;43(10):745–756.
52. Kühl M, Sheldahl LC, Park M, et al. The Wnt/Ca²⁺ pathway: a new vertebrate Wnt signaling pathway takes shape. *Trends Genet*. 2000;16(7):279–283.
53. McQuate A, Latorre-Esteves E, Barria A. A Wnt/calcium signaling cascade regulates neuronal excitability and trafficking of NMDARs. *Cell Rep*. 2017;21(1):60–69.
54. Hooper R, Zhang X, Webster M, et al. Novel protein kinase C-mediated control of Orai 1 function in invasive melanoma. *Mol Cell Biol*. 2015;35(16):2790–2798.
55. Wang HY, Malbon CC. Wnt signaling, Ca²⁺, and cyclic GMP: visualizing Frizzled functions. *Science*. 2003;300(5625):1529–1530.
56. Heit JJ, Apelqvist AA, Gu X, et al. Calcineurin/NFAT signalling regulates pancreatic beta-cell growth and function. *Nature*. 2006;443(7109):345–349.
57. Ishitani T, Kishida S, Hyodo-Miura J, et al. The TAK1-NLK mitogen-activated protein kinase cascade functions in the Wnt-5a/Ca²⁺ pathway to antagonize Wnt/beta-catenin signaling. *Mol Cell Biol*. 2003;23(1):131–139.
58. Ishitani T, Ninomiya-Tsuji J, Nagai S, et al. The TAK1-NLK-MAPK-related pathway antagonizes signalling between beta-catenin and transcription factor TCF. *Nature*. 1999;399(6738):798–802.
59. Petersen CP, Reddien PW. Wnt signaling and the polarity of the primary body axis. *Cell*. 2009;139(6):1056–1068.
60. Tanaka K, Okabayashi K, Asashima M, et al. The evolutionarily conserved porcupine gene family is involved in the processing of the Wnt family. *Eur J Biochem*. 2000;267(13):4300–4311.
61. Valenta T, Degirmenci B, Moor AE, et al. Wnt ligands secreted by subepithelial mesenchymal cells are essential for the survival of intestinal stem cells and gut homeostasis. *Cell Rep*. 2016;15(5):911–918.
62. Moon RT, Kohn AD, De Ferrari GV, et al. WNT and beta-catenin signalling: diseases and therapies. *Nat Rev Genet*. 2004;5(9):691–701.
63. Peradziry H, Kaplan NA, Podleschny M, et al. PTK7/Otk interacts with Wnts and inhibits canonical Wnt signalling. *EMBO J*. 2011;30(18):3729–3740.
64. Rubin EM, Guo Y, Tu K, et al. Wnt inhibitory factor 1 decreases tumorigenesis and metastasis in osteosarcoma. *Mol Cancer Therapeut*. 2010;9(3):731–741.
65. Zhao WN, Cheng C, Theriault KM, et al. A high-throughput screen for Wnt/ β -catenin signaling pathway modulators in human iPSC-derived neural progenitors. *J Biomol Screen*. 2012;17(9):1252–1263.
66. Liu Z, Xiu Y, Qiu F, et al. Canonical Wnt signaling drives *Myopia* development and can be pharmacologically modulated. *Invest Ophthalmol Vis Sci*. 2021;62(9):21.
67. Abu-Remaileh M, Gerson A, Farago M, et al. Oct-3/4 regulates stem cell identity and cell fate decisions by modulating Wnt/ β -catenin signalling. *EMBO J*. 2010;29(19):3236–3248.
68. Kawano Y, Kypta R. Secreted antagonists of the Wnt signalling pathway. *J Cell Sci*. 2003;116(Pt 13):2627–2634.
69. Ho HH, Susman MW, Bikoff JB, et al. Wnt5a-Ror-Dishevelled signaling constitutes a core developmental pathway that controls tissue morphogenesis. *Proc Natl Acad Sci U S A*. 2012;109(11):4044–4051.
70. Susman MW, Karuna EP, Kunz RC, et al. Kinesin superfamily protein Kif26b links Wnt5a-Ror signaling to the control of cell and tissue behaviors in vertebrates. *Elife*. 2017;6:e26509.
71. Karuna EP, Susman MW, Ho HH. Quantitative live-cell reporter assay for noncanonical Wnt activity. *Bio Protoc*. 2018;8(6):e2762.
72. Mills KM, Szczerkowski JLA, Habib SJ. Wnt ligand presentation and reception: from the stem cell niche to tissue engineering. *Open Biol*. 2017;7(8):170140.
73. Rocheleau CE, Downs WD, Lin R, et al. Wnt signaling and an APC-related gene specify endoderm in early *C. elegans* embryos. *Cell*. 1997;90(4):707–716.
74. Bänziger C, Soldini D, Schütt C, et al. Wntless, a conserved membrane protein dedicated to the secretion of Wnt proteins from signaling cells. *Cell*. 2006;125(3):509–522.
75. Kjellén L, Lindahl U. Proteoglycans: structures and interactions. *Annu Rev Biochem*. 1991;60:443–475.
76. Lin X, Perrimon N. Dally cooperates with *Drosophila* Frizzled 2 to transduce Wingless signalling. *Nature*. 1999;400(6741):281–284.
77. Wodarz A, Nusse R. Mechanisms of Wnt signaling in development. *Annu Rev Cell Dev Biol*. 1998;14:59–88.
78. Huang HC, Klein PS. The Frizzled family: receptors for multiple signal transduction pathways. *Genome Biol*. 2004;5(7):234.
79. Park WJ, Liu J, Adler PN. The frizzled gene of *Drosophila* encodes a membrane protein with an odd number of transmembrane domains. *Mech Dev*. 1994;45(2):127–137.
80. Wang Y, Macke JP, Abella BS, et al. A large family of putative transmembrane receptors homologous to the product of the *Drosophila* tissue polarity gene *frizzled*. *J Biol Chem*. 1996;271(8):4468–4476.
81. Fredriksson R, Lagerström MC, Lundin LG, et al. The G-protein-coupled receptors in the human genome form five main families. Phylogenetic analysis, paralogon groups, and fingerprints. *Mol Pharmacol*. 2003;63(6):1256–1272.
82. Vinson CR, Adler PN. Directional non-cell autonomy and the transmission of polarity information by the frizzled gene of *Drosophila*. *Nature*. 1987;329(6139):549–551.
83. Wong LL, Adler PN. Tissue polarity genes of *Drosophila* regulate the subcellular location for prehair initiation in pupal wing cells. *J Cell Biol*. 1993;123(1):209–221.
84. Adler PN. The genetic control of tissue polarity in *Drosophila*. *Bioessays*. 1992;14(11):735–741.

85. Zheng L, Zhang J, Carthew RW. Frizzled regulates mirror-symmetric pattern formation in the *Drosophila* eye. *Development*. 1995;121(9):3045–3055.
86. Gilmour DF. Familial exudative vitreoretinopathy and related retinopathies. *Eye*. 2015;29(1):1–14.
87. Yu H, Smallwood PM, Wang Y, et al. Frizzled 1 and frizzled 2 genes function in palate, ventricular septum and neural tube closure: general implications for tissue fusion processes. *Development*. 2010;137(21):3707–3717.
88. Ishikawa T, Tamai Y, Zorn AM, et al. Mouse Wnt receptor gene *Fzd5* is essential for yolk sac and placental angiogenesis. *Development*. 2001;128(1):25–33.
89. Liu C, Nathans J. An essential role for frizzled 5 in mammalian ocular development. *Development*. 2008;135(21):3567–3576.
90. Liu C, Bakeri H, Li T, et al. Regulation of retinal progenitor expansion by Frizzled receptors: implications for microphthalmia and retinal coloboma. *Hum Mol Genet*. 2012;21(8):1848–1860.
91. Wang Y, Badea T, Nathans J. Order from disorder: self-organization in mammalian hair patterning. *Proc Natl Acad Sci U S A*. 2006;103(52):19800–19805.
92. Hua ZL, Jeon S, Caterina MJ, et al. Frizzled 3 is required for the development of multiple axon tracts in the mouse central nervous system. *Proc Natl Acad Sci U S A*. 2014;111(29):E3005–E3014.
93. Wang Y, Guo N, Nathans J. The role of Frizzled 3 and Frizzled 6 in neural tube closure and in the planar polarity of inner-ear sensory hair cells. *J Neurosci*. 2006;26(8):2147–2156.
94. Nusse R, Clevers H. Wnt/ β -catenin signaling, disease, and emerging therapeutic modalities. *Cell*. 2017;169(6):985–999.
95. Tamai K, Zeng X, Liu C, et al. A mechanism for Wnt coreceptor activation. *Mol Cell*. 2004;13(1):149–156.
96. Wu G, Huang H, Garcia Abreu J, et al. Inhibition of GSK3 phosphorylation of β -catenin via phosphorylated PPPSPXS motifs of Wnt coreceptor LRP6. *PLoS One*. 2009;4(3):e4926.
97. Stamos JL, Chu ML, Enos MD, et al. Structural basis of GSK-3 inhibition by N-terminal phosphorylation and by the Wnt receptor LRP6. *Elife*. 2014;3:e01998.
98. Mao J, Wang J, Liu B, et al. Low-density lipoprotein receptor-related protein-5 binds to Axin and regulates the canonical Wnt signaling pathway. *Mol Cell*. 2001;7(4):801–809.
99. Pinson KI, Brennan J, Monkley S, et al. An LDL-receptor-related protein mediates Wnt signalling in mice. *Nature*. 2000;407(6803):535–538.
100. Kato M, Patel MS, Levasseur R, et al. *Cbfa1*-independent decrease in osteoblast proliferation, osteopenia, and persistent embryonic eye vascularization in mice deficient in *Lrp5*, a Wnt coreceptor. *J Cell Biol*. 2002;157(2):303–314.
101. Fujino T, Asaba H, Kang MJ, et al. Low-density lipoprotein receptor-related protein 5 (LRP5) is essential for normal cholesterol metabolism and glucose-induced insulin secretion. *Proc Natl Acad Sci U S A*. 2003;100(1):229–234.
102. de Lau WBM, Snel B, Clevers HC. The R-spondin protein family. *Genome Biol*. 2012;13(3):242.
103. Kamata T, Katsube K, Michikawa M, et al. *R-spondin*, a novel gene with thrombospondin type 1 domain, was expressed in the dorsal neural tube and affected in *Wnts* mutants. *Biochim Biophys Acta*. 2004;1676(1):51–62.
104. Kazanskaya O, Glinka A, del Barco Barrantes I, et al. R-Spondin 2 is a secreted activator of Wnt/ β -catenin signaling and is required for *Xenopus* myogenesis. *Dev Cell*. 2004;7(4):525–534.
105. Carmon KS, Gong X, Lin Q, et al. R-spondins function as ligands of the orphan receptors LGR4 and LGR5 to regulate Wnt/ β -catenin signaling. *Proc Natl Acad Sci U S A*. 2011;108(28):11452–11457.
106. Tocci JM, Felcher CM, García Solá ME, et al. R-spondin-mediated WNT signaling potentiation in mammary and breast cancer development. *IUBMB Life*. 2020;72(8):1546–1559.
107. Zebisch M, Xu Y, Krastev C, et al. Structural and molecular basis of ZNRF3/RNF43 transmembrane ubiquitin ligase inhibition by the Wnt agonist R-spondin. *Nat Commun*. 2013;4:2787.
108. Araki T, Milbrandt J. ZNRF proteins constitute a family of presynaptic E3 ubiquitin ligases. *J Neurosci*. 2003;23(28):9385–9394.
109. Hao HX, Xie Y, Zhang Y, et al. ZNRF₃ promotes Wnt receptor turnover in an R-spondin-sensitive manner. *Nature*. 2012;485(7397):195–200.
110. Lu W, Yamamoto V, Ortega B, et al. Mammalian Ryk is a Wnt coreceptor required for stimulation of neurite outgrowth. *Cell*. 2004;119(1):97–108.
111. Hovens CM, Stacker SA, Andres AC, et al. RYK, a receptor tyrosine kinase-related molecule with unusual kinase domain motifs. *Proc Natl Acad Sci U S A*. 1992;89(24):11818–11822.
112. Dura JM, Preat T, Tully T. Identification of linotte, a new gene affecting learning and memory in *Drosophila melanogaster*. *J Neurogenet*. 2007;21(4):307–320.
113. Yoshikawa S, McKinnon RD, Kokel M, et al. Wnt-mediated axon guidance via the *Drosophila* Derailed receptor. *Nature*. 2003;422(6932):583–588.
114. Inoue T, Oz HS, Wiland D, et al. *C. elegans* LIN-18 is a ryk ortholog and functions in parallel to LIN-17/frizzled in Wnt signaling. *Cell*. 2004;118(6):795–806.
115. Menck K, Heinrichs S, Baden C, et al. The WNT/ROR pathway in cancer: from signaling to therapeutic intervention. *Cells*. 2021;10(1):142.
116. Masiakowski P, Carroll RD. A novel family of cell surface receptors with tyrosine kinase-like domain. *J Biol Chem*. 1992;267(36):26181–26190.
117. Saldanha J, Singh J, Mahadevan D. Identification of a Frizzled-like cysteine rich domain in the extracellular region of developmental receptor tyrosine kinases. *Protein Sci*. 1998;7(7):1632–1635.
118. Matsuda T, Nomi M, Ikeya M, et al. Expression of the receptor tyrosine kinase genes, *Ror1* and *Ror2*, during mouse development. *Mech Dev*. 2001;105(1–2):153–156.
119. Oldridge M, Fortuna AM, Maringa M, et al. Dominant mutations in *ROR2*, encoding an orphan receptor tyrosine kinase, cause brachydactyly type B. *Nat Genet*. 2000;24(3):275–278.
120. Afzal AR, Rajab A, Fenske CD, et al. Recessive Robinow syndrome, allelic to dominant brachydactyly type B, is caused by mutation of *ROR2*. *Nat Genet*. 2000;25(4):419–422.
121. van Bokhoven H, Celli J, Kayserili H, et al. Mutation of the gene encoding the *ROR2* tyrosine kinase causes autosomal recessive Robinow syndrome. *Nat Genet*. 2000;25(4):423–426.
122. Petrova IM, Malessy MJ, Verhaagen J, et al. Wnt signaling through the Ror receptor in the nervous system. *Mol Neurobiol*. 2014;49(1):303–315.
123. Broome HE, Rassenti LZ, Wang HY, et al. *ROR1* is expressed on hematogones (non-neoplastic human B-lymphocyte precursors) and a minority of precursor-B acute lymphoblastic leukemia. *Leuk Res*. 2011;35(10):1390–1394.
124. Oishi I, Takeuchi S, Hashimoto R, et al. Spatio-temporally regulated expression of receptor tyrosine kinases, mRor1, mRor2, during mouse development: implications in development and function of the nervous system. *Gene Cell*. 1999;4(1):41–56.
125. Eubelen M, Bostaille N, Cabochette P, et al. A molecular mechanism for Wnt ligand-specific signaling. *Science*. 2018;361(6403):eaat1178.

126. Posokhova E, Shukla A, Seaman S, et al. GPR124 functions as a WNT7-specific coactivator of canonical β -catenin signaling. *Cell Rep.* 2015;10(2):123–130.
127. Vallon M, Yuki K, Nguyen TD, et al. A RECK-WNT7 receptor-ligand interaction enables isoform-specific regulation of Wnt bioavailability. *Cell Rep.* 2018;25(2):339–349.e9.
128. Chang J, Mancuso MR, Maier C, et al. Gpr124 is essential for blood-brain barrier integrity in central nervous system disease. *Nat Med.* 2017;23(4):450–460.
129. Ozawa M, Baribault H, Kemler R. The cytoplasmic domain of the cell adhesion molecule uvomorulin associates with three independent proteins structurally related in different species. *EMBO J.* 1989;8(6):1711–1717.
130. Wieschaus E, Nüsslein-Volhard C, Jürgens G. Mutations affecting the pattern of the larval cuticle in *Drosophila melanogaster*: III. Zygotic loci on the X-chromosome and fourth chromosome. *Wilehm Roux Arch Dev Biol.* 1984;193(5):296–307.
131. Riggelman B, Schedl P, Wieschaus E. Spatial expression of the *Drosophila* segment polarity gene *armadillo* is posttranscriptionally regulated by *wingless*. *Cell.* 1990;63(3):549–560.
132. Huber O, Korn R, McLaughlin J, et al. Nuclear localization of β -catenin by interaction with transcription factor LEF-1. *Mech Dev.* 1996;59(1):3–10.
133. Huber AH, Nelson WJ, Weis WI. Three-dimensional structure of the *Armadillo* repeat region of β -catenin. *Cell.* 1997;90(5):871–882.
134. Graham TA, Weaver C, Mao F, et al. Crystal structure of a β -catenin/Tcf complex. *Cell.* 2000;103(6):885–896.
135. Zhurinsky J, Shtutman M, Ben-Ze'ev A. Plakoglobin and β -catenin: protein interactions, regulation and biological roles. *J Cell Sci.* 2000;113(Pt 18):3127–3139.
136. Eklof Spink K, Fridman SG, Weis WI. Molecular mechanisms of β -catenin recognition by adenomatous polyposis coli revealed by the structure of an APC- β -catenin complex. *EMBO J.* 2001;20(22):6203–6212.
137. Huber AH, Weis WI. The structure of the β -catenin/E-cadherin complex and the molecular basis of diverse ligand recognition by β -catenin. *Cell.* 2001;105(3):391–402.
138. Xing Y, Clements WK, Kimelman D, et al. Crystal structure of a β -catenin/axin complex suggests a mechanism for the β -catenin destruction complex. *Genes Dev.* 2003;17(22):2753–2764.
139. Ha NC, Tonzuka T, Stamos JL, et al. Mechanism of phosphorylation-dependent binding of APC to β -catenin and its role in β -catenin degradation. *Mol Cell.* 2004;15(4):511–521.
140. Hrculak D, Kolar M, Strnad H, et al. TCF/LEF transcription factors: an update from the Internet resources. *Cancers.* 2016;8(7):70.
141. van de Wetering M, Cavallo R, Dooijes D, et al. *Armadillo* coactivates transcription driven by the product of the *Drosophila* segment polarity gene *DTCF*. *Cell.* 1997;88(6):789–799.
142. Atcha FA, Syed A, Wu B, et al. A unique DNA binding domain converts T-cell factors into strong Wnt effectors. *Mol Cell Biol.* 2007;27(23):8352–8363.
143. Sharma M, Castro-Piedras I, Simmons Jr GE, et al. Dishevelled: a masterful conductor of complex Wnt signals. *Cell Signal.* 2018;47:52–64.
144. Schwarz-Romond T, Fiedler M, Shibata N, et al. The DIX domain of Dishevelled confers Wnt signaling by dynamic polymerization. *Nat Struct Mol Biol.* 2007;14(6):484–492.
145. Moon RT, Shah K. Developmental biology: signalling polarity. *Nature.* 2002;417(6886):239–240.
146. Umbhauer M, Djiane A, Goisset C, et al. The C-terminal cytoplasmic Lys-thr-X-X-X-Trp motif in frizzled receptors mediates Wnt/ β -catenin signalling. *EMBO J.* 2000;19(18):4944–4954.
147. Wong HC, Bourdelas A, Krauss A, et al. Direct binding of the PDZ domain of Dishevelled to a conserved internal sequence in the C-terminal region of Frizzled. *Mol Cell.* 2003;12(5):1251–1260.
148. Pan WJ, Pang SZ, Huang T, et al. Characterization of function of three domains in dishevelled-1: DEP domain is responsible for membrane translocation of dishevelled-1. *Cell Res.* 2004;14(4):324–330.
149. Boligala GP, Yang MV, van Wunnik JC, et al. Nuclear Dishevelled: an enigmatic role in governing cell fate and Wnt signaling. *Biochim Biophys Acta Mol Cell Res.* 2022;1869(10):119305.
150. Mochizuki T, Saijoh Y, Tsuchiya K, et al. Cloning of inv, a gene that controls left/right asymmetry and kidney development. *Nature.* 1998;395(6698):177–181.
151. Simons M, Gloy J, Ganner A, et al. Inversin, the gene product mutated in nephronophthisis type II, functions as a molecular switch between Wnt signaling pathways. *Nat Genet.* 2005;37(5):537–543.
152. Cai Y, Stafford LJ, Bryan BA, et al. G-protein-activated phospholipase C- β , new partners for cell polarity proteins Par 3 and Par 6. *Oncogene.* 2005;24(26):4293–4300.
153. Izzi L, Attisano L. Ubiquitin-dependent regulation of TGF β signaling in cancer. *Neoplasia.* 2006;8(8):677–688.
154. Zhu H, Kavsak P, Abdollah S, et al. A SMAD ubiquitin ligase targets the BMP pathway and affects embryonic pattern formation. *Nature.* 1999;400(6745):687–693.
155. Lin X, Liang M, Feng XH. Smurf2 is a ubiquitin E3 ligase mediating proteasome-dependent degradation of Smad 2 in transforming growth factor- β signaling. *J Biol Chem.* 2000;275(47):36818–36822.
156. Sahai E, Garcia-Medina R, Pouyssegur J, et al. Smurf1 regulates tumor cell plasticity and motility through degradation of RhoA leading to localized inhibition of contractility. *J Cell Biol.* 2007;176(1):35–42.
157. Ozdamar B, Bose R, Barrios-Rodiles M, et al. Regulation of the polarity protein Par 6 by TGF β receptors controls epithelial cell plasticity. *Science.* 2005;307(5715):1603–1609.
158. Gubb D, Green C, Huen D, et al. The balance between isoforms of the prickly LIM domain protein is critical for planar polarity in *Drosophila* imaginal discs. *Genes Dev.* 1999;13(17):2315–2327.
159. Dawid IB, Toyama R, Taira M. LIM domain proteins. *C R Acad Sci III.* 1995;318(3):295–306.
160. Sweede M, Ankem G, Chutvirasakul B, et al. Structural and membrane binding properties of the prickly PET domain. *Biochemistry.* 2008;47(51):13524–13536.
161. Veeman MT, Slusarski DC, Kaykas A, et al. Zebrafish prickly, a modulator of noncanonical Wnt/Fz signaling, regulates gastrulation movements. *Curr Biol.* 2003;13(8):680–685.
162. Chougule A, Lapraz F, Földi I, et al. The *Drosophila* actin nucleator DAAM is essential for left-right asymmetry. *PLoS Genet.* 2020;16(4):e1008758.
163. Lebreton G, Géminard C, Lapraz F, et al. Molecular to organismal chirality is induced by the conserved myosin 1D. *Science.* 2018;362(6417):949–952.
164. Carlsson L, Nyström LE, Sundkvist I, et al. Actin polymerizability is influenced by profilin, a low molecular weight protein in non-muscle cells. *J Mol Biol.* 1977;115(3):465–483.
165. Tilney LG. The polymerization of actin. III. Aggregates of nonfilamentous actin and its associated proteins: a storage form of actin. *J Cell Biol.* 1976;69(1):73–89.
166. Karlsson R, Dräber P. Profilin-A master coordinator of actin and microtubule organization in mammalian cells. *J Cell Physiol.* 2021;236(10):7256–7265.
167. Pinto-Costa R, Sousa MM. Profilin as a dual regulator of actin and microtubule dynamics. *Cytoskeleton.* 2020;77(3–4):76–83.

168. Nejedla M, Sadi S, Sulimenko V, et al. Profilin connects actin assembly with microtubule dynamics. *Mol Biol Cell*. 2016; 27(15):2381–2393.
169. Sato A, Khadka DK, Liu W, et al. Profilin is an effector for Daam1 in non-canonical Wnt signaling and is required for vertebrate gastrulation. *Development*. 2006;133(21):4219–4231.
170. Nguyen LK, Kholodenko BN, von Kriegsheim A. Rac 1 and RhoA: networks, loops and bistability. *Small GTPases*. 2018; 9(4):316–321.
171. Kotelevets L, Chastre E. Rac 1 signaling: from intestinal homeostasis to colorectal cancer metastasis. *Cancers*. 2020; 12(3):665.
172. Mosaddeghzadeh N, Ahmadian MR. The RHO family GTPases: mechanisms of regulation and signaling. *Cells*. 2021;10(7): 1831.
173. Vigil D, Cherfils J, Rossman KL, et al. Ras superfamily GEFs and GAPs: validated and tractable targets for cancer therapy? *Nat Rev Cancer*. 2010;10(12):842–857.
174. Mehta P, Wavreille AS, Justiniano SE, et al. LyGDI, a novel SHIP-interacting protein, is a negative regulator of Fc γ R-mediated phagocytosis. *PLoS One*. 2011;6(6):e21175.
175. Machacek M, Hodgson L, Welch C, et al. Coordination of Rho GTPase activities during cell protrusion. *Nature*. 2009; 461(7260):99–103.
176. Jacobs M, Hayakawa K, Swenson L, et al. The structure of dimeric ROCK I reveals the mechanism for ligand selectivity. *J Biol Chem*. 2006;281(1):260–268.
177. Narumiya S, Thumkeo D. Rho signaling research: history, current status and future directions. *FEBS Lett*. 2018;592(11): 1763–1776.
178. Julian L, Olson MF. Rho-associated coiled-coil containing kinases (ROCK):structure, regulation, and functions. *Small GTPases*. 2014;5:e29846.
179. Ishizaki T, Maekawa M, Fujisawa K, et al. The small GTP-binding protein Rho binds to and activates a 160 kDa Ser/Thr protein kinase homologous to myotonic dystrophy kinase. *EMBO J*. 1996;15(8):1885–1893.
180. Coleman ML, Sahai EA, Yeo M, et al. Membrane blebbing during apoptosis results from caspase-mediated activation of ROCK I. *Nat Cell Biol*. 2001;3(4):339–345.
181. Sebbagh M, Renvoizé C, Hamelin J, et al. Caspase-3-mediated cleavage of ROCK I induces MLC phosphorylation and apoptotic membrane blebbing. *Nat Cell Biol*. 2001;3(4):346–352.
182. Leung T, Chen XQ, Manser E, et al. The p160 RhoA-binding kinase ROK alpha is a member of a kinase family and is involved in the reorganization of the cytoskeleton. *Mol Cell Biol*. 1996;16(10):5313–5327.
183. Murakami T, Ishikawa H. Stress fibers *in situ* in proximal tubules of the rat kidney. *Cell Struct Funct*. 1991;16(3): 231–240.
184. Schiller HB, Fässler R. Mechanosensitivity and compositional dynamics of cell-matrix adhesions. *EMBO Rep*. 2013;14(6): 509–519.
185. Bode AM, Dong Z. The functional contrariety of JNK. *Mol Carcinog*. 2007;46(8):591–598.
186. Grynberg K, Ma FY, Nikolic-Paterson DJ. The JNK signaling pathway in renal fibrosis. *Front Physiol*. 2017;8:829.
187. Avruch J. MAP kinase pathways: the first twenty years. *Biochim Biophys Acta*. 2007;1773(8):1150–1160.
188. Patterson KI, Brummer T, O'Brien PM, et al. Dual-specificity phosphatases: critical regulators with diverse cellular targets. *Biochem J*. 2009;418(3):475–489.
189. Zeke A, Misheva M, Reményi A, et al. JNK signaling: regulation and functions based on complex protein-protein partnerships. *Microbiol Mol Biol Rev*. 2016;80(3):793–835.
190. Tournier C, Hess P, Yang DD, et al. Requirement of JNK for stress-induced activation of the cytochrome c-mediated death pathway. *Science*. 2000;288(5467):870–874.
191. Kumar A, Singh UK, Kini SG, et al. JNK pathway signaling: a novel and smarter therapeutic targets for various biological diseases. *Future Med Chem*. 2015;7(15):2065–2086.
192. Angel P, Karin M. The role of Jun, Fos and the AP-1 complex in cell-proliferation and transformation. *Biochim Biophys Acta*. 1991;1072(2–3):129–157.
193. Kovary K, Bravo R. The jun and fos protein families are both required for cell cycle progression in fibroblasts. *Mol Cell Biol*. 1991;11(9):4466–4472.
194. Pfarr CM, Mechta F, Spyrou G, et al. Mouse JunD negatively regulates fibroblast growth and antagonizes transformation by *ras*. *Cell*. 1994;76(4):747–760.
195. Schreiber M, Kolbus A, Piu F, et al. Control of cell cycle progression by c-Jun is p53 dependent. *Genes Dev*. 1999;13(5): 607–619.
196. Shaulian E, Karin M. AP-1 as a regulator of cell life and death. *Nat Cell Biol*. 2002;4(5):E131–E136.
197. Katiyar S, Casimiro MC, Dettin L, et al. *C-jun* inhibits mammary apoptosis *in vivo*. *Mol Biol Cell*. 2010;21(23):4264–4274.
198. Ham J, Eilers A, Whitfield J, et al. c-Jun and the transcriptional control of neuronal apoptosis. *Biochem Pharmacol*. 2000;60(8):1015–1021.
199. Kuan CY, Yang DD, Roy DRS, et al. The Jnk 1 and Jnk 2 protein kinases are required for regional specific apoptosis during early brain development. *Neuron*. 1999;22(4):667–676.
200. Hernandez-Valladares M, Kim T, Kannan B, et al. Structural characterization of a capping protein interaction motif defines a family of actin filament regulators. *Nat Struct Mol Biol*. 2010;17(4):497–503.
201. Miyatake K, Kusakabe M, Takahashi C, et al. ERK7 regulates ciliogenesis by phosphorylating the actin regulator CapZIP in cooperation with Dishevelled. *Nat Commun*. 2015;6:6666.
202. Iwasaki T, Murata-Hori M, Ishitobi S, et al. Diphosphorylated MRLC is required for organization of stress fibers in interphase cells and the contractile ring in dividing cells. *Cell Struct Funct*. 2001;26(6):677–683.
203. Ikebe M, Hartshorne DJ. Phosphorylation of smooth muscle myosin at two distinct sites by myosin light chain kinase. *J Biol Chem*. 1985;260(18):10027–10031.
204. Ikebe M, Koretz J, Hartshorne DJ. Effects of phosphorylation of light chain residues threonine 18 and serine 19 on the properties and conformation of smooth muscle myosin. *J Biol Chem*. 1988;263(13):6432–6437.
205. Watanabe T, Hosoya H, Yonemura S. Regulation of myosin II dynamics by phosphorylation and dephosphorylation of its light chain in epithelial cells. *Mol Biol Cell*. 2007;18(2): 605–616.
206. Peng Y, Chen Z, Chen Y, et al. ROCK isoforms differentially modulate cancer cell motility by mechanosensing the substrate stiffness. *Acta Biomater*. 2019;88:86–101.
207. Goode BL, Eck MJ. Mechanism and function of formins in the control of actin assembly. *Annu Rev Biochem*. 2007;76: 593–627.
208. Paul AS, Pollard TD. The role of the FH1 domain and profilin in formin-mediated actin-filament elongation and nucleation. *Curr Biol*. 2008;18(1):9–19.
209. Lammers M, Rose R, Scrima A, et al. The regulation of mDia1 by autoinhibition and its release by Rho*GTP. *EMBO J*. 2005; 24(23):4176–4187.
210. Liu D, Fu X, Wang Y, et al. Protein diaphanous homolog 1 (Diaph1) promotes myofibroblastic activation of hepatic stellate cells by regulating Rab5a activity and TGF β receptor endocytosis. *FASEB J*. 2020;34(6):7345–7359.
211. Fukami K. Structure, regulation, and function of phospholipase C isozymes. *J Biochem*. 2002;131(3):293–299.
212. Harlan JE, Hajduk PJ, Yoon HS, et al. Pleckstrin homology domains bind to phosphatidylinositol-4, 5-bisphosphate. *Nature*. 1994;371(6493):168–170.

213. Essen LO, Perisic O, Cheung R, et al. Crystal structure of a mammalian phosphoinositide-specific phospholipase C delta. *Nature*. 1996;380(6575):595–602.
214. Bill CA, Vines CM, Phospholipase C. *Adv Exp Med Biol*. 2020; 1131:215–242.
215. Ellis MV, U S, Katan M. Mutations within a highly conserved sequence present in the X region of phosphoinositide-specific phospholipase C-delta 1. *Biochem J*. 1995;307(Pt 1):69–75.
216. Essen LO, Perisic O, Lynch DE, et al. A ternary metal binding site in the C2 domain of phosphoinositide-specific phospholipase C-delta 1. *Biochemistry*. 1997;36(10):2753–2762.
217. Hokin MR, Hokin LE. Enzyme secretion and the incorporation of P^{32} into phospholipids of pancreas slices. *J Biol Chem*. 1953;203(2):967–977.
218. Streb H, Irvine RF, Berridge MJ, et al. Release of Ca^{2+} from a nonmitochondrial intracellular store in pancreatic acinar cells by inositol-1, 4, 5-trisphosphate. *Nature*. 1983;306(5938):67–69.
219. Sun MK, Alkon DL. Pharmacology of protein kinase C activators: cognition-enhancing and antidementic therapeutics. *Pharmacol Ther*. 2010;127(1):66–77.
220. Rosse C, Linch M, Kermorgant S, et al. PKC and the control of localized signal dynamics. *Nat Rev Mol Cell Biol*. 2010;11(2):103–112.
221. Mochly-Rosen D, Das K, Grimes KV. Protein kinase C, an elusive therapeutic target? *Nat Rev Drug Discov*. 2012;11(12):937–957.
222. Manning G, Whyte DB, Martinez R, et al. The protein kinase complement of the human genome. *Science*. 2002;298(5600):1912–1934.
223. Cooke M, Magimaidas A, Casado-Medrano V, et al. Protein kinase C in cancer: the top five unanswered questions. *Mol Carcinog*. 2017;56(6):1531–1542.
224. Ono Y, Fujii T, Igarashi K, et al. Phorbol ester binding to protein kinase C requires a cysteine-rich zinc-finger-like sequence. *Proc Natl Acad Sci U S A*. 1989;86(13):4868–4871.
225. Mochly-Rosen D. Localization of protein kinases by anchoring proteins: a theme in signal transduction. *Science*. 1995; 268(5208):247–251.
226. Murphy NP, Binti Ahmad Mokhtar AM, Mott HR, et al. Molecular subversion of Cdc 42 signalling in cancer. *Biochem Soc Trans*. 2021;49(3):1425–1442.
227. Johnson DI, Pringle JR. Molecular characterization of CDC42, a *Saccharomyces cerevisiae* gene involved in the development of cell polarity. *J Cell Biol*. 1990;111(1):143–152.
228. Munemitsu S, Innis MA, Clark R, et al. Molecular cloning and expression of a G25K cDNA, the human homolog of the yeast cell cycle gene *CDC42*. *Mol Cell Biol*. 1990;10(11):5977–5982.
229. Erickson JW, Zhang CJ, Kahn RA, et al. Mammalian Cdc 42 is a brefeldin A-sensitive component of the Golgi apparatus. *J Biol Chem*. 1996;271(43):26850–26854.
230. Michaelson D, Silletti J, Murphy G, et al. Differential localization of Rho GTPases in live cells: regulation by hypervariable regions and RhoGDI binding. *J Cell Biol*. 2001;152:111–126.
231. Farhan H, Hsu VW. Cdc 42 and cellular polarity: emerging roles at the Golgi. *Trends Cell Biol*. 2016;26(4):241–248.
232. Luna A, Matas OB, Martínez-Menárguez JA, et al. Regulation of protein transport from the Golgi complex to the endoplasmic reticulum by CDC42 and N-WASP. *Mol Biol Cell*. 2002; 13(3):866–879.
233. Etienne-Manneville S. Cdc 42: the centre of polarity. *J Cell Sci*. 2004;117(Pt 8):1291–1300.
234. Lu MS, Drubin DG. Cdc 42 GTPase regulates ESCRTs in nuclear envelope sealing and ER remodeling. *J Cell Biol*. 2020;219(8): e201910119.
235. Hodge RG, Ridley AJ. Regulating Rho GTPases and their regulators. *Nat Rev Mol Cell Biol*. 2016;17(8):496–510.
236. Salat-Canela C, Carmona M, Martín-García R, et al. Stress-dependent inhibition of polarized cell growth through unbalancing the GEF/GAP regulation of Cdc 42. *Cell Rep*. 2021;37(5):109951.
237. Lisman J, Yasuda R, Raghavachari S. Mechanisms of CaMKII action in long-term potentiation. *Nat Rev Neurosci*. 2012; 13(3):169–182.
238. Schulman H, Greengard P. Ca^{2+} -dependent protein phosphorylation system in membranes from various tissues, and its activation by calcium-dependent regulator. *Proc Natl Acad Sci U S A*. 1978;75(11):5432–5436.
239. Tombes RM, Faison MO, Turbeville JM. Organization and evolution of multifunctional Ca^{2+} /CaM-dependent protein kinase genes. *Gene*. 2003;322:17–31.
240. Chao LH, Stratton MM, Lee IH, et al. A mechanism for tunable autoinhibition in the structure of a human Ca^{2+} /calmodulin-dependent kinase II holoenzyme. *Cell*. 2011;146(5):732–745.
241. Bayer KU, Schulman H. CaM kinase: still inspiring at 40. *Neuron*. 2019;103(3):380–394.
242. Burgos JI, Yeves AM, Barrena JP, et al. Nitric oxide and CaMKII: critical steps in the cardiac contractile response to IGF-1 and swim training. *J Mol Cell Cardiol*. 2017;112:16–26.
243. Beckendorf J, van den Hoogenhof MMG, Backs J. Physiological and unappreciated roles of CaMKII in the heart. *Basic Res Cardiol*. 2018;113(4):29.
244. Feng N, Anderson ME. CaMKII is a nodal signal for multiple programmed cell death pathways in heart. *J Mol Cell Cardiol*. 2017;103:102–109.
245. Weinreuter M, Kreusser MM, Beckendorf J, et al. CaM Kinase II mediates maladaptive post-infarct remodeling and pro-inflammatory chemoattractant signaling but not acute myocardial ischemia/reperfusion injury. *EMBO Mol Med*. 2014;6(10):1231–1245.
246. Shibuya H, Iwata H, Masuyama N, et al. Role of TAK1 and TAB1 in BMP signaling in early *Xenopus* development. *EMBO J*. 1998; 17(4):1019–1028.
247. Yamaguchi K, Shirakabe K, Shibuya H, et al. Identification of a member of the MAPKKK family as a potential mediator of TGF-beta signal transduction. *Science*. 1995;270(5244): 2008–2011.
248. Bosman MCJ, Schepers H, Jaques J, et al. The TAK1-NF- κ B axis as therapeutic target for AML. *Blood*. 2014;124(20): 3130–3140.
249. Xu YR, Lei CQ. TAK1-TABs complex: a central signalosome in inflammatory responses. *Front Immunol*. 2020;11:608976.
250. Tan L, Gurbani D, Weisberg EL, et al. Structure-guided development of covalent TAK1 inhibitors. *Bioorg Med Chem*. 2017;25(3):838–846.
251. Zonneville J, Wong V, Limoge M, et al. TAK1 signaling regulates p53 through a mechanism involving ribosomal stress. *Sci Rep*. 2020;10:2517.
252. Omori E, Matsumoto K, Sanjo H, et al. TAK1 is a master regulator of epidermal homeostasis involving skin inflammation and apoptosis. *J Biol Chem*. 2006;281(28):19610–19617.
253. Totzke J, Gurbani D, Raphemot R, et al. Takinib, a selective TAK1 inhibitor, broadens the therapeutic efficacy of TNF- α inhibition for cancer and autoimmune disease. *Cell Chem Biol*. 2017;24(8):1029–1039.e7.
254. Park SH, Kwak JA, Jung SH, et al. Piperidylmethoxychalcone improves immune-mediated acute liver failure via inhibiting TAK1 activity. *Exp Mol Med*. 2017;49(11):e392.
255. Coulombe P, Meloche S. Atypical mitogen-activated protein kinases: structure, regulation and functions. *Biochim Biophys Acta*. 2007;1773(8):1376–1387.
256. Choi KW, Benzer S. Rotation of photoreceptor clusters in the developing drosophila eye requires the *nemo* gene. *Cell*. 1994;78(1):125–136.

257. Brott BK, Pinsky BA, Erikson RL. Nlk is a murine protein kinase related to Erk/MAP kinases and localized in the nucleus. *Proc Natl Acad Sci U S A*. 1998;95(3):963–968.
258. Li M, Wang H, Huang T, et al. TAB2 scaffolds TAK1 and NLK in repressing canonical Wnt signaling. *J Biol Chem*. 2010;285(18):13397–13404.
259. Yamada M, Ohnishi J, Ohkawara B, et al. NARF, an Nemo-like kinase (NLK)-associated ring finger protein regulates the ubiquitylation and degradation of T cell factor/lymphoid enhancer factor (TCF/LEF). *J Biol Chem*. 2006;281(30):20749–20760.
260. Chen J, Lin Q, Ni T, et al. NLK interacts with 14-3-3 ζ to restore the expression of E-cadherin. *Oncol Rep*. 2020;43(6):1845–1852.
261. Chen X, Zhou Y, Wan Y, et al. The expression of NLK is functionally associated with colorectal cancers (CRC). *J Cancer*. 2021;12(23):7088–7100.
262. Liu R, Khalil H, Lin SJ, et al. Nemo-like kinase (NLK) is a pathological signaling effector in the mouse heart. *PLoS One*. 2016;11(10):e0164897.
263. Creamer TP. Calcineurin. *Cell Commun Signal*. 2020;18:137.
264. Wang JH, Desai R. A brain protein and its effect on the CA²⁺- and protein modulator-activated cyclic nucleotide phosphodiesterase. *Biochem Biophys Res Commun*. 1976;72(3):926–932.
265. Rusnak F, Mertz P. Calcineurin: form and function. *Physiol Rev*. 2000;80(4):1483–1521.
266. Klee CB, Ren H, Wang X. Regulation of the calmodulin-stimulated protein phosphatase, calcineurin. *J Biol Chem*. 1998;273(22):13367–13370.
267. Tarasova EO, Gaydukov AE, Balezina OP. Calcineurin and its role in synaptic transmission. *Biochemistry Mosc*. 2018;83(6):674–689.
268. Parra V, Rothermel BA. Calcineurin signaling in the heart: the importance of time and place. *J Mol Cell Cardiol*. 2017;103:121–136.
269. Cruciat CM, Niehrs C. Secreted and transmembrane Wnt inhibitors and activators. *Cold Spring Harbor Perspect Biol*. 2013;5(3):a015081.
270. Hoang B, Moos Jr M, Vukicevic S, et al. Primary structure and tissue distribution of FRZB, a novel protein related to *Drosophila* frizzled, suggest a role in skeletal morphogenesis. *J Biol Chem*. 1996;271(42):26131–26137.
271. Chong JM, Üren A, Rubin JS, et al. Disulfide bond assignments of secreted frizzled-related protein-1 provide insights about Frizzled homology and netrin modules. *J Biol Chem*. 2002;277(7):5134–5144.
272. Leyns L, Bouwmeester T, Kim SH, et al. Frzb-1 is a secreted antagonist of Wnt signaling expressed in the Spemann organizer. *Cell*. 1997;88(6):747–756.
273. Lin K, Wang S, Julius MA, et al. The cysteine-rich frizzled domain of Frzb-1 is required and sufficient for modulation of Wnt signaling. *Proc Natl Acad Sci U S A*. 1997;94(21):11196–11200.
274. Wang S, Krinks M, Lin K, et al. Frzb, a secreted protein expressed in the Spemann organizer, binds and inhibits Wnt-8. *Cell*. 1997;88(6):757–766.
275. Bafico A, Gazit A, Pramila T, et al. Interaction of frizzled related protein (FRP) with Wnt ligands and the frizzled receptor suggests alternative mechanisms for FRP inhibition of Wnt signaling. *J Biol Chem*. 1999;274(23):16180–16187.
276. Bi Y, Huang J, He Y, et al. Wnt antagonist SFRP3 inhibits the differentiation of mouse hepatic progenitor cells. *J Cell Biochem*. 2009;108(1):295–303.
277. Bhat RA, Stauffer B, Komm BS, et al. Structure-function analysis of secreted frizzled-related protein-1 for its Wnt antagonist function. *J Cell Biochem*. 2007;102(6):1519–1528.
278. Lopez-Rios J, Esteve P, Ruiz JM, et al. The Netrin-related domain of Sfrp1 interacts with Wnt ligands and antagonizes their activity in the anterior neural plate. *Neural Dev*. 2008;3:19.
279. Hsieh JC, Kodjabachian L, Rebbert ML, et al. A new secreted protein that binds to Wnt proteins and inhibits their activities. *Nature*. 1999;398(6726):431–436.
280. Glinka A, Wu W, Delius H, et al. Dickkopf-1 is a member of a new family of secreted proteins and functions in head induction. *Nature*. 1998;391(6665):357–362.
281. Niehrs C. Function and biological roles of the Dickkopf family of Wnt modulators. *Oncogene*. 2006;25(57):7469–7481.
282. Pandur P, Läsche M, Eisenberg LM, et al. Wnt-11 activation of a non-canonical Wnt signalling pathway is required for cardiogenesis. *Nature*. 2002;418(6898):636–641.
283. Ellies DL, Viviano B, McCarthy J, et al. Bone density ligand, Sclerostin, directly interacts with LRP5 but not LRP5G171V to modulate Wnt activity. *J Bone Miner Res*. 2006;21(11):1738–1749.
284. Itasaki N, Jones CM, Mercurio S, et al. Wise, a context-dependent activator and inhibitor of Wnt signalling. *Development*. 2003;130(18):4295–4305.
285. Lintern KB, Guidato S, Rowe A, et al. Characterization of wise protein and its molecular mechanism to interact with both Wnt and BMP signals. *J Biol Chem*. 2009;284(34):23159–23168.
286. Paszty C, Turner CH, Robinson MK. Sclerostin: a gem from the genome leads to bone-building antibodies. *J Bone Miner Res*. 2010;25(9):1897–1904.
287. Guidato S, Itasaki N. Wise retained in the endoplasmic reticulum inhibits Wnt signaling by reducing cell surface LRP6. *Dev Biol*. 2007;310(2):250–263.
288. Ohazama A, Johnson EB, Ota MS, et al. Lrp4 modulates extracellular integration of cell signaling pathways in development. *PLoS One*. 2008;3(12):e4092.
289. Li X, Zhang Y, Kang H, et al. Sclerostin binds to LRP5/6 and antagonizes canonical Wnt signaling. *J Biol Chem*. 2005;280(20):19883–19887.
290. Seménov M, Tamai K, He X. SOST is a ligand for LRP5/LRP6 and a Wnt signaling inhibitor. *J Biol Chem*. 2005;280(29):26770–26775.
291. Bouwmeester T, Kim S, Sasai Y, et al. *Cerberus* is a head-inducing secreted factor expressed in the anterior endoderm of Spemann's organizer. *Nature*. 1996;382(6592):595–601.
292. Aykul S, Martinez-Hackert E. New ligand binding function of human *Cerberus* and role of proteolytic processing in regulating ligand-receptor interactions and antagonist activity. *J Mol Biol*. 2016;428(3):590–602.
293. Biben C, Stanley E, Fabri L, et al. Murine *Cerberus* homologue mCer-1: a candidate anterior patterning molecule. *Dev Biol*. 1998;194(2):135–151.
294. Piccolo S, Agius E, Leyns L, et al. The head inducer *Cerberus* is a multifunctional antagonist of Nodal, BMP and Wnt signals. *Nature*. 1999;397(6721):707–710.
295. Zhu W, Shiojima I, Ito Y, et al. IGFBP-4 is an inhibitor of canonical Wnt signalling required for cardiogenesis. *Nature*. 2008;454(7202):345–349.
296. Ueno K, Hirata H, Majid S, et al. IGFBP-4 activates the Wnt/beta-catenin signaling pathway and induces M-CAM expression in human renal cell carcinoma. *Int J Cancer*. 2011;129(10):2360–2369.
297. Neufeld KL. Nuclear APC. *Adv Exp Med Biol*. 2009;656:13–29.
298. Spink KE, Polakis P, Weis WI. Structural basis of the Axin-adenomatous polyposis coli interaction. *EMBO J*. 2000;19(10):2270–2279.
299. Su Y, Fu C, Ishikawa S, et al. APC is essential for targeting phosphorylated beta-catenin to the SCFbeta-TrCP ubiquitin ligase. *Mol Cell*. 2008;32(5):652–661.
300. Day CL, Alber T. Crystal structure of the amino-terminal coiled-coil domain of the APC tumor suppressor. *J Mol Biol*. 2000;301(1):147–156.

301. Watanabe T, Wang S, Noritake J, et al. Interaction with IQGAP1 links APC to Rac 1, Cdc 42, and actin filaments during cell polarization and migration. *Dev Cell*. 2004;7(6): 871–883.
302. Jimbo T, Kawasaki Y, Koyama R, et al. Identification of a link between the tumour suppressor APC and the kinesin superfamily. *Nat Cell Biol*. 2002;4(4):323–327.
303. Smits R, Kielman MF, Breukel C, et al. Apc1638T: a mouse model delineating critical domains of the adenomatous polyposis coli protein involved in tumorigenesis and development. *Genes Dev*. 1999;13(10):1309–1321.
304. Half E, Bercovich D, Rozen P. Familial adenomatous polyposis. *Orphanet J Rare Dis*. 2009;4:22.
305. Zeng L, Fagotto F, Zhang T, et al. The mouse *Fused* locus encodes axin, an inhibitor of the Wnt signaling pathway that regulates embryonic axis formation. *Cell*. 1997;90(1): 181–192.
306. Mazzoni SM, Fearon ER. *AXIN1* and *AXIN2* variants in gastrointestinal cancers. *Cancer Lett*. 2014;355(1):1–8.
307. Mallick A, Taylor SKB, Ranawade A, et al. Axin family of scaffolding proteins in development: lessons from *C. elegans*. *J Dev Biol*. 2019;7(4):20.
308. Kishida M, Koyama S, Kishida S, et al. Axin prevents Wnt-3a-induced accumulation of beta-catenin. *Oncogene*. 1999; 18(4):979–985.
309. Hsu W, Zeng L, Costantini F. Identification of a domain of Axin that binds to the serine/threonine protein phosphatase 2A and a self-binding domain. *J Biol Chem*. 1999;274(6): 3439–3445.
310. Rui Y, Xu Z, Lin S, et al. Axin stimulates p53 functions by activation of HIPK₂ kinase through multimeric complex formation. *EMBO J*. 2004;23(23):4583–4594.
311. Dao DY, Yang X, Chen D, et al. Axin 1 and Axin 2 are regulated by TGF- and mediate cross-talk between TGF- and Wnt signaling pathways. *Ann N Y Acad Sci*. 2007;1116:82–99.
312. Embi N, Rylatt DB, Cohen P. Glycogen synthase kinase-3 from rabbit skeletal muscle. Separation from cyclic-AMP-dependent protein kinase and phosphorylase kinase. *Eur J Biochem*. 1980;107(2):519–527.
313. Rylatt DB, Aitken A, Bilham T, et al. Glycogen synthase from rabbit skeletal muscle. Amino acid sequence at the sites phosphorylated by glycogen synthase kinase-3, and extension of the N-terminal sequence containing the site phosphorylated by phosphorylase kinase. *Eur J Biochem*. 1980;107(2): 529–537.
314. Lin J, Song T, Li C, et al. GSK-3 β in DNA repair, apoptosis, and resistance of chemotherapy, radiotherapy of cancer. *Biochim Biophys Acta Mol Cell Res*. 2020;1867(5):118659.
315. Woodgett JR. Molecular cloning and expression of glycogen synthase kinase-3/factor A. *EMBO J*. 1990;9(8):2431–2438.
316. Kaidanovich-Beilin O, Lipina TV, Takao K, et al. Abnormalities in brain structure and behavior in GSK-3 alpha mutant mice. *Mol Brain*. 2009;2:35.
317. Hoeflich KP, Luo J, Rubie EA, et al. Requirement for glycogen synthase kinase-3 beta in cell survival and NF-kappaB activation. *Nature*. 2000;406(6791):86–90.
318. MacAulay K, Doble BW, Patel S, et al. Glycogen synthase kinase 3 alpha-specific regulation of murine hepatic glycogen metabolism. *Cell Metabol*. 2007;6(4):329–337.
319. Roach PJ. Glycogen and its metabolism. *Curr Mol Med*. 2002; 2(2):101–120.
320. Leung-Hagesteijn C, Mahendra A, Naruszewicz I, et al. Modulation of integrin signal transduction by ILKAP, a protein phosphatase 2C associating with the integrin-linked kinase, ILK1. *EMBO J*. 2001;20(9):2160–2170.
321. Ding VW, Chen RH, McCormick F. Differential regulation of glycogen synthase kinase 3 β by insulin and Wnt signaling. *J Biol Chem*. 2000;275(42):32475–32481.
322. Beurel E, Grieco SF, Jope RS. Glycogen synthase kinase-3 (GSK3): regulation, actions, and diseases. *Pharmacol Ther*. 2015;148:114–131.
323. Klein PS, Melton DA. A molecular mechanism for the effect of lithium on development. *Proc Natl Acad Sci U S A*. 1996; 93(16):8455–8459.
324. Watcharasil P, Bijur GN, Song L, et al. Glycogen synthase kinase-3 beta (GSK3beta) binds to and promotes the actions of p53. *J Biol Chem*. 2003;278(49):48872–48879.
325. Souder DC, Anderson RM. An expanding GSK3 network: implications for aging research. *GeroScience*. 2019;41(4): 369–382.
326. Janovská P, Normant E, Miskin H, et al. Targeting casein kinase 1 (CK1) in hematological cancers. *Int J Mol Sci*. 2020; 21(23):9026.
327. Knippschild U, Krüger M, Richter J, et al. The CK1 family: contribution to cellular stress response and its role in carcinogenesis. *Front Oncol*. 2014;4:96.
328. Carmel G, Leichus B, Cheng X, et al. Expression, purification, crystallization, and preliminary X-ray analysis of casein kinase-1 from *Schizosaccharomyces pombe*. *J Biol Chem*. 1994; 269(10):7304–7309.
329. Gietzen KF, Virshup DM. Identification of inhibitory autophosphorylation sites in casein kinase I ϵ . *J Biol Chem*. 1999; 274(45):32063–32070.
330. Fulcher LJ, Sapkota GP. Functions and regulation of the serine/threonine protein kinase CK1 family: moving beyond promiscuity. *Biochem J*. 2020;477(23):4603–4621.
331. Marin O, Bustos VH, Cesaro L, et al. A noncanonical sequence phosphorylated by casein kinase 1 in beta-catenin may play a role in casein kinase 1 targeting of important signaling proteins. *Proc Natl Acad Sci U S A*. 2003;100(18): 10193–10200.
332. Dumaz N, Milne DM, Meek DW. Protein kinase CK1 is a p53-threonine 18 kinase which requires prior phosphorylation of serine 15. *FEBS Lett*. 1999;463(3):312–316.
333. Elyada E, Pribluda A, Goldstein RE, et al. CK1 α ablation highlights a critical role for p53 in invasiveness control. *Nature*. 2011;470(7334):409–413.
334. Fulcher LJ, He Z, Mei L, et al. FAM83D directs protein kinase CK1 α to the mitotic spindle for proper spindle positioning. *EMBO Rep*. 2019;20(9):e47495.
335. Fuchs SY, Spiegelman VS, Kumar KG. The many faces of beta-TrCP E3 ubiquitin ligases: reflections in the magic mirror of cancer. *Oncogene*. 2004;23(11):2028–2036.
336. Kitagawa M, Hatakeyama S, Shirane M, et al. An F-box protein, FWD1, mediates ubiquitin-dependent proteolysis of beta-catenin. *EMBO J*. 1999;18(9):2401–2410.
337. Bai C, Sen P, Hofmann K, et al. SKP1 connects cell cycle regulators to the ubiquitin proteolysis machinery through a novel motif, the F-box. *Cell*. 1996;86(2):263–274.
338. Karin M, Ben-Neriah Y. Phosphorylation meets ubiquitination: the control of NF-[kappa]B activity. *Annu Rev Immunol*. 2000; 18:621–663.
339. Guardavaccaro D, Kudo Y, Boulaire J, et al. Control of meiotic and mitotic progression by the F box protein β -Trcp1 *in vivo*. *Dev Cell*. 2003;4(6):799–812.
340. Margottin-Goguet F, Hsu JY, Loktev A, et al. Prophase destruction of Emi 1 by the SCF $^{\beta$ TrCP/slimb ubiquitin ligase activates the anaphase promoting complex to allow progression beyond prometaphase. *Dev Cell*. 2003;4(6):813–826.
341. Mayo MW, Baldwin AS. The transcription factor NF-kappaB: control of oncogenesis and cancer therapy resistance. *Biochim Biophys Acta*. 2000;1470(2):M55–M62.
342. Wlodarchak N, Xing Y. PP2A as a master regulator of the cell cycle. *Crit Rev Biochem Mol Biol*. 2016;51(3):162–184.
343. O'Connor CM, Perl A, Leonard D, et al. Therapeutic targeting of PP2A. *Int J Biochem Cell Biol*. 2018;96:182–193.

344. Groves MR, Hanlon N, Turowski P, et al. The structure of the protein phosphatase 2A PR65/a subunit reveals the conformation of its 15 tandemly repeated HEAT motifs. *Cell*. 1999; 96(1):99–110.
345. Longin S, Zwaenepoel K, Louis JV, et al. Selection of protein phosphatase 2A regulatory subunits is mediated by the C terminus of the catalytic subunit. *J Biol Chem*. 2007;282(37): 26971–26980.
346. Mitra A, Menezes ME, Pannell LK, et al. DNAJB6 chaperones PP2A mediated dephosphorylation of GSK3 β to downregulate β -catenin transcription target, osteopontin. *Oncogene*. 2012; 31(41):4472–4483.
347. Lowe M, Gonatas NK, Warren G. The mitotic phosphorylation cycle of the cis-Golgi matrix protein GM130. *J Cell Biol*. 2000; 149(2):341–356.
348. Verbinen I, Vaneynde P, Reynhout S, et al. Protein Phosphatase 2A (PP2A) mutations in brain function, development, and neurologic disease. *Biochem Soc Trans*. 2021;49(4): 1567–1588.
349. Torrent L, Ferrer I. PP2A and alzheimer disease. *Curr Alzheimer Res*. 2012;9(2):248–256.
350. Nader CP, Cidem A, Verrills NM, et al. Protein phosphatase 2A (PP2A): a key phosphatase in the progression of chronic obstructive pulmonary disease (COPD) to lung cancer. *Respir Res*. 2019;20:222.
351. Chen G, Courey AJ. Groucho/TLE family proteins and transcriptional repression. *Gene*. 2000;249(1–2):1–16.
352. Daniels DL, Weis WI. Beta-catenin directly displaces Groucho/TLE repressors from Tcf/Lef in Wnt-mediated transcription activation. *Nat Struct Mol Biol*. 2005;12(4):364–371.
353. Chodaparambil JV, Pate KT, Hepler MRD, et al. Molecular functions of the TLE tetramerization domain in Wnt target gene repression. *EMBO J*. 2014;33(7):719–731.
354. Chen G, Fernandez J, Mische S, et al. A functional interaction between the histone deacetylase Rpd3 and the corepressor groucho in *Drosophila* development. *Genes Dev*. 1999;13(17): 2218–2230.
355. Yuzugullu H, Benhaj K, Ozturk N, et al. Canonical Wnt signaling is antagonized by noncanonical Wnt5a in hepatocellular carcinoma cells. *Mol Cancer*. 2009;8:90.
356. Guo X, Wang XF. Signaling cross-talk between TGF- β /BMP and other pathways. *Cell Res*. 2009;19(1):71–88.
357. Shi Y, Massagué J. Mechanisms of TGF- β signaling from cell membrane to the nucleus. *Cell*. 2003;113(6):685–700.
358. He G, Shi Y, Lim J, et al. Differential involvement of Wnt signaling in Bmp regulation of cancellous versus periosteal bone growth. *Bone Res*. 2017;5:17016.
359. Zhang F, Song J, Zhang H, et al. Wnt and BMP signaling crosstalk in regulating dental stem cells: implications in dental tissue engineering. *Genes Dis*. 2016;3(4):263–276.
360. Luo W, Zhang L, Huang B, et al. BMP9-initiated osteogenic/odontogenic differentiation of mouse tooth germ mesenchymal cells (TGMCS) requires Wnt/ β -catenin signalling activity. *J Cell Mol Med*. 2021;25(5):2666–2678.
361. Zhang L, Luo Q, Shu Y, et al. Transcriptomic landscape regulated by the 14 types of bone morphogenetic proteins (BMPs) in lineage commitment and differentiation of mesenchymal stem cells (MSCs). *Genes Dis*. 2019;6(3): 258–275.
362. Zhang H, Wang J, Deng F, et al. Canonical Wnt signaling acts synergistically on BMP9-induced osteo/odontoblastic differentiation of stem cells of dental apical papilla (SCAPs). *Biomaterials*. 2015;39:145–154.
363. Luo Q, Kang Q, Si W, et al. Connective tissue growth factor (CTGF) is regulated by Wnt and bone morphogenetic proteins signaling in osteoblast differentiation of mesenchymal stem cells. *J Biol Chem*. 2004;279(53):55958–55968.
364. Millar SE. Smad 7: licensed to kill β -catenin. *Dev Cell*. 2006; 11(3):274–276.
365. Tang Y, Liu Z, Zhao L, et al. Smad 7 stabilizes beta-catenin binding to E-cadherin complex and promotes cell-cell adhesion. *J Biol Chem*. 2008;283(35):23956–23963.
366. Kamiya N, Kobayashi T, Mochida Y, et al. Wnt inhibitors Dkk 1 and Sost are downstream targets of BMP signaling through the type IA receptor (BMPRIA) in osteoblasts. *J Bone Miner Res*. 2010;25(2):200–210.
367. Tang N, Song WX, Luo J, et al. BMP-9-induced osteogenic differentiation of mesenchymal progenitors requires functional canonical Wnt/ β -catenin signalling. *J Cell Mol Med*. 2009;13(8B):2448–2464.
368. Zhang M, Yan Y, Lim YB, et al. BMP-2 modulates beta-catenin signaling through stimulation of Lrp5 expression and inhibition of beta-TrCP expression in osteoblasts. *J Cell Biochem*. 2009; 108(4):896–905.
369. Fischer L, Boland G, Tuan RS. Wnt-3A enhances bone morphogenetic protein-2-mediated chondrogenesis of murine C3H10T1/2 mesenchymal cells. *J Biol Chem*. 2002;277(34): 30870–30878.
370. Zhang R, Oyajobi BO, Harris SE, et al. Wnt/ β -catenin signaling activates bone morphogenetic protein 2 expression in osteoblasts. *Bone*. 2013;52(1):145–156.
371. Rodríguez-Carballo E, Ulsamer A, Susperregui ARG, et al. Conserved regulatory motifs in osteogenic gene promoters integrate cooperative effects of canonical Wnt and BMP pathways. *J Bone Miner Res*. 2011;26(4):718–729.
372. Liu Z, Tang Y, Qiu T, et al. A dishevelled-1/Smad 1 interaction couples WNT and bone morphogenetic protein signaling pathways in uncommitted bone marrow stromal cells. *J Biol Chem*. 2006;281(25):17156–17163.
373. Hoppler S, Moon RT. BMP-2/-4 and Wnt-8 cooperatively pattern the *Xenopus* mesoderm. *Mech Dev*. 1998;71(1–2):119–129.
374. Wu M, Chen G, Li YP. TGF- β and BMP signaling in osteoblast, skeletal development, and bone formation, homeostasis and disease. *Bone Res*. 2016;4:16009.
375. Morrell NW, Bloch DB, ten Dijke P, et al. Targeting BMP signalling in cardiovascular disease and anaemia. *Nat Rev Cardiol*. 2016;13(2):106–120.
376. Luo K. Signaling cross talk between TGF- β /smad and other signaling pathways. *Cold Spring Harbor Perspect Biol*. 2017; 9(1):a022137.
377. Eger A, Stockinger A, Park J, et al. Beta-Catenin and TGF β signalling cooperate to maintain a mesenchymal phenotype after FosER-induced epithelial to mesenchymal transition. *Oncogene*. 2004;23(15):2672–2680.
378. Nawshad A, Medici D, Liu CC, et al. TGF β 3 inhibits E-cadherin gene expression in palate medial-edge epithelial cells through a Smad 2-Smad 4-LEF $_1$ transcription complex. *J Cell Sci*. 2007;120(Pt 9):1646–1653.
379. Zhou B, Liu Y, Kahn M, et al. Interactions between β -catenin and transforming growth factor- β signaling pathways mediate epithelial-mesenchymal transition and are dependent on the transcriptional co-activator cAMP-response element-binding protein (CREB)-binding protein (CBP). *J Biol Chem*. 2012; 287(10):7026–7038.
380. Jian H, Shen X, Liu I, et al. Smad 3-dependent nuclear translocation of beta-catenin is required for TGF- β 1-induced proliferation of bone marrow-derived adult human mesenchymal stem cells. *Genes Dev*. 2006;20(6):666–674.
381. Edeling M, Ragi G, Huang S, et al. Developmental signalling pathways in renal fibrosis: the roles of Notch, Wnt and Hedgehog. *Nat Rev Nephrol*. 2016;12(7):426–439.
382. Wang H, Zang C, Liu XS, et al. The role of Notch receptors in transcriptional regulation. *J Cell Physiol*. 2015;230(5): 982–988.

383. Sprinzak D, Blacklow SC. Biophysics of Notch signaling. *Annu Rev Biophys.* 2021;50:157–189.
384. Hayward P, Kalmar T, Arias AM. Wnt/Notch signalling and information processing during development. *Development.* 2008;135(3):411–424.
385. Fujimura S, Jiang Q, Kobayashi C, et al. Notch 2 activation in the embryonic kidney depletes nephron progenitors. *J Am Soc Nephrol.* 2010;21(5):803–810.
386. Galceran J, Sustmann C, Hsu SC, et al. LEF1-mediated regulation of Delta-like 1 links Wnt and Notch signaling in somitogenesis. *Genes Dev.* 2004;18(22):2718–2723.
387. Borggrefe T, Lauth M, Zwijsen A, et al. The Notch intracellular domain integrates signals from Wnt, Hedgehog, TGF β /BMP and hypoxia pathways. *Biochim Biophys Acta.* 2016;1863(2):303–313.
388. Jin YH, Kim H, Ki H, et al. Beta-catenin modulates the level and transcriptional activity of Notch 1/NICD through its direct interaction. *Biochim Biophys Acta.* 2009;1793(2):290–299.
389. Kwon C, Cheng P, King IN, et al. Notch post-translationally regulates β -catenin protein in stem and progenitor cells. *Nat Cell Biol.* 2011;13(10):1244–1251.
390. Foltz DR, Santiago MC, Berechid BE, et al. Glycogen synthase kinase-3 beta modulates Notch signaling and stability. *Curr Biol.* 2002;12(12):1006–1011.
391. Ann EJ, Kim HY, Seo MS, et al. Wnt5a controls Notch 1 signaling through CaMKII-mediated degradation of the SMRT corepressor protein. *J Biol Chem.* 2012;287(44):36814–36829.
392. Collu GM, Hidalgo-Sastre A, Acar A, et al. Dishevelled limits Notch signalling through inhibition of CSL. *Development.* 2012;139(23):4405–4415.
393. Pinto I, Duque M, Gonçalves J, et al. NRARP displays either pro- or anti-tumoral roles in T-cell acute lymphoblastic leukemia depending on Notch and Wnt signaling. *Oncogene.* 2020;39(5):975–986.
394. Rodilla V, Villanueva A, Obrador-Hevia A, et al. Jagged 1 is the pathological link between Wnt and Notch pathways in colorectal cancer. *Proc Natl Acad Sci U S A.* 2009;106(15):6315–6320.
395. Fendler A, Bauer D, Busch J, et al. Inhibiting WNT and NOTCH in renal cancer stem cells and the implications for human patients. *Nat Commun.* 2020;11:929.
396. Zhou J, Cheng P, Youn JI, et al. Notch and wingless signaling cooperate in regulation of dendritic cell differentiation. *Immunity.* 2009;30(6):845–859.
397. Germar K, Dose M, Konstantinou T, et al. T-cell factor 1 is a gatekeeper for T-cell specification in response to Notch signaling. *Proc Natl Acad Sci U S A.* 2011;108(50):20060–20065.
398. MacGrogan D, Münch J, de la Pompa JL. Notch and interacting signalling pathways in cardiac development, disease, and regeneration. *Nat Rev Cardiol.* 2018;15(11):685–704.
399. Adams JM, Jafar-Nejad H. The roles of Notch signaling in liver development and disease. *Biomolecules.* 2019;9(10):608.
400. Canalis E. Notch in skeletal physiology and disease. *Osteoporos Int.* 2018;29(12):2611–2621.
401. Ma S, Meng Z, Chen R, et al. The hippo pathway: biology and pathophysiology. *Annu Rev Biochem.* 2019;88:577–604.
402. Zhao B, Li L, Tumaneng K, et al. A coordinated phosphorylation by lats and CK1 regulates YAP stability through SCF(β -TRCP). *Genes Dev.* 2010;24(1):72–85.
403. Liu CY, Zha ZY, Zhou X, et al. The hippo tumor pathway promotes TAZ degradation by phosphorylating a phosphodegron and recruiting the SCF(β -TRCP) E3 ligase. *J Biol Chem.* 2010;285(48):37159–37169.
404. Xu F, Wang YL, Chang JJ, et al. Mammalian sterile 20-like kinase 1/2 inhibits the Wnt/ β -catenin signalling pathway by directly binding casein kinase 1 ϵ . *Biochem J.* 2014;458(1):159–169.
405. Imajo M, Miyatake K, Imura A, et al. A molecular mechanism that links Hippo signalling to the inhibition of Wnt/ β -catenin signalling. *EMBO J.* 2012;31(5):1109–1122.
406. Piccolo S, Dupont S, Cordenonsi M. The biology of YAP/TAZ: hippo signaling and beyond. *Physiol Rev.* 2014;94(4):1287–1312.
407. Jiang L, Li J, Zhang C, et al. YAP-mediated crosstalk between the Wnt and Hippo signaling pathways (Review). *Mol Med Rep.* 2020;22(5):4101–4106.
408. Park HW, Kim YC, Yu B, et al. Alternative Wnt signaling activates YAP/TAZ. *Cell.* 2015;162(4):780–794.
409. Heallen T, Zhang M, Wang J, et al. Hippo pathway inhibits Wnt signaling to restrain cardiomyocyte proliferation and heart size. *Science.* 2011;332(6028):458–461.
410. Carballo GB, Honorato JR, de Lopes GPF, et al. A highlight on Sonic hedgehog pathway. *Cell Commun Signal.* 2018;16:11.
411. Jia J, Tong C, Wang B, et al. Hedgehog signalling activity of Smoothened requires phosphorylation by protein kinase A and casein kinase I. *Nature.* 2004;432(7020):1045–1050.
412. Chen Y, Sasai N, Ma G, et al. Sonic Hedgehog dependent phosphorylation by CK1 α and GRK2 is required for ciliary accumulation and activation of smoothened. *PLoS Biol.* 2011;9(6):e1001083.
413. Shi Q, Li S, Li S, et al. Hedgehog-induced phosphorylation by CK1 sustains the activity of Ci/Gli activator. *Proc Natl Acad Sci U S A.* 2014;111(52):E5651–E5660.
414. Zhou Q, Kalderon D. Hedgehog activates fused through phosphorylation to elicit a full spectrum of pathway responses. *Dev Cell.* 2011;20(6):802–814.
415. Takenaka K, Kise Y, Miki H. GSK3 β positively regulates Hedgehog signaling through Sufu in mammalian cells. *Biochem Biophys Res Commun.* 2007;353(2):501–508.
416. He J, Sheng T, Stelzer AA, et al. Suppressing Wnt signaling by the hedgehog pathway through sFRP-1. *J Biol Chem.* 2006;281(47):35598–35602.
417. Meng X, Poon R, Zhang X, et al. Suppressor of fused negatively regulates β -catenin signaling. *J Biol Chem.* 2001;276(43):40113–40119.
418. Ebeid M, Huh SH. FGF signaling: diverse roles during cochlear development. *BMB Rep.* 2017;50(10):487–495.
419. Katoh M, Katoh M. Cross-talk of WNT and FGF signaling pathways at GSK3 β to regulate beta-catenin and SNAIL signaling cascades. *Cancer Biol Ther.* 2006;5(9):1059–1064.
420. Cohen ED, Wang Z, Lepore JJ, et al. Wnt/ β -catenin signaling promotes expansion of Isl-1-positive cardiac progenitor cells through regulation of FGF signaling. *J Clin Invest.* 2007;117(7):1794–1804.
421. Volckaert T, De Langhe SP. Wnt and FGF mediated epithelial-mesenchymal crosstalk during lung development. *Dev Dynam.* 2015;244(3):342–366.
422. Sutkeviciute I, Clark LJ, White AD, et al. PTH/PTHrP receptor signaling, allostery, and structures. *Trends Endocrinol Metab.* 2019;30(11):860–874.
423. Kakar S, Einhorn TA, Vora S, et al. Enhanced chondrogenesis and Wnt signaling in PTH-treated fractures. *J Bone Miner Res.* 2007;22(12):1903–1912.
424. Tobimatsu T, Kaji H, Sowa H, et al. Parathyroid hormone increases beta-catenin levels through Smad 3 in mouse osteoblastic cells. *Endocrinology.* 2006;147(5):2583–2590.

425. Romero G, Sneddon WB, Yang Y, et al. Parathyroid hormone receptor directly interacts with dishevelled to regulate β -catenin signaling and osteoclastogenesis. *J Biol Chem.* 2010; 285(19):14756–14763.
426. Cabrae R, Dubuquoy C, Caüzac M, et al. Insulin activates hepatic Wnt/ β -catenin signaling through stearyl-CoA desaturase 1 and porcupine. *Sci Rep.* 2020;10:5186.
427. Ugi S, Imamura T, Ricketts W, et al. Protein phosphatase 2A forms a molecular complex with Shc and regulates Shc tyrosine phosphorylation and downstream mitogenic signaling. *Mol Cell Biol.* 2002;22(7):2375–2387.
428. Zwaenepoel K, Goris J, Erneux C, et al. Protein phosphatase 2A PR130/B''alpha 1 subunit binds to the SH2 domain-containing inositol polyphosphate 5-phosphatase 2 and prevents epidermal growth factor (EGF)-induced EGF receptor degradation sustaining EGF-mediated signaling. *FASEB J.* 2010; 24(2):538–547.